

LINEAR EQUATIONS OVER FREE LIE ALGEBRAS

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Alaa Altassan
School of Mathematics

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Alaa Altassan

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In this thesis, we study equations of the form $[x_1, u_1] + [x_2, u_2] + \dots + [x_k, u_k] = 0$ over a free Lie algebra L , where $k > 1$ and the coefficients $u_1, u_2, \dots, u_k$ belong to L . The starting point of this research is a paper [22], in which the authors embarked on a systematic study of very concrete linear equations over free Lie algebras. They focused on the given equations in the case where $k = 2$. We generalise and develop a number of the results on equations with two variables to equations with an arbitrary number of indeterminates. Most of the results refer to the case where the coefficients coincide with the free generators of L . Throughout our research, we study some features of the solution space of these equations such as the homogenous structure and the fine homogenous structure. The main achievement in this work is that we give a detailed description of the solution space. Then we obtain explicit bases for some specific fine homogeneous components of the solution space, in particular, we give a basis for the “multilinear” fine homogenous component. Moreover, we generalise earlier results on commutator calculus using the “language” of free Lie algebras and apply them to determine the radical and the coordinate algebra of the solution space of the given equations.

Declaration

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Chapter 1

Introduction

Throughout the history of mathematics and among generations of mathematicians, studying equations has played an important role in both applied and pure mathematical disciplines. In the former, equations are used to describe heat, motion and several natural phenomena, whilst in the latter, for example, identities appear in the definitions of several algebraic structures, and the equivalence notion in mathematical logic, implicitly, holds the meaning of equality. Thus, every mathematician has used equations directly or indirectly in their research. According to È. Daniyarova, V. Remeslennikov states that “No Equations $\Rightarrow$ No Mathematics” [6].

When it comes to studying equations over algebraic structures other than fields, groups would be one of the first examples to appear. A theoretical framework for studying equations over groups (called Algebraic Geometry over groups) has been developed by several group theorists. This has led to studying equations over other algebraic systems such as rings and algebras. Indeed, many of the developments that have reached the study of equations over these more complicated systems were inspired by group theoretic results. A list of papers on equations over groups can be found in [10].

The theory of equations in free Lie algebras is still in its early stages. Motivated by considerable advances in the theory of equations over groups, and in particular over free groups, in the last two decades or so, a systematic approach towards a general theory of equations over Lie algebras was developed by È. Daniyarova. An account

of her work can be found in the survey [16]. More specific problems about equations over free metabelian Lie algebras were studied in [8] and [9], and the paper [11] deals with systems of equations over free Lie algebras. A detailed survey of the results in [8, 9, 11] is given in [16]. Also, a certain progress is achieved on equations over Lie algebras in [4], [5] and [7]. Equations in simple Lie algebras were considered in [2].

A remarkable contribution in developing the theory of equations over free groups, particularly, and over other algebraic structures in general, was made by V. Remeslennikov. In 2007, a paper on equations over free Lie algebras [22] was published as a result of his work with R. Stöhr. This research was carried out while Remeslennikov was visiting the University of Manchester. The aim of the paper [22] is to begin the study of very concrete equations over free Lie algebras, in particular the equation $[x_1, u_1] + [x_2, u_2] = 0$, where u_1 and u_2 are elements in a free Lie algebra L , and each x_i is an indeterminate. This paper dealt exclusively with the case of two indeterminates. Despite its apparent simplicity, the study of this equation raised many problems that need to be solved. Some interesting questions determined the starting point of our research: Can we generalise the results in [22] to more general equations with $k > 2$ variables? What about exploiting some of the features of free Lie algebras to give shorter and more transparent proofs? More questions have appeared during our work and have been answered throughout the research.

Let L be a free Lie algebra over an arbitrary field K . In this thesis, we study linear equations of the form

$$[x_1, u_1] + [x_2, u_2] + \cdots + [x_k, u_k] = 0, \quad (1.1)$$

where $u_1, u_2, \dots, u_k \in L$ and $x_1, x_2, \dots, x_k$ are indeterminates. This research is a continuation of the investigations in [22] which focused on Eq. (1.1) in the case where $k = 2$. In the present work we generalise some of the results in [22] to equations with an arbitrary finite number of variables. Our main results refer to the case where the coefficients are precisely the free generators of L . In this case we obtain a detailed description of the solution space in terms of generators. Moreover, we exhibit a basis of the “multilinear” fine homogeneous component of this space.

More specifically, for the greater part of our work we will assume that the coefficients $u_1, u_2, \dots, u_k$ in (1.1) form an independent set, that is, they form a set of free generators for the Lie subalgebra they generate. At the beginning of Chapter 3, we introduce the notion of the associated linear map of Eq. (1.1) which allows us to identify the solution space of the equation with the kernel of this linear map. In Section 3.2 we first consider the main case where L is of rank k and the coefficients $u_1, u_2, \dots, u_k$ form a free generating set of L . In this situation, we obtain a formula for the dimension of the homogeneous components of the solution space of (1.1) (Theorem 3.1). Then by considering that $u_1, u_2, \dots, u_k$ are free generators of L , but the rank of L may be greater than k , we show that all the solutions of (1.1) are inner solutions, that is, they belong to the subalgebra generated by $u_1, u_2, \dots, u_k$ in L (Theorem 3.3). Finally, we consider the case where $\{u_1, u_2, \dots, u_k\}$ is an arbitrary independent set in L . In this case we prove that the space of inner solutions is of finite codimension in the full solution space of (1.1) (Theorem 3.5). For $k = 2$, all results in this section were obtained in [22]. In our present work we generalise these results from [22] to arbitrary k . For Theorem 3.5 this is easily done by adopting the original proof to the more general situation, but for Theorems 3.1 and 3.3 we offer genuinely new proofs. In particular, in the proof of Theorem 3.3, we have been able to replace the highly technical argument used in [22] by a shorter and more elegant proof based on a powerful lemma of Shirshov's. In the following Section 3.3 we return to the situation where the coefficients $u_1, u_2, \dots, u_k \in L$ in (1.1) are free generators of L . We exhibit the fine homogeneous structure of the solution space, and in the main case where the rank of L is k we give formulae for the dimensions of its fine homogeneous components. Moreover, we consider the action of the symmetric group of degree k on the free Lie algebra of rank k , and its action on the solution space of equation (1.1). The result in Theorem 3.7 helps in the process of obtaining bases of the homogenous components of the solution space in Chapter 4. Finally, methods of obtaining concrete solutions are discussed in Section 3.4.

In the first part of Chapter 4, we reinterpret the solution space in the main case as the kernel of the linear map $L \otimes L_1 \rightarrow L$, where L_1 is the degree one homogeneous component of L (the span of the free generators). This point of view will be exploited

in the following sections which contain our main results. In Section 4.2 we exhibit spanning sets of the kernels of the maps $L_{n-1} \otimes L_1 \rightarrow L_n$, see Theorem 4.3. In its strongest form this requires restrictions on the characteristic of the field K , but it is valid in full generality over fields of characteristic zero. In a corollary to this theorem, we point out how to obtain generating sets (as a K -space) for the homogeneous components of the solution space over an arbitrary field K . Later, in Section 4.3, we exhibit bases of some specific fine homogenous components of the solution space in arbitrary characteristic (in particular, we give a basis for the multilinear fine homogeneous component). In the final two sections of this chapter, we use the spanning sets that are given in Section 4.2 to write explicit spanning sets of the homogeneous components of the solution space of (1.1), and construct bases for the homogenous components of degree 3, 4 and 5.

The results in Chapter 3 and Chapter 4 form the most important achievements in this thesis and the majority of their content have been published in [1] (a joint paper with R. Stöhr).

In 1933, commutator calculus in combinatorial group theory was one of the major research interests of Philip Hall [14]. He introduced this area of research, and worked with commutators which can be formed from a free generating set of a free group [23]. Later, in 1950, Marshall Hall continued the study of these commutators in the case of free groups and free Lie rings. This led him to construct a basis of free Lie algebras, called a Hall basis (see [13]). In fact basic commutators (basis elements) in free groups are totally ordered. In 1960, H. V. Waldinger proved that this ordering is preserved by multiplication (i.e., “under the process of forming the commutator with the same basic commutator on both sides of an inequality” [29]). Throughout Chapter 5, we state and generalise Waldinger’s results in which an element d in the PBW-basis of the universal enveloping algebra A of a free Lie algebra L acts on an element in a Hall basis $\mathcal{H}$ of L . Also, we consider the case when d acts on an arbitrary non-zero element in L . In order to generalise the theorem that was given by Waldinger in [29], we define a total ordering on the PBW-basis of A . Later, as an application of the achieved results and under some conditions, we show that the subspace A_m of A and

$e \circ A_m$ have the same dimension, where A_m is the degree m homogenous component of the universal envelope A of L and e is a standard Lie monomial in $\mathcal{H}$. At this stage, it perhaps seems to the reader that we are far from the equations' world. Indeed, these results built the steps that helped us to complete our research. We exploit what we have obtained in this chapter to prove the main results in Chapter 6.

As our aim is to continue working concretely on equations over free Lie algebras, for the most part of this thesis we naturally use tools from linear algebra and abstract algebra (specifically free Lie algebras). In Chapter 6, we introduce the most essential notions and give basic definitions as far as is needed in the context of this thesis. In Section 6.3, we use the results from the previous chapter, and a particular solution of Eq. (1.1), which has been obtained in Section 6.2, to determine the radical and the coordinate algebra of the solution space of the given equation (when the coefficients $u_1, u_2, \dots, u_k$ are free generators of L). Finally, as an application, we determine the radical and the coordinate algebra in the case of bilinear equations over free Lie algebras. Most of the generalisations in the last two sections in this chapter are obtained by adopting arguments similar to the ones that are used in [22]. However, the new solutions that we found and Waldinger-type results are the key players in the way of proving Theorem 6.12, and consequently the rest of this chapter.

To sum up, we organise the material of this thesis as follows. In Chapter 2, we set up notation and recall some relevant facts about free Lie algebras. Throughout Chapter 3, we study the structure of the solution space of Eq. (1.1) and prove several results in different cases, as discussed. Reinterpreting the solution space of our equation, in Chapter 4, allows us to find a spanning set of the solution space and bases for particular fine homogenous components. Waldinger-type results are given in Chapter 5 and have been applied to prove several results in Chapter 6. In the latter chapter, radicals and coordinate algebras are discussed and bilinear equations are considered.

The main sources of information in this thesis are [1] and [22]. For more information about equations over several types of groups and algebras, we refer the reader to [10].

Chapter 2

Preliminaries

Our goal in this chapter is to give the background that helps to understand the new results in the thesis. For this, we begin by giving basic definitions, setting up notation and then recording some preliminary results. For more information, we refer the reader to [1], [15], [22] and [23].

2.1 Free Lie algebras

A *Lie algebra* over a field K is a vector space L with a bilinear multiplication

$$[,] : L \times L \rightarrow L$$

given by $(u, v) \mapsto [u, v]$ for all u and v in L , which satisfies the following identities:

- (i) $[u, u] = 0$,
- (ii) $[[u, v], w] + [[v, w], u] + [[w, u], v] = 0$,

$\forall u, v, w \in L$.

This multiplication is called the Lie product (or Lie bracket). The first condition (i) implies

$$[u, v] = -[v, u] \text{ (anticommutativity),} \tag{2.1}$$

but in general (2.1) does not imply (i). The second identity (ii) is called the *Jacobi identity*.

It is well-known that we can turn any associative algebra A over a field K into a Lie algebra by introducing a new multiplication

$$[u, v] = uv - vu, \quad \forall u, v \in A, \quad (2.2)$$

where uv denotes the usual product of the elements u and v in A . In fact, it is easy to verify that this new multiplication satisfies the two Lie identities (i) and (ii). According to Kevin McCrimmon, every mathematician should do this “directly, once and only once in their life” [20].

A *Lie subalgebra* of a Lie algebra L is a subspace that is closed under the Lie product. An *ideal* I in L is a subalgebra such that $[u, v] \in I$ for all $u \in L$ and $v \in I$. Note that we simply speak about ideals and do not distinguish between right ideals and left ideals. Indeed, it follows from the identity (2.1) that every ideal is two-sided [3]. The quotient of L by I is an algebra called the *factor algebra* which is usually denoted by L/I .

Let U_1, U_2, W_1 and W_2 be finite dimensional vector spaces over a field K such that $U_1 \cap U_2 = 0$. Then for the linear maps $\phi_1 : U_1 \rightarrow W_1$ and $\phi_2 : U_2 \rightarrow W_2$, we define the direct sum

$$\phi_1 \oplus \phi_2 : U_1 \oplus U_2 \rightarrow W_1 \oplus W_2$$

such that for any $(u_1, u_2) \in U_1 \oplus U_2$

$$(u_1, u_2)(\phi_1 \oplus \phi_2) = (u_1\phi_1, u_2\phi_2).$$

Let L and $\tilde{L}$ be Lie algebras over a field K . A *Lie homomorphism* is a linear map $\phi : L \rightarrow \tilde{L}$ that satisfies $[u, v]\phi = [u\phi, v\phi]$ for all u and v belonging to L . One of the famous examples of Lie homomorphisms is the adjoint homomorphism $\text{ad} : L \rightarrow \text{gl}(L)$ given by $u \mapsto \text{ad}_u$ for all $u \in L$, where $\text{gl}(L)$ is the general linear algebra (the Lie algebra of linear maps on L) and the linear map ad_u maps v into $v \text{ad}_u = [v, u]$ for all $v \in L$ [26].

Free Lie algebras are the largest Lie algebras in the sense that all other Lie algebras can be obtained as homomorphic images of the free ones. Indeed, for a Lie algebra

L generated by a set $\mathcal{A}$ over a field K , we say that L is *freely generated* by $\mathcal{A}$, if every map from $\mathcal{A}$ into an arbitrary Lie algebra $\tilde{L}$ over K , extends to a Lie algebra homomorphism from L into $\tilde{L}$. We write $L = L(\mathcal{A})$ to denote the free Lie algebra on a set $\mathcal{A}$. In the case where $\mathcal{A}$ is finite, say $|\mathcal{A}| = r$, the cardinality of $\mathcal{A}$ ($r \geq 1$) is called the rank of $L(\mathcal{A})$. Otherwise, the free Lie algebra is said to be of infinite rank.

Free Lie algebras have been a major topic of research since the 1930's. These objects are closely related to free groups. For a number of remarkable results on free groups there are similar ones on free Lie algebras. For example, the famous Nielsen-Schreier Theorem (Nielsen [21], Schreier [25]) on free groups says that every subgroup of a free group is free [18], and the celebrated Shirshov-Witt Theorem (Shirshov [27], Witt [30]) on free Lie algebras states that every subalgebra of a free Lie algebra is free [23].

In a free Lie algebra $L(\mathcal{A})$, we define Lie monomials of degree one to be the elements in $\mathcal{A}$. Then $[u, v]$ is a Lie monomial if each u and v are Lie monomials. Simply, *Lie monomials* are the monomials formed from free generators using Lie brackets. *Left-normed Lie products* are defined inductively by $[u_1] = u_1$ and $[u_1, \dots, u_{m-1}, u_m] = [[u_1, \dots, u_{m-1}], u_m]$, where $u_i \in L(\mathcal{A})$ for all $1 \leq i \leq m$. Thus we write

$$[u_1, u_2, u_3, \dots, u_m] = [\dots [[u_1, u_2], u_3], \dots, u_m].$$

A *left-normed Lie monomial* is a left-normed Lie product $[u_1, u_2, \dots, u_m]$ with $u_i \in \mathcal{A}$ for all i . It is well-known that the set of all left-normed Lie monomials spans the whole free Lie algebra L on $\mathcal{A}$.

Throughout our work we will make use of the following version of the identity (ii)

$$[u, v, w] = [u, w, v] + [u, [v, w]], \quad \forall u, v, w \in L.$$

As an immediate consequence, we have

$$[u_1, \dots, u_i, u_{i+1}, \dots, u_m] = [u_1, \dots, u_{i+1}, u_i, \dots, u_m] + [u_1, \dots, [u_i, u_{i+1}], \dots, u_m], \quad (2.3)$$

where $u_i \in L$ for all $1 < i < m$.

“Hall sets” in free groups were introduced by P. Hall. More than fifteen years later, M. Hall used the procedure in P. Hall’s work to construct bases of free Lie algebras [18]. A *Hall basis* $\mathcal{H}$ of $L(\mathcal{A})$ is a set consisting of standard Lie monomials that can be constructed inductively as follows. Let the free generating set be ordered in an arbitrary way. The standard Lie monomials of degree one are the elements in $\mathcal{A}$. Assume that all standard monomials of degree less than $n > 1$ have been defined and ordered. Then the standard monomials of degree n are of the form $[h, e]$ where $h, e \in \mathcal{H}$ such that $\deg(h) + \deg(e) = n$, $h > e$ and $\deg(h) = 1$ or, if $h = [h_1, h_2]$ with $h_1, h_2 \in \mathcal{H}$, then $h_2 \leq e$. We order the degree n elements arbitrarily and consider the elements of degree n to be larger than elements of lesser degree (see [3] and [22]). We denote the subset of $\mathcal{H}$ that contains all the elements of degree $n \geq 1$ by $\mathcal{H}_n$.

We have seen that any associative algebra can be turned into a Lie algebra, and we can furthermore construct free Lie algebras in the following way. Let A be the ring of non-commutative polynomials in r indeterminates $a_1, a_2, \dots, a_r$ (the free associative algebra on $\mathcal{A} = \{a_1, a_2, \dots, a_r\}$) with coefficients in a field K . Then A is a Lie algebra with respect to the operation (2.2). The Lie subalgebra L generated by $\mathcal{A}$ in A is the free Lie algebra on $\mathcal{A}$, and A is its universal envelope, that is every Lie homomorphism from L into an arbitrary associative algebra $\tilde{A}$ over K extends uniquely to an algebra homomorphism from A into $\tilde{A}$ [15, Theorem 7]. An element u in A such that u belongs to L is called a *Lie element*. For example, $a_1a_2 - a_2a_1 \in A$ is a Lie element, but a_1a_2 is not. Moreover, L is an A -module via the adjoint representation. For an arbitrary element u in L and any free generator a in the universal envelope A of L , the (right) module action of A on L is given by $u \circ a = [u, a]$. In general, $u \circ (a_1a_2 \dots a_m) = [u, a_1, a_2, \dots, a_m]$, where $a_i \in \mathcal{A}$ for $1 \leq i \leq m$. Note that this module action can be extended linearly. Moreover, if v is a Lie element in A , then $u \circ v = [u, v]$.

It is well-known that the set, say $\mathcal{D}$, of all different associative monomials that can be formed from the free generating set $\mathcal{A}$ together with the identity (the empty monomial of degree zero) form a basis of A . The famous Poincaré-Birkhoff-Witt Theorem provides another basis of A . A PBW-basis $\mathcal{P}$ of A that is derived from a

Hall basis $\mathcal{H}$ of L is defined to be the set that consists of the identity element in A and all the monomials d of the form $d = d_1 d_2 \dots d_s$, where $d_1 \leq d_2 \leq \dots \leq d_s$ and $d_i \in \mathcal{H}$ for all $1 \leq i \leq s$ [15].

2.2 Preliminary results on free Lie algebras

Throughout this thesis we let L denote the free Lie algebra with (finite or countably infinite) free generating set $\mathcal{A} = \{a_1, a_2, \dots\}$, $|\mathcal{A}| \geq 2$, over a field K , and A is its universal enveloping algebra. Recall that A is a graded algebra. We write

$$A = \bigoplus_{n \geq 0} A_n,$$

where A_n is the degree $n > 0$ homogenous component of A and $A_0 = K$. Indeed, A_n is a vector subspace of A with standard basis $\mathcal{D}_n \subset \mathcal{D}$. Moreover, as $\mathcal{P}$ gives another homogenous basis of A , the subset $\mathcal{P}_n$ which consists of all the elements $d = d_1 d_2 \dots d_s$ in $\mathcal{P}$ with $\sum_{i=1}^s \deg(d_i) = n$, $n \geq 1$, forms the PBW-basis of A_n . The grading of A induces a grading of L and thus

$$L = \bigoplus_{n \geq 1} L_n,$$

where L_n denotes the degree n *homogeneous component* of L . Any $u \in L_n$ is said to be a *homogenous element* of degree n . It is well-known that L_n is the span of all left-normed Lie monomials $[b_1, b_2, \dots, b_n]$ with $b_i \in \mathcal{A}$. Moreover, $\mathcal{H}_n$ forms a basis of L_n . If $\mathcal{A}$ is a finite set, say $\mathcal{A} = \{a_1, a_2, \dots, a_r\}$, and hence L the free Lie algebra of finite rank r , then the homogeneous components of L are finite-dimensional, and the dimension d_n of L_n (the cardinality of $\mathcal{H}_n$) is given by Witt's formula [18, Theorem 5.11]

$$d_n = \frac{1}{n} \sum_{d|n} \mu(d) r^{n/d} \quad (2.4)$$

where μ is the Möbius function which is defined for any positive integer d as follows.

$$\mu(d) = \begin{cases} 1, & \text{if } d = 1; \\ 0, & \text{if } d \text{ is not square-free;} \\ (-1)^s, & \text{otherwise,} \end{cases}$$

where s denotes the number of distinct primes appearing in d 's factorisation.

The *fine homogeneous components* of L are spanned by all left-normed Lie monomials of the same multidegree in the free generators. More precisely, a *composition* q of n in r parts ($q \models n$) is a sequence $q = (q_1, q_2, \dots, q_r)$ of non-negative integers such that $\sum_{j=1}^r q_j = n$. Note that our use of compositions is slightly different from the usual conventions: We do allow zero parts and assume throughout that the number of parts is r , unless otherwise stated. For a fixed composition $q = (q_1, q_2, \dots, q_r)$ of n , let L_q be the subspace of L generated by all Lie products of partial degree q_j with respect to a_j for $1 \leq j \leq r$. Then

$$L_n = \bigoplus_{q \models n} L_q.$$

The subspaces L_q are known as the fine homogeneous components of L . Note that, for $n > 1$, $L_q = 0$ if $q_j = n$ for some j . The dimension d_q of L_q with $q \models n$ is given by the fine Witt formula

$$d_q = \frac{1}{n} \sum_{d \mid \gcd(q_1, q_2, \dots, q_r)} \mu(d) \frac{(n/d)!}{(q_1/d)!(q_2/d)! \dots (q_r/d)!}$$

where $\gcd(q_1, q_2, \dots, q_r)$ denotes the greatest common divisor of $q_1, q_2, \dots, q_r$ [18, Theorem 5.11].

Let $u, v \in L$ and let $s \geq 0$ be an integer. For abbreviation, we write

$$[u, v^s] = [u, \underbrace{v, v, \dots, v}_{s \text{ times}}]$$

with the convention that $[u, v^0] = u$. Moreover, for $v_1, v_2, \dots, v_l \in L$, the convention is that $[u, v_1, v_2, \dots, v_l] = u$ if $l = 0$.

Now we record some important results on free Lie algebras. Let L be the free Lie algebra on a set $\mathcal{A}$. If we write $\mathcal{A}$ as a disjoint union of $\mathcal{B}$ and $\mathcal{C}$, then

$$L(\mathcal{A}) = L(\mathcal{B}) \bigoplus \text{Id}(\mathcal{C}),$$

where $L(\mathcal{B})$ is the free Lie subalgebra of $L(\mathcal{A})$ generated by $\mathcal{B}$ and $\text{Id}(\mathcal{C})$ denotes the ideal generated by $\mathcal{C}$ in $L(\mathcal{A})$. According to the Shirshov-Witt Theorem, $\text{Id}(\mathcal{C})$ is a free

Lie subalgebra of $L(\mathcal{A})$. The next theorem, which is known as the Lazard Elimination Theorem, gives a specific set of free generators of $\text{Id}(\mathcal{C})$.

Theorem 2.1 [3, Chap. II, Sec. 2, Proposition 10] *The elements $[c, b_1, \dots, b_s]$, with $c \in \mathcal{C}$, $b_i \in \mathcal{B}$ and $s \geq 0$, form a free generating set for the ideal $\text{Id}(\mathcal{C})$ as a free Lie algebra.*

For any free Lie algebra L , a subalgebra B is a *free factor* of L if L has a free generating set containing a free generating set of B [24]. The following lemma is an analogue to a result in group theory (see [12]).

Lemma 2.2 [24, Theorem 8.4] *Let B be a finitely generated subalgebra of the free Lie algebra L . Then there exists a subalgebra C of finite codimension in L such that B is a free factor of C .*

Chapter 3

The solution space of linear equations in L

In [22], the authors studied the equation (1.1) in the case where $k = 2$, that is for two indeterminates. Throughout this chapter, we generalise and develop some of the results obtained in that paper to linear equations with arbitrary many indeterminates. One of the main outcomes here is that all solutions of equation (3.2) are inner solutions. To prove this result, we exploit a famous lemma of Shirshov's to provide a more transparent proof as compared to the rather technical argument used in [22]. Most of the results in this chapter can be found in [1].

3.1 Linear equations over L

Let L be a free Lie algebra on $\mathcal{A} = \{a_1, a_2, \dots\}$ over a field K , where $|\mathcal{A}| \geq 2$. We study a specific linear equation of the following form

$$[x_1, u_1] + [x_2, u_2] + \dots + [x_k, u_k] = v, \quad (3.1)$$

where $x_1, x_2, \dots, x_k$ are indeterminates and $u_1, u_2, \dots, u_k, v \in L$. A solution of (3.1) over L is a sequence $s = (s_1, s_2, \dots, s_k)$, such that each s_i belongs to L and we get a true equality when we substitute x_i by s_i for $1 \leq i \leq k$. If $v = 0$, we get the equation (1.1) which represents the corresponding homogenous equation of (3.1). The general solution of the non-homogenous equation is

$$s = s_p + s_{ho},$$

where s_p is a particular solution of (3.1) and s_{ho} is the general solution of the homogeneous equation (1.1).

From now on we will focus on Eq. (1.1). We regard the solutions $(s_1, s_2, \dots, s_k)$ of this equation as elements of the direct sum

$$L^{\oplus k} = L^1 \oplus L^2 \oplus \dots \oplus L^k,$$

where $L^1, L^2, \dots, L^k$ are isomorphic copies of L . It is clear that the set of all solutions V is a subspace of $L^{\oplus k}$. Indeed, it is not hard to see that for $s, s' \in V$ and $\lambda, \lambda' \in K$, the element $\lambda s + \lambda' s' \in V$. Moreover, we introduce the coefficient vector $u = (u_1, u_2, \dots, u_k)$ of Eq. (1.1). Then the solution space V may be identified with the kernel of the linear map $\phi_u : L^{\oplus k} \rightarrow L$ defined by

$$(v_1, v_2, \dots, v_k)\phi_u = [v_1, u_1] + [v_2, u_2] + \dots + [v_k, u_k]$$

for all $(v_1, v_2, \dots, v_k) \in L^{\oplus k}$. We call ϕ_u the linear map associated with Eq. (1.1). We say that a solution $(s_1, s_2, \dots, s_k) \in V$ is an *inner solution* if each s_i belongs to the subalgebra generated by the coefficients $u_1, u_2, \dots, u_k$ in L . It is plain that the inner solutions form a subspace of V . We denote this subspace by V_{in} .

3.2 The homogeneous structure of the solution space

In this section, we consider Eq. (1.1) under certain special conditions on the coefficients. In the first two subsections we deal with the case where the coefficients are actually distinct free generators of L , that is $u = (a_1, a_2, \dots, a_k)$. In that case Eq. (1.1) turns into

$$[x_1, a_1] + [x_2, a_2] + \dots + [x_k, a_k] = 0, \quad (3.2)$$

and the associated map $\phi = \phi_u$ (we suppress the subscript u in this special case) is given by

$$(v_1, v_2, \dots, v_k)\phi = [v_1, a_1] + [v_2, a_2] + \dots + [v_k, a_k]. \quad (3.3)$$

Its restriction to $L_n^1 \oplus L_n^2 \oplus \cdots \oplus L_n^k$ maps this direct summand of $L^{\oplus k}$ into the homogeneous component L_{n+1} of L . We denote this restriction by ϕ_{n+1} :

$$\phi_{n+1} : L_n^1 \oplus L_n^2 \oplus \cdots \oplus L_n^k \rightarrow L_{n+1}. \quad (3.4)$$

It follows easily that the solution space V decomposes into the direct sum of its subspaces

$$V = \bigoplus_{n \geq 1} V_n, \quad \text{with } V_n = \ker \phi_{n+1}. \quad (3.5)$$

We call V_n the degree n homogeneous component of the solution space V .

Recall that free Lie algebras are graded algebras. Indeed, the grading of L induces a grading of V as follows. Let $(s_1, s_2, \dots, s_k)$ be a solution of Eq. (3.2). Since L is graded, we have

$$s_j = \sum_{n \geq 1} s_{j,n}, \quad s_{j,n} \in L_n.$$

By substituting each s_j in the equation (3.2), for $1 \leq j \leq k$, we get

$$\sum_{n \geq 1} ([s_{1,n}, a_1] + [s_{2,n}, a_2] + \cdots + [s_{k,n}, a_k]) = 0.$$

For a fixed n , $[s_{1,n}, a_1] + [s_{2,n}, a_2] + \cdots + [s_{k,n}, a_k] \in L_{n+1}$. Since L is a direct sum of its homogenous components, each $(s_{1,n}, s_{2,n}, \dots, s_{k,n})$ forms a solution by itself. Hence V decomposes as in (3.5), where $V_n = V \cap (L_n^1 \oplus L_n^2 \oplus \cdots \oplus L_n^k)$ is the space of all solutions of degree n in V .

3.2.1 $k = r$ and $u_1 = a_1, u_2 = a_2, \dots, u_k = a_k$

In this subsection, the linear map (3.4) is surjective for all $n \geq 1$. Indeed, as mentioned before, the homogeneous component L_{n+1} is spanned by all left-normed Lie monomials $[b_1, b_2, \dots, b_n, b_{n+1}]$ with $b_i \in \mathcal{A}$. In the case under consideration, any such Lie monomial is in the image of ϕ_{n+1} . Indeed, if $b_{n+1} = a_i$ for some i with $1 \leq i \leq k$, then for $(v_1, v_2, \dots, v_k)$ with $v_i = [b_1, b_2, \dots, b_n]$ and $v_j = 0$ for $j \neq i$ we have, in view of (3.3), $(v_1, v_2, \dots, v_k)\phi = [b_1, b_2, \dots, b_n, b_{n+1}]$. The dimension of the domain in (3.4) is kd_n , and the dimension of the codomain is d_{n+1} . Since this linear map is surjective, the dimension of the kernel is $kd_n - d_{n+1}$ (by the Rank-Nullity Theorem). We summarise our discussion in the following

Theorem 3.1 *Let L be the free Lie algebra of rank k over a field K with free generators $a_1, a_2, \dots, a_k$, and let V_n denote the degree n homogeneous component of the solution space of Eq. (3.2). Then*

$$\dim V_n = kd_n - d_{n+1}$$

for all $n \geq 1$, where d_n and d_{n+1} are as in (2.4) with $r = k$.

The following tables give the dimension of V_n in the case of equations in two variables for $1 \leq n \leq 24$, (Note that $\dim V_n$ for $1 \leq n \leq 15$ have been already calculated in [22]).

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$\dim V_n$	3	0	1	0	3	0	6	4	13	12	37	40	99	140	284

n	16	17	18	19	20	21	22	23	24
$\dim V_n$	450	888	1470	2811	4896	9159	16392	30574	55564

As it can be seen, even in the simplest case of equation (3.2) where $k = 2$, the dimension of V_n grows exponentially. This indicates that finding bases for V_n is not an easy task and it would be rather more complicated in equations with more than two variables.

3.2.2 $k < r$ and $u_1 = a_1, u_2 = a_2, \dots, u_k = a_k$

In this case we show that all solutions of Eq. (1.1) are inner solutions. The proof is based on a special case of a powerful lemma due to Shirshov. A set of elements in L is called *independent* if it is a free generating set of the Lie subalgebra of L it generates. Of course, any subalgebra of L is itself free by the famous Shirshov-Witt Theorem. Shirshov's Lemma is in essence the key ingredient of Shirshov's original proof of this theorem (see [27]). An English translation of this celebrated paper is now available in [28]. Recall that an element of L is called homogeneous if it is contained in some homogeneous component. For example, $[a_2, a_1, a_1] + [a_2, a_1, a_2] \in L_3$ is homogenous, but $[a_2, a_1, a_1] + [a_2, a_1]$ is not. A set $\mathcal{G}$ of homogeneous elements in L is *reduced* if

and only if no element $g \in \mathcal{G}$ belongs to the subalgebra generated by the set $\mathcal{G} \setminus \{g\}$ in L . We mention that in this definition of a reduced set it is not required that all the elements of the set belong to one and the same homogeneous component. Throughout our work, we use a special case of Shirshov's Lemma. The result reads as follows.

Lemma 3.2 [27, Shirshov] *Any reduced set of homogeneous elements in a free Lie algebra L is independent.*

Now we are ready for our theorem.

Theorem 3.3 *Let k be a positive integer and let L be the free Lie algebra over a field K of finite or countably infinite rank with more than k free generators $a_1, a_2, \dots, a_k, a_{k+1}, \dots$. Then all solutions of Eq. (3.2) in L are inner solutions.*

Proof. The free Lie algebra L can be written as a vector space direct sum

$$L = L(a_1, a_2, \dots, a_k) \oplus I(\{a_{k+1}, a_{k+2}, \dots\}) = L(\mathcal{B}) \oplus \text{Id}(\mathcal{C})$$

where $L(\mathcal{B})$ is the free Lie algebra on $\mathcal{B} = \{a_1, a_2, \dots, a_k\}$ and $\text{Id}(\mathcal{C})$ is the ideal generated by $\mathcal{C} = \{a_{k+1}, a_{k+2}, \dots\}$ in L .

It is sufficient to prove the assertion of the theorem for homogeneous solutions. Let $(s_1, s_2, \dots, s_k)$ be a non-zero homogeneous solution of (3.2) with $\deg s_i = n \geq 1$. Write $s_i = v_i + w_i$ with $v_i \in L(\mathcal{B})$ and $w_i \in \text{Id}(\mathcal{C})$, $i = 1, 2, \dots, k$. We need to show that $w_i = 0$ for all i . Let $\mathcal{T} = \{t_1, t_2, \dots, t_m\}$ be a basis of the K -span of $\{w_1, w_2, \dots, w_k\}$, and set $\mathcal{G} = \mathcal{B} \cup \mathcal{T}$. Note that the t_j are homogeneous elements with $\deg t_j = n$. We claim that the set $\mathcal{G}$ is reduced in the sense of Shirshov. Suppose this is not the case. Then one of the elements of $\mathcal{G}$, g say, belongs to the subalgebra generated by the remaining elements of $\mathcal{G}$. It is clear that g cannot be in $\mathcal{B}$. Indeed, the elements of $\mathcal{B}$ are of degree 1, and hence the only way of expressing them in terms of the remaining elements of $\mathcal{G}$ would be as a linear combination. But this is impossible since $\mathcal{G}$ is a linearly independent set. Now suppose that $g = t_j \in \mathcal{T}$. Then

$$t_j = \sum_l \alpha_l m_l \tag{3.6}$$

where each m_l is a Lie product of degree n in $\mathcal{G} \setminus \{t_j\}$ and $\alpha_l \in K$. None of the m_l can involve factors from both $\mathcal{B}$ and $\mathcal{T} \setminus \{t_j\}$ (since the degree of such monomials would be

strictly greater than n). Hence all the Lie monomials in (3.6) are either monomials in $\mathcal{B}$, and hence elements of $L(\mathcal{B})$, or they are actual elements of $\mathcal{T} \setminus \{t_j\}$. Moreover, since L is the direct sum of $L(\mathcal{B})$ and $\text{Id}(\mathcal{C})$, and $t_j \in \text{Id}(\mathcal{C})$, the $L(\mathcal{B})$ -part in (3.6) must be zero. But then the right-hand side turns into a linear combination of elements from $\mathcal{T} \setminus \{t_j\}$, and this gives a contradiction since $\mathcal{T}$ is a linearly independent set. Hence $\mathcal{G}$ is a reduced set, and then Shirshov's Lemma gives that $\mathcal{G}$ is independent. Recall that the $s_i = v_i + w_i$ are a solution of (3.2):

$$\sum_{i=1}^k [v_i, a_i] + \sum_{i=1}^k [w_i, a_i] = 0. \quad (3.7)$$

Since L is the direct sum of $L(\mathcal{B})$ and $\text{Id}(\mathcal{C})$, and since the first sum on the left-hand side of (3.7) is in $L(\mathcal{B})$ and the second sum is in $\text{Id}(\mathcal{C})$, it follows that either sum is zero, and hence both $(v_1, v_2, \dots, v_k)$ and $(w_1, w_2, \dots, w_k)$ are solutions of (3.2). In particular,

$$\sum_{i=1}^k [w_i, a_i] = 0. \quad (3.8)$$

Since $\mathcal{T}$ is a basis of the span of $w_1, w_2, \dots, w_k$, each w_i can be written as a linear combination

$$w_i = \sum_{j=1}^m \beta_{ij} t_j \quad (3.9)$$

for some $\beta_{ij} \in K$. Substituting this into (3.8) gives

$$\sum_{i=1}^k \sum_{j=1}^m \beta_{ij} [t_j, a_i] = 0.$$

Since $\mathcal{G}$ is an independent set, the Lie products $[t_j, a_i]$ with $1 \leq j \leq m$ and $1 \leq i \leq k$ are linearly independent. Consequently, all β_{ij} are zero. But then (3.9) implies that all w_i are zero as required. $\square$

Since the space of inner solutions in the setting of Theorem 3.3 is the same as the full solution space in the setting of Theorem 3.1, the following is an immediate consequence of the two theorems.

Corollary 3.4 *Under the assumptions of Theorem 3.3,*

$$\dim V_n = kd_n - d_{n+1}$$

where d_n and d_{n+1} are as in (2.4) with $r = k$.

3.2.3 $\{u_1, u_2, \dots, u_k\}$ is an arbitrary independent set

In the previous subsection, we have seen that all solutions of Eq. (3.2) are inner. This is no longer true if the coefficient set $\{u_1, u_2, \dots, u_k\}$ is an arbitrary independent set. For example, if L is free on $\{a_1, a_2, a_3\}$, then $x_1 = [a_3, a_2]$, $x_2 = -a_3$, $x_3 = a_2$ is a solution of the equation $[x_1, a_1] + [x_2, [a_2, a_1]] + [x_3, [a_3, a_1]] = 0$ which is not inner.

However, in the special case where the equation (1.1) contains two indeterminates

$$[x_1, u_1] + [x_2, u_2] = 0, \quad (3.10)$$

when the coefficients u_1 and u_2 are linearly independent elements in L , it has been shown in [22] that V_{in} is of finite codimension in V . The condition of linear independence was used in order to show that the set $\{u_1, u_2\}$ is reduced, and hence, by Shirshov's Lemma, it is independent.

In fact if the coefficients u_1 and u_2 are linearly dependent, say $u_2 = \alpha u_1$ where α is a non-zero scalar in K , we have

$$[x_1 + \alpha x_2, u_1] = 0.$$

Then any solution s of Eq. (3.10), in this case, is of the form $(\beta u_1 - \alpha v, v)$ such that $v \in L$ and $\beta \in K$. Therefore for any non-zero $v \notin Ku_1$, s is a non-inner solution of equation (3.10). Hence the inner solution space is of infinite codimension in the full solution space. To generalise the result in [22] to linear equations in more than two indeterminates, we replace the condition of linear independence by the natural assumption that the set $\{u_1, u_2, \dots, u_k\}$ is independent.

Theorem 3.5 *Let L be a free Lie algebra and let $\{u_1, u_2, \dots, u_k\}$ be an independent subset of L . Then the space V_{in} of inner solutions of Eq. (1.1) over the free Lie algebra L has finite codimension in the full solution space V .*

Proof. Let B be the subalgebra generated by $u_1, u_2, \dots, u_k$. Then, by Lemma 2.2, there exists a subalgebra C of finite codimension in L such that B is a free factor of C . Thus C is a free subalgebra of L with $C = L(u_1, u_2, \dots, u_k, \dots)$. Let W be a complement of C as a subspace in L . Thus $L = C \oplus W$, where W is of finite

dimension. Let $(s_1, s_2, \dots, s_k)$ be a solution of (1.1), with $s_i = s_{ic} + s_{iw}$, $1 \leq i \leq k$, $s_{ic} \in C$ and $s_{iw} \in W$. Then

$$\sum_{i=1}^k [s_{ic}, u_i] = - \sum_{i=1}^k [s_{iw}, u_i] \in [W, u_1] + [W, u_2] + \dots + [W, u_k].$$

Indeed, the map $\psi : C^{\oplus k} \longrightarrow C$ given by

$$(c_1, c_2, \dots, c_k) \mapsto [c_1, u_1] + [c_2, u_2] + \dots + [c_k, u_k] \quad (c_i \in C)$$

is linear and it maps $(s_{1c}, s_{2c}, \dots, s_{kc})$ into Y , where $Y = \text{im } \psi \cap ([W, u_1] + [W, u_2] + \dots + [W, u_k])$. Since W has a finite dimension, $\dim[W, u_i]$ is finite for all i , and hence Y is a finite-dimensional subspace. Now, since $\dim Y < \infty$, there exists a finite-dimensional subspace $D \subset C^{\oplus k}$ such that $D\psi = Y$, and thus the whole inverse image of Y in $C^{\oplus k}$ under this map is $D + \ker \psi$. Note that $\ker \psi$ is equal to the solution space of (1.1) over C . By Theorem 3.3, this solution space coincides with the solution space of (1.1) over B , so $\ker \psi = V_{in}$. Thus $(s_{1c}, s_{2c}, \dots, s_{kc}) \in D + V_{in}$, and hence $(s_1, s_2, \dots, s_k) \in D + V_{in} + W^{\oplus k}$. As a result, the full solution space V of the given equation is contained in $D + W^{\oplus k} + V_{in}$. As D and $W^{\oplus k}$ have finite dimensions, V_{in} is of finite codimension in V . $\square$

Note that, if we replace the independence condition by assuming that the coefficients are linearly independent as in [22] (a weaker condition), then the required result on V_{in} need not to be true. For example, let L be the free Lie algebra on $\mathcal{A} = \{a_1, a_2, a_3\}$ and put $u_1 = a_1$, $u_2 = a_2$ and $u_3 = [a_2, a_1]$ in the equation (1.1) with $k = 3$. Then u_1 , u_2 and u_3 are linearly independent but they do not form an independent set. Write $L(\mathcal{A}) = L(a_1, a_2) \oplus \text{Id}(a_3)$, where $\text{Id}(a_3)$ is a free Lie algebra freely generated by an infinite set (see Theorem 2.1). Then it is not difficult to see that any element of the form $(-[u, a_2], [u, a_1], u)$ is a non-inner solution of the equation (1.1), where u belongs to a free generating set of $\text{Id}(a_3)$. Thus V_{in} is of infinite codimension in the solution space V .

3.3 The fine homogeneous structure of the solution space

In this section, we return to the situation considered in Section 3.2.2, but we assume now that L is of finite rank. Thus L is free of rank r on $\mathcal{A} = \{a_1, a_2, \dots, a_r\}$, $2 \leq k \leq r$, and we study Eq. (3.2). In this case the associated linear map decomposes into the direct sum of its restrictions to direct sums of fine homogeneous components as follows: For any composition $q = (q_1, q_2, \dots, q_r) \models n+1$ ($n \geq 1$), in r parts, let $(L^{\oplus k})_q$ denote the direct sum

$$(L^{\oplus k})_q = L_{(q_1-1, q_2, \dots, q_r)}^1 \oplus L_{(q_1, q_2-1, \dots, q_r)}^2 \oplus \dots \oplus L_{(q_1, q_2, \dots, q_k-1, \dots, q_r)}^k,$$

of fine homogeneous components of total degree n with the convention that

$$L_{(q_1, \dots, q_i-1, \dots, q_r)}^i = 0 \text{ if } q_i = 0, \ 1 \leq i \leq k.$$

Then the restriction of the associated linear map ϕ of Eq. (3.2) to the subspace $(L^{\oplus k})_q$, say ϕ_q , maps this direct summand of $L^{\oplus k}$ into the direct summand L_q of L_{n+1} . Thus the linear map ϕ decomposes into the direct sum of its restrictions ϕ_q , that is

$$\phi = \bigoplus_{n=1}^{\infty} \bigoplus_{q \models n+1} \phi_q.$$

It follows that the solution space V is the direct sum of the kernels of these restrictions:

$$V = \bigoplus_{n=1}^{\infty} \bigoplus_{q \models n+1} V_q \quad \text{where} \quad V_q = \ker \phi_q.$$

We refer to those kernels as the fine homogeneous components of multidegree q of the solution space of Eq. (3.2). In particular, the degree n homogeneous component of V is the direct sum of the fine homogeneous components V_q where q runs over all compositions of $n+1$ in r parts.

Theorem 3.6 *Let L be a free Lie algebra of rank r , $2 \leq k \leq r$. Then the n th homogeneous component of the solution space of Eq. (3.2) decomposes into the direct sum of fine homogeneous components*

$$V_n = \bigoplus_{q \models n+1} V_q$$

where V_q is the kernel of the restriction of the associated linear map ϕ to the direct summand $(L^{\oplus k})_q$ of $L^{\oplus k}$. Moreover, if $k = r$, then, for $q = (q_1, q_2, \dots, q_k) \models n + 1$,

$$\dim V_q = d_{(q_1-1, q_2, \dots, q_k)} + d_{(q_1, q_2-1, \dots, q_k)} + \dots + d_{(q_1, q_2, \dots, q_k-1)} - d_{(q_1, q_2, \dots, q_k)}. \quad (3.11)$$

Proof. The first part of the theorem has already been discussed. For the case where $k = r$, the dimension of V_q can be determined by using the fine Witt formula in a way similar to the one that was used in Theorem 3.1. Recall that, for $q = (q_1, q_2, \dots, q_k) \models n + 1$, the subspace L_q is spanned by all left-normed Lie monomials of the form $[b_1, b_2, \dots, b_{n+1}]$, where $b_j \in \mathcal{A}$, such that q_i of the b_j are equal to a_i . As the rank of L is equal to k , each ϕ_q is surjective for $q \models n + 1$. Clearly, for any $[b_1, b_2, \dots, b_n, a_i]$ in L_q , there exists $v = (v_1, v_2, \dots, v_k) \in (L^{\oplus k})_q$ with $v_i = [b_1, b_2, \dots, b_n]$ $\in L_{(q_1, \dots, q_{i-1}, \dots, q_k)}$ and $v_j = 0$ for $j \neq i$ such that $v\phi_q = [b_1, b_2, \dots, b_n, a_i]$. The dimensions of the domain and the codomain of ϕ_q can be obtained by the fine Witt formula, and the result follows by applying the Rank-Nullity Theorem. $\square$

As an application of the above result we calculate the dimensions of some fine homogenous components V_q when $n = 1, 2, 3, 4, 5$ and $k = 3, 4, 5$. This can be found in Appendix A. Note that, for $n > 1$, $V_q = 0$ if $q_i = n + 1$ or $q_i = n$ for some $1 \leq i \leq k$. In Theorem 3.3, we have proved that all solutions of equation (3.2) are inner. Thus as a corollary of Theorem 3.6, if $r > k$, then $\dim V_q$ is obtained as in (3.11).

Let S_r denote the symmetric group of degree r . Then any permutation $\sigma \in S_r$ gives rise to an automorphism τ_σ of the free Lie algebra L given by $a_i\tau_\sigma = a_{i\sigma}$. We call τ_σ the permutation automorphism corresponding to σ . Permutation automorphisms permute the fine homogeneous components of L . In fact, for any composition $q = (q_1, q_2, \dots, q_r) \models n$ in r parts one has clearly

$$L_q\tau_\sigma = L_{q\sigma} \quad \text{where} \quad q\sigma = (q_{1\sigma^{-1}}, q_{2\sigma^{-1}}, \dots, q_{r\sigma^{-1}}).$$

Now let $k = r$ and suppose $(s_1, s_2, \dots, s_k)$ is a solution of Eq. (3.2), that is

$$[s_1, a_1] + [s_2, a_2] + \dots + [s_k, a_k] = 0.$$

Applying a permutation automorphism τ_σ to this equation gives

$$\begin{aligned} ([s_1, a_1] + [s_2, a_2] + \cdots + [s_k, a_k])\tau_\sigma &= [s_1\tau_\sigma, a_{1\sigma}] + [s_2\tau_\sigma, a_{2\sigma}] + \cdots + [s_k\tau_\sigma, a_{k\sigma}] \\ &= [s_{1\sigma^{-1}}\tau_\sigma, a_1] + [s_{2\sigma^{-1}}\tau_\sigma, a_2] + \cdots + [s_{k\sigma^{-1}}\tau_\sigma, a_k] \\ &= 0. \end{aligned}$$

Consequently, $(s_{1\sigma^{-1}}\tau_\sigma, s_{2\sigma^{-1}}\tau_\sigma, \dots, s_{k\sigma^{-1}}\tau_\sigma)$ too is a solution of (3.2). Moreover, if $(s_1, s_2, \dots, s_k) \in V_q$ for some $q \models n+1$, then $(s_{1\sigma^{-1}}\tau_\sigma, s_{2\sigma^{-1}}\tau_\sigma, \dots, s_{k\sigma^{-1}}\tau_\sigma) \in V_{q\sigma}$ and by Theorem 3.6, we have $\dim V_q = \dim V_{q\sigma}$. This yields the following.

Theorem 3.7 *Let L be a free Lie algebra of rank $k \geq 2$, let $\sigma \in S_k$ and τ_σ denote the corresponding permutation automorphism of L . Then:*

- (i) *If $(s_1, s_2, \dots, s_k)$ is a solution of Eq. (3.2) then $(s_{1\sigma^{-1}}\tau_\sigma, s_{2\sigma^{-1}}\tau_\sigma, \dots, s_{k\sigma^{-1}}\tau_\sigma)$ is also a solution of this equation.*
- (ii) *The restriction of the K -linear map $\psi_\sigma : L^{\oplus k} \rightarrow L^{\oplus k}$ given by*

$$\psi_\sigma : (s_1, s_2, \dots, s_k) \mapsto (s_{1\sigma^{-1}}\tau_\sigma, s_{2\sigma^{-1}}\tau_\sigma, \dots, s_{k\sigma^{-1}}\tau_\sigma)$$

to a fine homogeneous component V_q of the solution space of (3.2) is a K -space isomorphism between V_q and $V_{q\sigma}$ where for $q = (q_1, q_2, \dots, q_k) \models n+1$, $n \geq 1$, $q\sigma$ is defined as $q\sigma = (q_{1\sigma^{-1}}, q_{2\sigma^{-1}}, \dots, q_{k\sigma^{-1}})$.

3.4 Methods to find concrete solutions

Whenever the word equation is mentioned, the first question that naturally occurs is how can we solve it? Here we discuss two methods for obtaining some particular solutions of equation (3.2) over $L(\mathcal{A})$, where $\mathcal{A} = \{a_1, a_2, \dots, a_k\}$. The first one is to use the fact that L is embedded in its universal enveloping algebra A . This gives us some general solutions that are not necessarily linearly independent. On the other hand, in the second method, using Hall bases of free Lie algebras allows us to find bases of the homogenous components of the solution space. In both methods, we make use of the anticommutativity of the Lie product in L and the Jacobi identity.

3.4.1 Expressing Lie elements in the universal envelope of L

As we are working with homogenous equations over L , we can use the Lie identities to obtain solutions. Throughout this part, we use the following facts

- (i) $[u, u] = 0 \ \forall \ u \in L$.
- (ii) Any $u \in L$ can be expressed as a linear combination of associative monomials in the free generators (in A).

Recall that the set of all different associative monomials forms the standard basis of A . Let u be a homogenous element of degree n in L , and let u_A represent u as an element in the free associative algebra A . For example, if $u = [a_2, a_1]$, then $u_A = a_2a_1 - a_1a_2$. Then $u \circ u_A = 0$. Note that $u_A = u_A^{(1)}a_1 + u_A^{(2)}a_2 + \dots + u_A^{(k)}a_k$, where each $u_A^{(i)}$ is a unique homogenous polynomial of degree $n - 1$. Put $s_i = u \circ u_A^{(i)}$. Then $(s_1, s_2, \dots, s_k)$ is a solution of equation (3.2).

Theorem 3.8 *Let u be an element of degree n in a Hall basis $\mathcal{H}$ of L . Then the solution $(s_1, s_2, \dots, s_k)$, with $s_i = u \circ u_A^{(i)}$ for $1 \leq i \leq k$ is non-zero.*

The proof of the above theorem will be given later as an application of the results in Chapter 5. To give some concrete examples we need the following lemma.

Lemma 3.9 *Let $m \geq 1$. Then for $u, v \in L$, we have*

$$[v, u^m] = \sum_{i=0}^m (-1)^i \binom{m}{i} u^i v u^{m-i}.$$

Proof. See Appendix B.1. □

As a corollary of Lemma 3.9, we directly get

Corollary 3.10 *For all $u, v \in L$ and any integer $m \geq 1$, we have*

$$(i) \quad [v, u^m, v] = \sum_{i=0}^m (-1)^{i+1} \binom{m}{i} v u^i v u^{m-i} + \sum_{i=0}^m (-1)^i \binom{m}{i} u^i v u^{m-i} v.$$

$$\begin{aligned}
 \text{(ii) } [v, u^m, v^2] &= \sum_{i=0}^m (-1)^i \binom{m}{i} u^i v u^{m-i} v^2 + 2 \sum_{i=0}^m (-1)^{i+1} \binom{m}{i} v u^i v u^{m-i} v \\
 &\quad + \sum_{i=0}^m (-1)^i \binom{m}{i} v^2 u^i v u^{m-i}.
 \end{aligned}$$

In this section, we keep all our concrete examples simple by working on the free Lie algebra L generated by $\mathcal{A} = \{a, b\}$ (with $a < b$) over a field K . Therefore we consider the equation that is given in [22]

$$[x_1, a] + [x_2, b] = 0, \quad (3.12)$$

where $a, b \in \mathcal{A}$, and both x_1 and x_2 are indeterminates.

Two families of solutions were obtained by applying anticommutativity and the Jacobi identity in [22, Theorem 4.3] and [22, Lemma 4.2]. The first family of solutions are inside the fine homogenous component $V_{(2l,2)}$ with $l \geq 1$, and the second is in $V_{(l+m+n,3)}$, where l, m and n are non-negative integers with $l < m < n$. In the next proposition, we obtain new solutions in $V_{(2m,4)}$ for $m \geq 1$.

Proposition 3.11 *Let $m \geq 1$. Then $(s_{1,m}, s_{2,m})$ is a non-zero solution of Eq. (3.12), where*

$$s_{1,m} = \sum_{i=0}^{m-1} (-1)^{i+1} \binom{m}{i} [b, a^m, b^2, a^i, b, a^{m-i-1}].$$

If m is even, then

$$s_{2,m} = \sum_{i=1}^m (-1)^i \binom{m}{i} [b, a^m, b, a^i, b, a^{m-i}].$$

If m is odd, then

$$s_{2,m} = \sum_{i=1}^m (-1)^i \binom{m}{i} [b, a^m, b, a^i, b, a^{m-i}] + 2[b, a^m, b^2, a^m].$$

Proof. The result follows from the fact that $[u, u] = 0$ for all u in L and applying Corollary 3.10 (i). Indeed, we have

$$\begin{aligned}
 0 &= [[b, a^m, b], [b, a^m, b]] \\
 &= [b, a^m, b] \circ \left(\sum_{i=0}^m (-1)^{i+1} \binom{m}{i} ba^i ba^{m-i} + \sum_{i=0}^m (-1)^i \binom{m}{i} a^i ba^{m-i} b \right) \\
 &= [b, a^m, b] \circ \left(\sum_{i=0}^{m-1} (-1)^{i+1} \binom{m}{i} ba^i ba^{m-i} + \sum_{i=1}^m (-1)^i \binom{m}{i} a^i ba^{m-i} b \right. \\
 &\quad \left. + (-1)^{m+1} ba^m b + ba^m b \right)
 \end{aligned}$$

Note that the last two summands are equal to zero when m is even and they are equal to $2ba^m b$ in the case where m is odd. Hence $(s_{1,m}, s_{2,m})$ is a solution of the homogenous equation (3.12). Moreover, Theorem 3.8 implies that this solution is non-zero.

□

More series of solutions can be found in Appendix B.1.

3.4.2 Using Hall bases

For the equation (3.12), bases of V_1 , V_3 and V_5 have been found in [22]. In what follows we describe a general procedure for obtaining a basis of V_n . Then we apply it to construct a basis when $n = 7$.

Note that for any $(s_1, s_2, \dots, s_k) \in V_n \subseteq L_n^1 \oplus L_n^2 \oplus \dots \oplus L_n^k$, each component s_i can be written uniquely as a K -linear combination of elements in $\mathcal{H}_n$, that is

$$s_i = \sum_{h \in \mathcal{H}_n} \alpha_{i,h} h. \quad (3.13)$$

If we substitute each s_i in equation (3.2), we get

$$\sum_{h \in \mathcal{H}_n} \alpha_{1,h} [h, a_1] + \sum_{h \in \mathcal{H}_n} \alpha_{2,h} [h, a_2] + \dots + \sum_{h \in \mathcal{H}_n} \alpha_{k,h} [h, a_k] = 0.$$

By using the anticommutativity of the Lie product and the Jacobi identity, we rewrite each $[h, a_i] \in L_{n+1}$ in the standard form, that is as a K -linear combination of elements in $\mathcal{H}_{n+1}$. Then by collecting the coefficients of each standard Lie monomial in

$\mathcal{H}_{n+1}$, we obtain homogenous systems of linear equations in the indeterminates $\alpha_{i,h}$, $1 \leq i \leq k$. Finally, we solve these systems, substitute the values of the scalars $\alpha_{i,h}$ in (3.13) and get the required solution. Indeed, throughout this process, the equation (3.2) boils down to systems of linear equations with variables in K , and the desired solution can be obtained by solving these systems. As an application, we refer the reader to Appendix B.2, where we obtain a basis of the homogenous component V_7 of the solution space of equation (3.12).

When n is large, the task of rewriting the elements $[h, a_i]$ in the standard form is, generally, not easy. Moreover, the corresponding homogenous systems of linear equations in the scalars $\alpha_{i,h}$ are quite large. This becomes even more complicated when we work with equations in $k > 2$ variables. In the next chapter, we find spanning sets of the homogenous components of the solution space of Eq. (3.2). This result allows us to find more bases of V_n in the case of equations in more than two variables.

Chapter 4

Reinterpreting the solution space

4.1 Introduction

A large number of the material in this chapter is published in [1]. Throughout this part of the thesis, L denotes the free Lie algebra of rank $k > 1$ with free generators $a_1, a_2, \dots, a_k$, unless otherwise stated, over a field K . Recall that the degree 1 homogeneous component L_1 is the K -span of the free generators in L . The direct sum $L^{\oplus k} = L^1 \oplus L^2 \oplus \dots \oplus L^k$ is isomorphic (as K -space) to the tensor product $L \otimes L_1$ via the linear map

$$\theta : L^{\oplus k} \rightarrow L \otimes L_1$$

determined by

$$\theta : (u_1, u_2, \dots, u_k) \mapsto \sum_{i=1}^k u_i \otimes a_i$$

where $u_i \in L^i$. Let φ denote the linear map $L \otimes L_1 \rightarrow L$ given by $u \otimes v \mapsto [u, v]$ where $u \in L$ and $v \in L_1$. Then we have a commutative diagram

$$\begin{array}{ccc} L^{\oplus k} & & \\ \theta \downarrow & \searrow \phi & \\ L \otimes L_1 & \xrightarrow{\varphi} & L \end{array}$$

in which the diagonal map $\phi = \phi_u$ with $u = (a_1, a_2, \dots, a_k)$ is the linear map associated with Eq. (3.2). As θ is an isomorphism and $\theta\varphi = \phi$, it follows that θ maps the solution space of (3.2), that is the kernel of the associated map ϕ , isomorphically onto the kernel of the map φ . This allows us to identify the solution space of (3.2) with the kernel of

the linear map φ . The maps in the diagram respect the grading of L , and hence θ maps the homogeneous component V_{n-1} of the solution space of (3.2) isomorphically onto the kernel of the restriction of φ to the subspace $L_{n-1} \otimes L_1$ of $L \otimes L_1$. This restriction maps $L_{n-1} \otimes L_1$ onto L_n , and we write $\varphi_n : L_{n-1} \otimes L_1 \rightarrow L_n$ for the corresponding map. We summarise our discussion as follows.

Lemma 4.1 *For all $n \geq 2$, the isomorphism $\theta : L^{\oplus k} \rightarrow L \otimes L_1$ yields an isomorphism*

$$V_{n-1} \cong \ker \varphi_n$$

where $\varphi_n : L_{n-1} \otimes L_1 \rightarrow L_n$ is, for all $u \in L_{n-1}$ and $v \in L_1$, given by $u \otimes v \mapsto [u, v]$.

In the next section we will obtain an explicit splitting for the surjection map φ_n (if the field K is of characteristic coprime to n), and this in turn will allow us to exhibit a spanning set for the kernel. But first we need to introduce some more results on free Lie algebras.

Let T denote the tensor algebra on L_1 . Thus

$$T = \bigoplus_{n \geq 0} T_n \quad \text{where} \quad T_n = \underbrace{L_1 \otimes L_1 \otimes \cdots \otimes L_1}_{n \text{ times}}$$

with the convention that $T_0 = K$. For $v_1, v_2, \dots, v_m \in T$ it will be convenient to write $v_1 v_2 \dots v_m$ rather than $v_1 \otimes v_2 \otimes \dots \otimes v_m$. In fact, T can be identified with the free associative algebra on $a_1, a_2, \dots, a_k$. As we have seen before, T is the universal envelope of L . The restriction to the homogeneous components of L of the canonical embedding $L \rightarrow T$ gives embeddings $\nu_n : L_n \rightarrow T_n$ given by

$$[u_1, u_2, \dots, u_n] \nu_n = [u_1, u_2, \dots, u_n] \in T_n$$

where $u_i \in L_1$ and the Lie bracket on the right-hand side is the usual Lie bracket in the associative algebra T , i.e. $[u, v] = u \otimes v - v \otimes u$ for all $u, v \in T$. On the other hand, there are canonical projections $\rho_n : T_n \rightarrow L_n$ given by

$$(u_1 u_2 \dots u_n) \rho_n = [u_1, u_2, \dots, u_n] \in L_n,$$

where again $u_i \in L_1$. These two maps are related by Wever's formulae

$$\nu_n \rho_n = n, \tag{4.1}$$

that is, the composite of ν_n and ρ_n amounts to multiplication by n in L_n . Thus we have the following commutative diagram

$$\begin{array}{ccc} L_n & & \\ \nu_n \downarrow & \searrow n & \\ T_n & \xrightarrow{\rho_n} & L_n \end{array}$$

Moreover, for all $u_1, u_2, \dots, u_n \in L_1$ and $1 \leq k < n$

$$(u_1 u_2 \dots u_k [u_{k+1}, \dots, u_n]) \rho_n = [u_1, u_2, \dots, u_k, [u_{k+1}, \dots, u_n]]. \quad (4.2)$$

For example, if $n = 4$, then

$$\begin{aligned} (u_1 u_2 [u_3, u_4]) \rho_4 &= (u_1 u_2 u_3 u_4 - u_1 u_2 u_4 u_3) \rho_4 \\ &= [u_1, u_2, u_3, u_4] - [u_1, u_2, u_4, u_3] \\ &= [u_1, u_2, [u_3, u_4]]. \end{aligned}$$

For more details about (4.1) and (4.2), see [18, Theorem 5.16].

4.2 A spanning set for the solution space

Our first aim in this section is to find an explicit splitting for the map $\varphi_n : L_{n-1} \otimes L_1 \rightarrow L_n$, i.e. a map $\psi_n : L_n \rightarrow L_{n-1} \otimes L_1$ such that $\psi_n \varphi_n = 1$, where 1 denotes the identity map of L_n . We construct this map as follows. Let $\bar{\psi}_n : L_n \rightarrow L_{n-1} \otimes L_1$ be the composite of

$$\nu_n : L_n \rightarrow T_n = T_{n-1} \otimes L_1 \quad \text{and} \quad \rho_{n-1} \otimes 1 : T_{n-1} \otimes L_1 \rightarrow L_{n-1} \otimes L_1.$$

Observe that $\rho_n : T_n \rightarrow L_n$ factors through $L_{n-1} \otimes L_1$, i.e. it can be written as the composite of

$$\rho_{n-1} \otimes 1 : T_{n-1} \otimes L_1 \rightarrow L_{n-1} \otimes L_1 \quad \text{and} \quad \varphi_n : L_{n-1} \otimes L_1 \rightarrow L_n.$$

Then we have

$$\bar{\psi}_n \varphi_n = n.$$

Indeed, the following diagram is commutative

$$\begin{array}{ccc}
 L_n & \xrightarrow{\nu_n} & T_{n-1} \otimes L_1 \\
 \downarrow n & \swarrow \bar{\psi}_n & \searrow \rho_n \\
 L_n & \xleftarrow{\varphi_n} & L_{n-1} \otimes L_1 \\
 & & \downarrow \rho_{n-1} \otimes 1
 \end{array}$$

Therefore

$$\bar{\psi}_n \varphi_n = \nu_n(\rho_{n-1} \otimes 1) \varphi_n = \nu_n \rho_n = n.$$

Consequently, if the characteristic of K does not divide n , then $\psi_n = (1/n)\bar{\psi}_n$ is a splitting map for φ_n . Thus

$$L_{n-1} \otimes L_1 = \ker \varphi_n \oplus \text{im } \psi_n.$$

Indeed, any $u \in L_{n-1} \otimes L_1$ is uniquely written in the form

$$u = u(1 - \varphi_n \psi_n) + u \varphi_n \psi_n.$$

Therefore, we have that the kernel of φ_n is precisely

$$\ker \varphi_n = (L_{n-1} \otimes L_1)(1 - \varphi_n \psi_n). \quad (4.3)$$

In order to calculate this kernel more explicitly, we need the following result.

Lemma 4.2 *Let A be an arbitrary associative algebra. Then*

$$\begin{aligned}
 [c_1, c_2, \dots, c_m] &= (-1)^{m-1} c_m c_{m-1} \dots c_2 c_1 \\
 &\quad + \sum_{i=2}^{m-1} (-1)^{m-i} c_m c_{m-1} \dots c_{i+1} [c_1, c_2, \dots, c_{i-1}] c_i \\
 &\quad + [c_1, c_2, \dots, c_{m-1}] c_m
 \end{aligned}$$

for all $c_1, c_2, \dots, c_m \in A$, $m \geq 2$.

Proof. We prove this identity by induction on m . For $m = 2$, the result is trivial ($[c_1, c_2] = c_1 c_2 - c_2 c_1$). Suppose the result is true at $m - 1$, that is

$$\begin{aligned}
 [c_1, c_2, \dots, c_{m-1}] &= [c_1, c_2, \dots, c_{m-2}] c_{m-1} \\
 &\quad + \sum_{i=2}^{m-2} (-1)^{m-1-i} c_{m-1} c_{m-2} \dots c_{i+1} [c_1, c_2, \dots, c_{i-1}] c_i \\
 &\quad + (-1)^{m-2} c_{m-1} c_{m-2} \dots c_2 c_1.
 \end{aligned}$$

Then

$$\begin{aligned}
 [c_1, c_2, \dots, c_m] &= [c_1, c_2, \dots, c_{m-1}]c_m - c_m[c_1, c_2, \dots, c_{m-1}] \\
 &= [c_1, c_2, \dots, c_{m-1}]c_m - c_m[c_1, c_2, \dots, c_{m-2}]c_{m-1} \\
 &\quad - \sum_{i=2}^{m-2} (-1)^{m-1-i} c_m c_{m-1} c_{m-2} \dots c_{i+1} [c_1, c_2, \dots, c_{i-1}] c_i \\
 &\quad - (-1)^{m-2} c_m c_{m-1} c_{m-2} \dots c_2 c_1 \\
 &= [c_1, c_2, \dots, c_{m-1}]c_m - (-1)^{m-2} c_m c_{m-1} c_{m-2} \dots c_2 c_1 \\
 &\quad - (-1)^{m-3} c_m c_{m-1} \dots c_3 c_1 c_2 - (-1)^{m-4} c_m c_{m-1} \dots c_4 [c_1, c_2] c_3 \\
 &\quad - (-1)^{m-5} c_m c_{m-1} \dots c_5 [c_1, c_2, c_3] c_4 - \dots \\
 &\quad - (-1)^{m-1-(m-4)} c_m c_{m-1} c_{m-2} c_{m-3} [c_1, c_2, \dots, c_{m-5}] c_{m-4} \\
 &\quad - (-1)^{m-1-(m-3)} c_m c_{m-1} c_{m-2} [c_1, c_2, \dots, c_{m-4}] c_{m-3} \\
 &\quad - (-1)^{m-1-(m-2)} c_m c_{m-1} [c_1, c_2, \dots, c_{m-3}] c_{m-2} \\
 &\quad - c_m [c_1, c_2, \dots, c_{m-2}] c_{m-1} \\
 &= [c_1, c_2, \dots, c_{m-1}]c_m + \sum_{i=2}^{m-1} (-1)^{m-i} c_m c_{m-1} \dots c_{i+1} [c_1, c_2, \dots, c_{i-1}] c_i \\
 &\quad + (-1)^{m-1} c_m c_{m-1} \dots c_2 c_1,
 \end{aligned}$$

as required. □

Now let $u_1, u_2, \dots, u_n \in L_1$. Then, by using (4.1), Lemma 4.2 with $m = n - 1$ and (4.2) we get

$$\begin{aligned}
 &([u_1, u_2, \dots, u_{n-1}] \otimes u_n) \varphi_n \psi_n \\
 &= \frac{1}{n} [u_1, u_2, \dots, u_n] \bar{\psi}_n \\
 &= \frac{1}{n} [u_1, u_2, \dots, u_n] (\rho_{n-1} \otimes 1) \\
 &= \frac{1}{n} ([u_1, u_2, \dots, u_{n-1}] u_n - u_n [u_1, u_2, \dots, u_{n-1}]) (\rho_{n-1} \otimes 1)
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{n} \left([u_1, u_2, \dots, u_{n-1}] u_n \right. \\
 &\quad - u_n ((-1)^{n-2} u_{n-1} u_{n-2} \dots u_2 u_1 \\
 &\quad + \sum_{i=2}^{n-2} (-1)^{n-1-i} u_{n-1} u_{n-2} \dots u_{i+1} [u_1, u_2, \dots, u_{i-1}] u_i \\
 &\quad \left. + [u_1, u_2, \dots, u_{n-2}] u_{n-1} \right) (\rho_{n-1} \otimes 1) \\
 &= \frac{1}{n} \left([u_1, u_2, \dots, u_{n-1}] \rho_{n-1} \otimes u_n \right. \\
 &\quad + (-1)^{n-1} (u_n u_{n-1} u_{n-2} \dots u_2) \rho_{n-1} \otimes u_1 \\
 &\quad + \sum_{i=2}^{n-2} (-1)^{n-i} (u_n u_{n-1} u_{n-2} \dots u_{i+1} [u_1, u_2, \dots, u_{i-1}]) \rho_{n-1} \otimes u_i \\
 &\quad \left. - (u_n [u_1, u_2, \dots, u_{n-2}]) \rho_{n-1} \otimes u_{n-1} \right) \\
 &= \frac{1}{n} \left((n-1) [u_1, u_2, \dots, u_{n-1}] \otimes u_n \right. \\
 &\quad + (-1)^{n-1} [u_n, u_{n-1}, u_{n-2}, \dots, u_2] \otimes u_1 \\
 &\quad + \sum_{i=2}^{n-1} (-1)^{n-i} [u_n, u_{n-1}, u_{n-2}, \dots, u_{i+1}, [u_1, u_2, \dots, u_{i-1}]] \otimes u_i \\
 &\quad \left. + [u_1, u_2, \dots, u_{n-1}] \otimes u_n \right).
 \end{aligned}$$

Hence

$$\begin{aligned}
 &([u_1, u_2, \dots, u_{n-1}] \otimes u_n) (1 - \varphi_n \psi_n) \\
 &= \frac{1}{n} \left((-1)^n [u_n, u_{n-1}, u_{n-2}, \dots, u_2] \otimes u_1 \right. \\
 &\quad + \sum_{i=2}^{n-1} (-1)^{n+1-i} [u_n, u_{n-1}, u_{n-2}, \dots, u_{i+1}, [u_1, u_2, \dots, u_{i-1}]] \otimes u_i \\
 &\quad \left. + [u_1, u_2, \dots, u_{n-1}] \otimes u_n \right).
 \end{aligned}$$

In view of (4.3) we can now state our first main result.

Theorem 4.3 *Let L be the free Lie algebra of rank $k > 1$ with free generators $a_1, a_2, \dots, a_k$ over a field K . If the characteristic of K does not divide $n \geq 2$, then the*

kernel of the linear map $\varphi_n : L_{n-1} \otimes L_1 \rightarrow L_n$ is spanned by the elements

$$\begin{aligned} w_n(u_1, u_2, \dots, u_{n-1}, u_n) &= (-1)^n [u_n, u_{n-1}, \dots, u_2] \otimes u_1 \\ &\quad + \sum_{i=2}^{n-1} (-1)^{n+1-i} [u_n, u_{n-1}, \dots, u_{i+1}, [u_1, u_2, \dots, u_{i-1}]] \otimes u_i \\ &\quad + [u_1, u_2, \dots, u_{n-1}] \otimes u_n \end{aligned}$$

with $u_i \in \{a_1, a_2, \dots, a_k\}$.

Let W be a free $\mathbb{Z}$ -module, where $\mathbb{Z}$ denotes the ring of integers. For a free $\mathbb{Z}$ -submodule U in W , the set $\text{Is}_W(U) = \{w \in W : nw \in U \text{ for some integer } n \geq 1\}$ is called the isolator of U in W . Note that $U \subset \text{Is}_W(U)$, and U is said to be isolated if $\text{Is}_W(U) = U$ [17].

In order to obtain a description of $\ker \varphi_n$ in the case where K is a field of arbitrary characteristic, let L be the free Lie algebra on $a_1, a_2, \dots, a_k$ over the field of rational numbers $\mathbb{Q}$, and consider the free Lie ring $\mathfrak{L}$ (the free Lie algebra over $\mathbb{Z}$) on $a_1, a_2, \dots, a_k$ as a subring of L . Then the elements $w_n(u_1, u_2, \dots, u_{n-1}, u_n)$ with $u_i \in \{a_1, a_2, \dots, a_k\}$ are in $\mathfrak{L}_{n-1} \otimes \mathfrak{L}_1$, and they belong to the kernel of the restriction $\Phi_n : \mathfrak{L}_{n-1} \otimes \mathfrak{L}_1 \rightarrow \mathfrak{L}_n$ of φ_n to $\mathfrak{L}_{n-1} \otimes \mathfrak{L}_1$. Since $\mathfrak{L}_n$ is a free $\mathbb{Z}$ -module, for any positive integer m , if $mv \in \ker \Phi_n$, then v itself belongs to this kernel. Therefore the kernel of Φ_n is the isolator of the $\mathbb{Z}$ -span of the elements $w_n(u_1, u_2, \dots, u_{n-1}, u_n)$ in $\mathfrak{L}_{n-1} \otimes \mathfrak{L}_1$. But once we have the kernel of Φ_n , we get the kernel of φ_n over an arbitrary field K by tensoring with K as $\mathbb{Z}$ -modules. Hence the following is a consequence of Theorem 4.3.

Corollary 4.4 *The kernel of the canonical map $\Phi_n : \mathfrak{L}_{n-1} \otimes \mathfrak{L}_1 \rightarrow \mathfrak{L}_n$ is the isolator of the $\mathbb{Z}$ -span of the elements $w_n(u_1, u_2, \dots, u_{n-1}, u_n)$ with $u_i \in \{a_1, a_2, \dots, a_k\}$ in $\mathfrak{L}_{n-1} \otimes \mathfrak{L}_1$. For the free Lie algebra L over an arbitrary field K , the kernel of the canonical map $\varphi_n : L_{n-1} \otimes L_1 \rightarrow L_n$ is $\ker \Phi_n \otimes K$.*

For small n the elements $w_n = w_n(u_1, u_2, \dots, u_n)$ are as follows:

$$\begin{aligned}
 w_2 &= u_1 \otimes u_2 + u_2 \otimes u_1, \\
 w_3 &= [u_1, u_2] \otimes u_3 + [u_3, u_1] \otimes u_2 - [u_3, u_2] \otimes u_1 \\
 &= [u_1, u_2] \otimes u_3 + [u_3, u_1] \otimes u_2 + [u_2, u_3] \otimes u_1, \\
 w_4 &= [u_1, u_2, u_3] \otimes u_4 + [u_4, [u_1, u_2]] \otimes u_3 - [u_4, u_3, u_1] \otimes u_2 + [u_4, u_3, u_2] \otimes u_1 \\
 &= [u_1, u_2, u_3] \otimes u_4 - [u_1, u_2, u_4] \otimes u_3 - [u_4, u_3, u_1] \otimes u_2 + [u_4, u_3, u_2] \otimes u_1.
 \end{aligned}$$

These can be used to get bases of $\ker \varphi_n$ for $n = 2, 3$. We have $\dim \ker \varphi_2 = \binom{k+1}{2}$ and the elements

$$a_i \otimes a_i \quad (1 \leq i \leq k); \quad a_i \otimes a_j + a_j \otimes a_i \quad (1 \leq i < j \leq k) \quad (4.4)$$

form a basis of $\ker \varphi_2$. Also $\dim \ker \varphi_3 = \binom{k}{3}$, and the elements

$$[a_i, a_j] \otimes a_m + [a_m, a_i] \otimes a_j + [a_j, a_m] \otimes a_i \quad (1 \leq i < j < m \leq k) \quad (4.5)$$

form a basis of $\ker \varphi_3$. We mention that the basis elements $a_i \otimes a_i \in \ker \varphi_2$ are not in the span of the elements $w_2(u_1, u_2)$ with $u_1, u_2 \in \{a_1, a_2, \dots, a_k\}$ if K is a field of characteristic 2. This shows that taking the isolator in Corollary 4.4 is essential. This can also be seen in the case where $n = 4$. The value of $w_4(u_1, u_2, u_3, u_4)$ if exactly two of the u_i are equal to a_1 and the other two are equal to a_2 is either 0 or $\pm 2([a_1, a_2, a_1] \otimes a_2 - [a_1, a_2, a_2] \otimes a_1)$. Hence, in characteristic 2, $[a_1, a_2, a_1] \otimes a_2 - [a_1, a_2, a_2] \otimes a_1$ is not in the span of the elements $w_4(u_1, u_2, u_3, u_4)$, but it is clearly in the kernel of Φ_4 .

4.3 Bases of particular fine homogeneous components of the solution space

In [22] some series of solutions of Eq. (3.2) with $k = 2$ were exhibited, and a complete description of the degree n homogeneous components V_n of the solution space were given for $n \leq 5$. As to fine homogeneous components, a complete description of $V_{(n-1,2)} \subseteq V_n$ can be extracted from [22]. Indeed, in that paper, Theorem 4.2 gives that $V_{(n-1,2)} = 0$ if n is even and that $\dim V_{(n-1,2)} = 1$ if n is odd. Moreover, a non-zero solution belonging to $V_{(n-1,2)}$ for odd $n \geq 3$ is given in [22, Theorem 4.3]. It appears

that this was so far the only instance where a complete description of a reasonable class of fine homogeneous components of the solution space has been obtained. Our main result adds to this knowledge. It refers to the fine homogenous components V_q , where $q = (q_1, q_2, \dots, q_k)$ with $q_i = q_j = 1$ for some $1 \leq i, j \leq k$. In particular, we find a basis of the multilinear fine homogeneous component $V_{(1,1,\dots,1)}$ of the solution space of equation (3.2) with $k \geq 2$.

To achieve our aim, we need the following result. Let L be the free Lie algebra of rank n , and let $\mathcal{F}_n$ be the set of all possible well-defined functions from the set $\{1, 2, \dots, n\}$ into itself. Then any function $\alpha \in \mathcal{F}_n$ gives rise to an endomorphism τ_α of the free Lie algebra L given by $a_i \tau_\alpha = a_{i\alpha}$. We call τ_α the function endomorphism corresponding to α . In particular, consider all the function endomorphisms corresponding to α that map L onto the Lie subalgebra $L(a_1, a_2, \dots, a_k)$ where $k \leq n$. Then for a composition $q = (q_1, q_2, \dots, q_k)$ of $n > 1$ in k non-zero parts, we define $\mathcal{F}_{n,k,q}$ to be the set of all distinct well-defined functions α from $\{1, 2, \dots, n\}$ onto $\{1, 2, \dots, k\}$ such that $|\{j\alpha^{-1}\}| = q_j$, $1 \leq j \leq k$. Fix a subscript i for some $1 \leq i \leq k$ and let $q_i = 1$. Consider the set $\mathcal{M}_i^{(n,q)} = \{[a_i, a_{2\alpha}, \dots, a_{n\alpha}] : \alpha \in \mathcal{F}_{n,k,q}, 1\alpha = i\}$. In other words, the set $\mathcal{M}_i^{(n,q)}$ consists of all left-normed Lie monomials of degree n in which each free generator a_j occurs with multiplicity $q_j \geq 1$ for $j \neq i$ and the free generator a_i occurs once in the first position (i.e. as a first factor in the given Lie monomials). Then there are

$$\frac{(n-1)!}{q_1! \dots q_{i-1}! q_{i+1}! \dots q_k!}$$

distinct elements in this set. According to the fine Witt formula, this number is equal to the dimension the fine homogenous component $L_{(q_1, q_2, \dots, q_k)}$ in L_n with $q_i = 1$ for some i . Moreover, the set $\mathcal{M}_i^{(n,q)}$ is linearly independent. Indeed, if we expand the Lie monomials $[a_i, a_{2\alpha}, \dots, a_{n\alpha}]$ in the universal enveloping algebra A of L , then for each $\alpha \in \mathcal{F}_{n,k,q}$ with $i\alpha^{-1} = 1$ and $|\{j\alpha^{-1}\}| = q_j$, $1 \leq j \leq k$, the associative monomial $a_i a_{2\alpha} \dots a_{n\alpha}$ in the expansion of $[a_i, a_{2\alpha}, \dots, a_{n\alpha}]$ appears exactly once with coefficient 1. Therefore, for all $\alpha \in \mathcal{F}_{n,k,q}$, the set that contains the expansions of $[a_i, a_{2\alpha}, \dots, a_{n\alpha}]$ in A is linearly independent, and thus its inverse image in L is linearly independent too. Hence the following lemma holds.

Lemma 4.5 *Let $q = (q_1, q_2, \dots, q_k)$ be a composition of $n > 1$ such that $q_i = 1$ for some $1 \leq i \leq k$. Then the set $\mathcal{M}_i^{(n,q)}$ is a K -basis for the fine homogeneous component L_q in L_n .*

The fine homogeneous components of the solution space in our new interpretation, i.e. as the kernel of $\varphi : L \otimes L_1 \rightarrow L$, are defined in the obvious way. For a composition $q = (q_1, q_2, \dots, q_k) \models n$, the fine homogeneous component $(L \otimes L_1)_q$ of $L \otimes L_1$ is the span of all elements $[b_1, b_2, \dots, b_{n-1}] \otimes b_n$ with $b_i \in \mathcal{A}$ and exactly q_j of the b_i being equal to a_j , and the fine homogeneous component $(\ker \varphi)_q$ of the solution space of (3.2) is the intersection $(\ker \varphi)_q = \ker \varphi \cap (L \otimes L_1)_q$. It is plain that $(\ker \varphi)_q$ is the image of V_q under the isomorphism θ . The main result of this section is as follows.

Theorem 4.6 *Let L be the free Lie algebra of rank $k > 1$ with free generators $a_1, a_2, \dots, a_k$ over a field K . Then the elements*

$$w_n(a_1, a_{2\alpha}, a_{3\alpha}, \dots, a_{(n-1)\alpha}, a_k) \quad \text{with } \alpha \in \mathcal{F}_{n,k,q} \quad (4.6)$$

such that $1\alpha^{-1} = 1$ and $k\alpha^{-1} = n$ form a K -basis of the fine homogeneous component $(\ker \varphi)_q$ of the solution space of (3.2) with $q = (1, q_2, \dots, q_{k-1}, 1) \models n$.

Proof. By Theorem 3.6, we have

$$\dim(\ker \varphi)_{(1, q_2, \dots, q_{k-1}, 1)} = \binom{n-2}{q_2 \dots q_{k-1}},$$

which is precisely the number of the elements (4.6). We will work with the free Lie ring $\mathfrak{L}$ of rank $k > 1$ with free generators $a_1, a_2, \dots, a_k$. In view of Corollary 4.4 it is sufficient to show that the elements (4.6) are linearly independent in $\mathfrak{L}_{n-1} \otimes \mathfrak{L}_1$, and that their span is isolated in this free $\mathbb{Z}$ -module. Let π denote the projection map $\mathfrak{L}_{n-1} \otimes \mathfrak{L}_1 \rightarrow \mathfrak{L}_{n-1} \otimes \langle a_k \rangle$ that is induced by the projection of the free $\mathbb{Z}$ -module $\mathfrak{L}_1$ onto $\langle a_k \rangle$. Then we have

$$w_n(a_1, a_{2\alpha}, a_{3\alpha}, \dots, a_{(n-1)\alpha}, a_k)\pi = [a_1, a_{2\alpha}, a_{3\alpha}, \dots, a_{(n-1)\alpha}] \otimes a_k. \quad (4.7)$$

By Lemma 4.5, the elements $[a_1, a_{2\alpha}, a_{3\alpha}, \dots, a_{(n-1)\alpha}]$ where α runs over all functions in $\mathcal{F}_{n-1, k-1, q}$ such that $1\alpha^{-1} = 1$ and $q = (1, q_2, \dots, q_{k-1})$ form a basis of the fine homogeneous component of multidegree q in the free Lie ring of rank $k-1$ with free

generators $a_1, a_2, \dots, a_{k-1}$. It follows that the elements (4.7) can be included into a basis of $\mathfrak{L}_{n-1} \otimes \langle a_k \rangle$ and therefore the elements (4.6) are linearly independent in $\mathfrak{L}_{n-1} \otimes \mathfrak{L}_1$. Since $\mathfrak{L}_{n-1} \otimes \langle a_k \rangle$ is a direct summand of $\mathfrak{L}_{n-1} \otimes \mathfrak{L}_1$, the elements (4.7) can be included into a basis of this tensor product. Thus their span is isolated in $\mathfrak{L}_{n-1} \otimes \mathfrak{L}_1$ and this implies that the span of the elements (4.6) is isolated too. Indeed, if we assume that this is not true, then there exists $u \in \mathfrak{L}_{n-1} \otimes \mathfrak{L}_1$ such that mu belongs to the span of the elements (4.6) for some integer $m > 1$, but not u . Thus

$$mu = \sum_{\alpha \in \mathcal{F}_{n,k,q}} \lambda_\alpha w_n(a_1, a_{2\alpha}, \dots, a_{(n-1)\alpha}, a_k), \quad 1\alpha^{-1} = 1, \quad k\alpha^{-1} = n. \quad (4.8)$$

By applying the projection π on the both sides of (4.8), we get

$$mu\pi = \sum_{\alpha \in \mathcal{F}_{n,k,q}} \lambda_\alpha [a_1, a_{2\alpha}, \dots, a_{(n-1)\alpha}] \otimes a_k, \quad 1\alpha^{-1} = 1, \quad k\alpha^{-1} = n.$$

Since $[a_1, a_{2\alpha}, \dots, a_{(n-1)\alpha}] \otimes a_k$ are distinct linearly independent elements in $\mathfrak{L}_{n-1} \otimes \mathfrak{L}_1$, m divides λ_α for all $\alpha \in \mathcal{F}_{n,k,q}$ such that $1\alpha^{-1} = 1$ and $k\alpha^{-1} = n$. Therefore u belongs to the span of the elements (4.6), and this contradicts our assumption. Hence the result follows. □

The multilinear fine homogeneous component for the free Lie algebra L of rank k is the one determined by the composition $(1, 1, \dots, 1) \models k$. In the case of the multilinear fine homogenous component of degree n in L , Lemma 4.5 with $k = n$ reads as follows:

Lemma 4.7 [23, 5.6.2] *Let $n > 1$. Then for a fixed integer i such that $i \leq n$, the set*

$$\mathcal{M}_i^{(n)} = \{[a_i, a_{2\sigma}, \dots, a_{n\sigma}] : \sigma \in S_n, \quad 1\sigma = i\}$$

is a K -basis for the multilinear fine homogeneous component $L_{(1,1,\dots,1)}$ of total degree n in L .

As a consequence of Theorem 4.6 and Lemma 4.7, a basis of the multilinear fine homogenous component of $\ker \varphi_n$ can be easily obtained.

Corollary 4.8 *Let L be the free Lie algebra of rank $k > 1$ with free generators $a_1, a_2, \dots, a_k$ over a field K . Then the elements*

$$w_k(a_1, a_{2\sigma}, a_{3\sigma}, \dots, a_{(k-1)\sigma}, a_k) \quad \text{with } \sigma \in S_{k-1} \text{ such that } 1\sigma = 1 \quad (4.9)$$

form a K -basis of the multilinear fine homogeneous component $(\ker \varphi)_{(1,1,\dots,1)}$ of the solution space of (3.2).

4.4 Explicit spanning sets of the homogenous components of V

Let L be the free Lie algebra on $\mathcal{A} = \{a_1, a_2, \dots, a_n\}$. In Section 4.2, we found a spanning set of $\ker \varphi_n$. In this part of the thesis, we write an explicit spanning set of the homogenous components V_{n-1} of the solution space of Eq. (3.2), $n \geq 2$. Note that any element u in the domain of $\theta^{-1} : L \otimes L_1 \rightarrow L^{\oplus k}$ can be written uniquely as a K -linear combination of simple tensors, that is

$$\begin{aligned} u &= \sum_{h \in \mathcal{H}} \sum_{1 \leq i \leq n} \lambda_{h,i} (h \otimes a_i) \\ &= \sum_{h \in \mathcal{H}} \lambda_{h,1} (h \otimes a_1) + \sum_{h \in \mathcal{H}} \lambda_{h,2} (h \otimes a_2) + \dots + \sum_{h \in \mathcal{H}} \lambda_{h,n} (h \otimes a_n). \end{aligned}$$

Thus

$$u\theta^{-1} = \left(\sum_{h \in \mathcal{H}} \lambda_{h,1} h, \sum_{h \in \mathcal{H}} \lambda_{h,2} h, \dots, \sum_{h \in \mathcal{H}} \lambda_{h,n} h \right).$$

Indeed, any $u \in L \otimes L_1$ is written in the following form

$$\begin{aligned} u &= u_1 \otimes \sum_{i=1}^n \lambda_{1,i} a_i + u_2 \otimes \sum_{i=1}^n \lambda_{2,i} a_i + \dots + u_m \otimes \sum_{i=1}^n \lambda_{m,i} a_i \\ &= \sum_{i=1}^m \lambda_{i,1} (u_i \otimes a_1) + \sum_{i=1}^m \lambda_{i,2} (u_i \otimes a_2) + \dots + \sum_{i=1}^m \lambda_{i,n} (u_i \otimes a_n), \end{aligned}$$

where $u_1, u_2, \dots, u_m \in L$ for some $m \geq 1$. Therefore

$$u\theta^{-1} = \left(\sum_{i=1}^m \lambda_{i,1} u_i, \sum_{i=1}^m \lambda_{i,2} u_i, \dots, \sum_{i=1}^m \lambda_{i,n} u_i \right).$$

Let $\alpha \in \mathcal{F}_n$ and let its corresponding function endomorphism τ_α be defined as before. We rewrite the spanning set in Theorem 4.3 as follows:

Corollary 4.9 *Let L be the free Lie algebra of rank $n > 1$ with free generators $a_1, a_2, \dots, a_n$ over a field K . If the characteristic of K does not divide n , then the kernel of the linear map $\varphi_n : L_{n-1} \otimes L_1 \rightarrow L_n$ is spanned by the elements*

$$w_n(a_1, a_2, \dots, a_{n-1}, a_n)\tau_\alpha = w_n(a_{1\alpha}, a_{2\alpha}, \dots, a_{(n-1)\alpha}, a_{n\alpha}), \quad (4.10)$$

where $\alpha \in \mathcal{F}_n$.

Note that if $\alpha \in S_n$ in Corollary 4.9, then (4.10) is a spanning set of the multilinear fine homogenous component $(\ker \varphi)_{(1,1,\dots,1)} \subset \ker \varphi_n$. To write an explicit spanning set of V_{n-1} , it is enough to find a spanning set for all V_q , $q = (q_1, q_2, \dots, q_n) \models n$ with $q_i \geq 0$. Now let

$$g_1 = (-1)^n [a_n, a_{n-1}, \dots, a_2],$$

$$g_i = (-1)^{n-i+1} [a_n, a_{n-1}, \dots, a_{i+1}, [a_1, \dots, a_{i-1}]], \quad 2 \leq i \leq n-1,$$

$$g_n = [a_1, a_2, \dots, a_{n-1}].$$

Then if the elements (4.10) span $(\ker \varphi)_q$ where $q \models n$, then the fine homogenous component $V_q \subset V_{n-1}$ is spanned by the elements

$$(w_n(a_{1\alpha}, a_{2\alpha}, \dots, a_{n\alpha}))\theta^{-1} = \left(\sum_{1\alpha^{-1}} g_{1\alpha^{-1}}\tau_\alpha, \sum_{2\alpha^{-1}} g_{2\alpha^{-1}}\tau_\alpha, \dots, \sum_{n\alpha^{-1}} g_{n\alpha^{-1}}\tau_\alpha \right),$$

where $\alpha \in \mathcal{F}_n$ such that $q_j = |\{j\alpha^{-1}\}|$ with the convention that $q_j = 0$ and $g_{j\alpha^{-1}} = 0$ if j is not in the image of α , $1 \leq j \leq n$. For instance, let $n = 3$ and let $\alpha \in \mathcal{F}_3$ such that $1 \mapsto 1$, $2 \mapsto 2$ and $3 \mapsto 1$. Then

$$\begin{aligned} w_3(a_1, a_2, a_3)\tau_\alpha &= (g_1 \otimes a_1 + g_2 \otimes a_2 + g_3 \otimes a_3)\tau_\alpha \\ &= g_1\tau_\alpha \otimes a_1 + g_2\tau_\alpha \otimes a_2 + g_3\tau_\alpha \otimes a_1 \end{aligned}$$

is an element in the spanning set (4.10) of $(\ker \varphi)_q$ where $q = (2, 1, 0)$. Thus

$$(w_3(a_{1\alpha}, a_{2\alpha}, a_{3\alpha}))\theta^{-1} = (g_1\tau_\alpha + g_3\tau_\alpha, g_2\tau_\alpha, 0)$$

is an element in a spanning set of $V_{(2,1,0)}$.

4.5 Bases of some homogenous components of $\ker \varphi$

In this section, as it is more convenient to talk about finding concrete solutions of degree n rather than solutions of degree $n-1$, we write $\ker \varphi_{n+1}$ and V_n instead of $\ker \varphi_n$

and V_{n-1} , respectively. As we have already obtained a spanning set of $\ker \varphi_{n+1}$, it is enough now to find the right number of linearly independent elements to get a basis of this kernel. This number of course is equal to $\dim \ker \varphi_{n+1}$. Since $\ker \varphi_{n+1} \cong_{\theta} V_n$ and both θ and θ^{-1} have been explicitly expressed, it is sufficient to write a basis of either $\ker \varphi_{n+1}$ or V_n . Thus in this section we list bases of $\ker \varphi_{n+1}$, where $n = 3, 4, 5$. Bases of $\ker \varphi_2$ and $\ker \varphi_3$ are given in (4.4) and (4.5), respectively (see Section 4.2).

Recall that the symmetric group of degree k acts on the free Lie algebra of rank k via the permutation automorphism τ_{β} where $\beta \in S_k$. Permutation automorphisms permute the fine homogeneous components of L , and therefore they permute the fine homogeneous components of the solution space of Eq. (3.2). Moreover, as we have seen before, S_k acts on the set $\mathcal{C} = \{q : q \models n\}$, where the action is given by $q\beta = (q_{1\beta^{-1}}, q_{2\beta^{-1}}, \dots, q_{k\beta^{-1}})$ for all $\beta \in S_k$. For a fixed composition q of n , let $\mathcal{C}_q$ denote the orbit of q under this action, and let $\{q^{(1)}, q^{(2)}, \dots, q^{(l)}\}$ be the set of all distinct entries in q where $l \leq k$. If m_i is the multiplicity of $q^{(i)}$ in q for $1 \leq i \leq l$, then the number of all compositions of n that have the same entries as q (including q itself) is equal to

$$|\mathcal{C}_q| = \binom{k}{m_1 m_2 \dots m_l} = \frac{k!}{m_1! m_2! \dots m_l!},$$

where $\sum_{i=1}^l m_i = k$. We denote the stabilizer of q in S_k by $\text{Stab}_{S_k}(q)$. Let $\mathcal{T}_q$ denote a right transversal of $\text{Stab}_{S_k}(q)$ in S_k . Then whenever we have a basis of L_q , we obtain a basis of all $L_{q'}$ such that $q' \in \mathcal{C}_q$, by applying the permutation automorphisms τ_{β} on the elements of the given basis of L_q where $\beta \in \mathcal{T}_q$ and $q\beta = q'$.

Similarly, we use these results to obtain bases of V_q where $q \models n+1$. Let $n \geq 1$ be fixed. We construct a basis of the solution space V_n of equation (3.2) where $k \geq 2$. Indeed, as we will see in the following examples, it is sufficient to work out this in the case when the equation consists of $k \leq n+1$ indeterminates. Recall that V_n decomposes into the direct sum of its fine homogenous components V_q where $q \models n+1$ in k parts. Thus for a fixed q , if we find a basis of V_q , then we can obtain bases for all other fine homogenous components $V_{q'}$ with $q' \in \mathcal{C}_q$ by applying the permutation automorphisms τ_{β} (Theorem 3.7), where $\beta \in \mathcal{T}_q$. Therefore, for a fixed $n \geq 1$, it is enough to find bases for all fine homogenous components of multidegree $q \models n+1$ such

that q runs through representatives of the orbits $\mathcal{C}_q$.

Let $Sym\{i_1, i_2, \dots, i_l\}$ denote the symmetric group in the letters $i_1, i_2, \dots, i_l$ for some integer $l \geq 2$. In what follows, for each $q = (q_1, q_2, \dots, q_k) \models n+1$ with $q_j \geq 0$ for all $1 \leq j \leq k$, we list a basis of $(\ker \varphi)_q$ when $n = 3, 4, 5$. The dimension of each $(\ker \varphi)_q$ ($\cong V_q$) can be found in Appendix A.

The case $n = 3$

Let i and j be fixed such that $1 \leq i < j \leq k$. In the case where $q_i = q_j = 2$ and $q_m = 0$ for $m \neq i, j$, the set consisting of the element

$$[a_i, a_j, a_j] \otimes a_i + [a_j, a_i, a_i] \otimes a_j,$$

forms a basis of the one dimensional space $(\ker \varphi)_q$.

For fixed i, j and m with $1 \leq i < j < m \leq k$, if $q_i = 2, q_j = q_m = 1$ and $q_s = 0$ for $s \notin \{i, j, m\}$, then a basis of $\bigoplus_{\beta} (\ker \varphi)_{q\beta}$ consists of the elements

$$([a_{i\beta}, a_{m\beta}, a_{j\beta}] + [a_{i\beta}, a_{j\beta}, a_{m\beta}]) \otimes a_{i\beta} + [a_{m\beta}, a_{i\beta}, a_{i\beta}] \otimes a_{j\beta} + [a_{j\beta}, a_{i\beta}, a_{i\beta}] \otimes a_{m\beta},$$

where $\beta \in \{1, (mi), (ji)\} \subset Sym\{i, j, m\}$.

Let $q_i = q_j = q_m = q_s = 1$ be fixed such that $1 \leq i < j < m < s \leq k$ and $q_t = 0$ for $t \notin \{i, j, m, s\}$. The elements $w_4(a_i, a_{j\sigma}, a_{m\sigma}, a_s)$, by Corollary 4.8, form a basis of the multilinear fine homogenous component of the solution space where $\sigma \in Sym\{j, m\}$.

The case $n = 4$

Let i, j, m be fixed with $1 \leq i < j < m \leq k$. In the case where $q_i = 3, q_j = q_m = 1$ and $q_s = 0$ for $s \notin \{i, j, m\}$, the following elements form a basis of $\bigoplus_{\beta} (\ker \varphi)_{q\beta}$

$$\begin{aligned} & ([a_{m\beta}, a_{i\beta}, a_{i\beta}, a_{j\beta}] - [a_{j\beta}, a_{i\beta}, a_{i\beta}, a_{m\beta}] + [a_{j\beta}, a_{i\beta}, [a_{m\beta}, a_{i\beta}]]) \otimes a_{i\beta} \\ & - [a_{m\beta}, a_{i\beta}, a_{i\beta}, a_{i\beta}] \otimes a_{j\beta} + [a_{j\beta}, a_{i\beta}, a_{i\beta}, a_{i\beta}] \otimes a_{m\beta}, \end{aligned}$$

where $\beta \in \{1, (mi), (ji)\} \subset \text{Sym}\{i, j, m\}$.

For fixed i, j and m such that $1 \leq i < j < m \leq k$, if $q_i = 1$, $q_j = q_m = 2$ and $q_s = 0$ for $s \notin \{i, j, m\}$, then the elements

$$\begin{aligned} & [a_{j\beta}, a_{m\beta}, a_{m\beta}, a_{j\beta}] \otimes a_{i\beta} \\ & + ([a_{j\beta}, a_{i\beta}, a_{m\beta}, a_{m\beta}] - [a_{j\beta}, a_{m\beta}, a_{m\beta}, a_{i\beta}]) \otimes a_{j\beta} \\ & + ([a_{j\beta}, a_{i\beta}, [a_{j\beta}, a_{m\beta}]] - [a_{j\beta}, a_{i\beta}, a_{m\beta}, a_{j\beta}]) \otimes a_{m\beta}, \end{aligned}$$

form a basis of $\bigoplus_{\beta} (\ker \varphi)_{q\beta}$, where $\beta \in \{1, (mi), (ji)\} \subset \text{Sym}\{i, j, m\}$.

For fixed i, j, m and s with $1 \leq i < j < m < s \leq k$, when $q_i = q_m = q_s = 1$, $q_j = 2$ and $q_t = 0$ for $t \notin \{i, j, m, s\}$, a basis of the direct sum of the fine homogenous components of multidegree $q\beta$ in $\ker \varphi$ consists of the elements

$$\begin{aligned} & - [a_{s\beta}, a_{j\beta}, a_{j\beta}, a_{m\beta}] \otimes a_{i\beta} + [a_{s\beta}, a_{j\beta}, a_{j\beta}, a_{i\beta}] \otimes a_{m\beta} \\ & + ([a_{i\beta}, a_{m\beta}, [a_{s\beta}, a_{j\beta}]] - [a_{i\beta}, a_{m\beta}, a_{j\beta}, a_{s\beta}]) \otimes a_{j\beta} \\ & + [a_{i\beta}, a_{m\beta}, a_{j\beta}, a_{j\beta}] \otimes a_{s\beta}, \end{aligned}$$

$$\begin{aligned} & - [a_{s\beta}, a_{j\beta}, a_{m\beta}, a_{j\beta}] \otimes a_{i\beta} + [a_{i\beta}, a_{j\beta}, [a_{s\beta}, a_{j\beta}]] \otimes a_{m\beta} \\ & + ([a_{s\beta}, a_{j\beta}, a_{m\beta}, a_{i\beta}] - [a_{i\beta}, a_{j\beta}, a_{m\beta}, a_{s\beta}]) \otimes a_{j\beta} \\ & + [a_{i\beta}, a_{j\beta}, a_{m\beta}, a_{j\beta}] \otimes a_{s\beta}, \end{aligned}$$

and

$$\begin{aligned} & - [a_{s\beta}, a_{m\beta}, a_{j\beta}, a_{j\beta}] \otimes a_{i\beta} - [a_{i\beta}, a_{j\beta}, a_{j\beta}, a_{s\beta}] \otimes a_{m\beta} \\ & + ([a_{i\beta}, a_{j\beta}, [a_{s\beta}, a_{m\beta}]] + [a_{s\beta}, a_{m\beta}, a_{j\beta}, a_{i\beta}]) \otimes a_{j\beta} \\ & + [a_{i\beta}, a_{j\beta}, a_{j\beta}, a_{m\beta}] \otimes a_{s\beta}, \end{aligned}$$

where $\beta \in \{(ij), (mj), (sj)\}$.

For fixed i, j, m, s and t with $1 \leq i < j < m < s < t \leq k$, a basis of the multilinear fine homogenous component consists of the following elements

$$\begin{aligned}
 w_5(a_i, a_{j\sigma}, a_{m\sigma}, a_{s\sigma}, a_t) = & -[a_t, a_{s\sigma}, a_{m\sigma}, a_{j\sigma}] \otimes a_i + [a_t, a_{s\sigma}, a_{m\sigma}, a_i] \otimes a_{j\sigma} \\
 & - [a_t, a_{s\sigma}, [a_i, a_{j\sigma}]] \otimes a_{m\sigma} - [a_i, a_{j\sigma}, a_{m\sigma}, a_t] \otimes a_{s\sigma} \\
 & + [a_i, a_{j\sigma}, a_{m\sigma}, a_{s\sigma}] \otimes a_t,
 \end{aligned}$$

where $\sigma \in \text{Sym}\{j, m, s\}$.

The case $n = 5$

Let i and j be fixed such that $1 \leq i < j \leq k$. When $q_i = 4$, $q_j = 2$ and $q_m = 0$ for $m \notin \{i, j\}$, the following elements

$$\begin{aligned}
 & ([a_{i\beta}, a_{j\beta}, a_{i\beta}, a_{i\beta}, a_{j\beta}] + [a_{i\beta}, a_{j\beta}, a_{i\beta}, [a_{i\beta}, a_{j\beta}]]) \otimes a_{i\beta} \\
 & - [a_{i\beta}, a_{j\beta}, a_{i\beta}, a_{i\beta}, a_{i\beta}] \otimes a_{j\beta},
 \end{aligned}$$

form a basis of $\bigoplus_{\beta} (\ker \varphi)_{q\beta}$, where $\beta \in \text{Sym}\{i, j\}$.

For fixed $q_i = q_j = 3$, where $1 \leq i < j \leq k$ and $q_m = 0$ for $m \notin \{i, j\}$, the element

$$\begin{aligned}
 & ([a_i, a_j, a_j, [a_i, a_j]] + [a_i, a_j, a_j, a_i, a_j] + [a_i, a_j, a_i, a_j, a_j]) \otimes a_i \\
 & + (-[a_i, a_j, a_j, a_i, a_i] + [a_i, a_j, a_i, [a_i, a_j]] - [a_i, a_j, a_i, a_j, a_i]) \otimes a_j,
 \end{aligned}$$

forms a basis of the one dimensional space $(\ker \varphi)_q$.

Let i, j and m be fixed with $1 \leq i < j < m \leq k$. If $q_i = 4$, $q_j = q_m = 1$ and $q_s = 0$ for $s \notin \{i, j, m\}$, then the following elements give us a basis of $\bigoplus_{\beta} (\ker \varphi)_{q\beta}$

$$\begin{aligned}
 & ([a_{m\beta}, a_{i\beta}, a_{i\beta}, [a_{j\beta}, a_{i\beta}]] - [a_{m\beta}, a_{i\beta}, a_{i\beta}, a_{i\beta}, a_{j\beta}] \\
 & + [a_{j\beta}, a_{i\beta}, a_{i\beta}, [a_{m\beta}, a_{i\beta}]] - [a_{j\beta}, a_{i\beta}, a_{i\beta}, a_{i\beta}, a_{m\beta}]) \otimes a_{i\beta} \\
 & + [a_{m\beta}, a_{i\beta}, a_{i\beta}, a_{i\beta}, a_{i\beta}] \otimes a_{j\beta} \\
 & + [a_{j\beta}, a_{i\beta}, a_{i\beta}, a_{i\beta}a_{i\beta}] \otimes a_{m\beta},
 \end{aligned}$$

where $\beta \in \{1, (mi), (ji)\} \subset \text{Sym}\{i, j, m\}$.

Whenever $q_i = 3$, $q_j = 2$, $q_m = 1$ and $q_s = 0$ for $s \notin \{i, j, m\}$, for fixed i, j and m such that $1 \leq i < j < m \leq k$, the elements

$$\begin{aligned} & ([a_{m\beta}, a_{j\beta}, a_{i\beta}, [a_{j\beta}, a_{i\beta}]] - [a_{m\beta}, a_{j\beta}, a_{i\beta}, a_{i\beta}, a_{j\beta}] + [a_{j\beta}, a_{i\beta}, a_{i\beta}, [a_{m\beta}, a_{j\beta}]]) \otimes a_{i\beta} \\ & + ([a_{m\beta}, a_{j\beta}, a_{i\beta}, a_{i\beta}, a_{i\beta}] - [a_{j\beta}, a_{i\beta}, a_{i\beta}, a_{i\beta}, a_{m\beta}]) \otimes a_{j\beta} \\ & + [a_{j\beta}, a_{i\beta}, a_{i\beta}, a_{i\beta} a_{j\beta}] \otimes a_{m\beta}, \end{aligned}$$

and

$$\begin{aligned} & ([a_{m\beta}, a_{i\beta}, a_{j\beta}, [a_{j\beta}, a_{i\beta}]] - [a_{m\beta}, a_{i\beta}, a_{j\beta}, a_{i\beta}, a_{j\beta}] - [a_{j\beta}, a_{i\beta}, a_{i\beta}, a_{j\beta}, a_{m\beta}]) \otimes a_{i\beta} \\ & + ([a_{m\beta}, a_{i\beta}, a_{j\beta}, a_{i\beta}, a_{i\beta}] + [a_{j\beta}, a_{i\beta}, a_{i\beta}, [a_{m\beta}, a_{i\beta}]]) \otimes a_{j\beta} \\ & + [a_{j\beta}, a_{i\beta}, a_{i\beta}, a_{j\beta}, a_{i\beta}] \otimes a_{m\beta}, \end{aligned}$$

form a basis of $\bigoplus_{\beta} (\ker \varphi)_{q\beta}$ where $\beta \in \text{Sym}\{i, j, m\}$.

Let i, j and m be fixed where $1 \leq i < j < m \leq k$. If $q_i = q_j = q_m = 2$ and $q_s = 0$ for $s \notin \{i, j, m\}$, we get a basis of $(\ker \varphi)_q$ consisting of four elements

$$\begin{aligned} & [a_i, a_j, a_m, a_m, a_j] \otimes a_i - [a_i, a_j, a_m, a_m, a_i] \otimes a_j + [a_i, a_j, a_m, [a_i, a_j]] \otimes a_m, \\ & [a_i, a_m, a_j, a_j, a_m] \otimes a_i + [a_i, a_m, a_j, [a_i, a_m]] \otimes a_j - [a_i, a_m, a_j, a_j, a_i] \otimes a_m, \\ & [a_m, a_j, a_i, [a_m, a_j]] \otimes a_i - [a_m, a_j, a_i, a_i, a_m] \otimes a_j + [a_m, a_j, a_i, a_i, a_j] \otimes a_m, \end{aligned}$$

and

$$\begin{aligned} & ([a_j, a_m, a_m, a_i, a_j] + [a_j, a_m, a_m, [a_i, a_j]]) \otimes a_i \\ & + ([a_i, a_j, a_i, a_m, a_m] - [a_j, a_m, a_m, a_i, a_i]) \otimes a_j \\ & + ([a_i, a_j, a_i, [a_j a_m]] - [a_i, a_j, a_i, a_m, a_j]) \otimes a_m. \end{aligned}$$

For fixed $q_i = q_m = q_s = 1$, $q_j = 3$ where $1 \leq i < j < m < s \leq k$ and $q_t = 0$ for $t \notin \{i, j, m, s\}$, the elements

$$\begin{aligned} & - [a_{j\beta}, a_{s\beta}, a_{m\beta}, a_{j\beta}, a_{j\beta}] \otimes a_{i\beta} - [a_{j\beta}, a_{i\beta}, a_{j\beta}, a_{m\beta}, a_{j\beta}] \otimes a_{s\beta} \\ & + [a_{j\beta}, a_{i\beta}, a_{j\beta}, [a_{j\beta}, a_{s\beta}]] \otimes a_{m\beta} + ([a_{j\beta}, a_{s\beta}, a_{m\beta}, a_{j\beta}, a_{i\beta}] \\ & + [a_{j\beta}, a_{i\beta}, a_{j\beta}, a_{m\beta}, a_{s\beta}] + [a_{j\beta}, a_{s\beta}, a_{m\beta}, [a_{j\beta}, a_{i\beta}]]) \otimes a_{j\beta}, \end{aligned}$$

$$\begin{aligned}
& - [a_{j\beta}, a_{s\beta}, a_{j\beta}, a_{j\beta}, a_{m\beta}] \otimes a_{i\beta} - [a_{m\beta}, a_{i\beta}, a_{j\beta}, a_{j\beta}, a_{j\beta}] \otimes a_{s\beta} \\
& + [a_{j\beta}, a_{s\beta}, a_{j\beta}, a_{j\beta}, a_{i\beta}] \otimes a_{m\beta} + ([a_{m\beta}, a_{i\beta}, a_{j\beta}, [a_{j\beta}, a_{s\beta}]] \\
& + [a_{m\beta}, a_{i\beta}, a_{j\beta}, a_{j\beta}, a_{s\beta}] + [a_{j\beta}, a_{s\beta}, a_{j\beta}, [a_{m\beta}, a_{i\beta}]]) \otimes a_{j\beta}, \\
& - [a_{m\beta}, a_{s\beta}, a_{j\beta}, a_{j\beta}, a_{j\beta}] \otimes a_{i\beta} - [a_{j\beta}, a_{i\beta}, a_{j\beta}, a_{j\beta}, a_{m\beta}] \otimes a_{s\beta} \\
& + [a_{j\beta}, a_{i\beta}, a_{j\beta}, a_{j\beta}, a_{s\beta}] \otimes a_{m\beta} + ([a_{m\beta}, a_{s\beta}, a_{j\beta}, a_{j\beta}, a_{i\beta}] \\
& + [a_{j\beta}, a_{i\beta}, a_{j\beta}, [a_{m\beta}, a_{s\beta}]] + [a_{m\beta}, a_{s\beta}, a_{j\beta}, [a_{j\beta}, a_{i\beta}]]) \otimes a_{j\beta},
\end{aligned}$$

and

$$\begin{aligned}
& - [a_{j\beta}, a_{s\beta}, a_{j\beta}, a_{m\beta}, a_{j\beta}] \otimes a_{i\beta} - [a_{j\beta}, a_{i\beta}, a_{m\beta}, a_{j\beta}, a_{j\beta}] \otimes a_{s\beta} \\
& + [a_{j\beta}, a_{s\beta}, a_{j\beta}, [a_{j\beta}, a_{i\beta}]] \otimes a_{m\beta} + ([a_{j\beta}, a_{s\beta}, a_{j\beta}, a_{m\beta}, a_{i\beta}] \\
& + [a_{j\beta}, a_{i\beta}, a_{m\beta}, a_{j\beta}, a_{s\beta}] + [a_{j\beta}, a_{i\beta}, a_{m\beta}, [a_{j\beta}, a_{s\beta}]]) \otimes a_{j\beta},
\end{aligned}$$

form a basis of the direct sum of the fine homogenous components of multidegree $q\beta$ in $\ker \varphi$, where $\beta \in \{(ij), (mj), (sj), 1\} \subset \text{Sym}\{i, j, m, s\}$.

Let i, j, m and s be fixed such that $1 \leq i < j < m < s \leq k$. If $q_i = q_s = 1$, $q_j = q_m = 2$ and $q_t = 0$ for $t \notin \{i, j, m, s\}$, a basis of $\bigoplus_{\beta} (\ker \varphi)_{q\beta}$ consists of the elements

$$\begin{aligned}
& - [a_{j\beta}, a_{s\beta}, a_{m\beta}, a_{m\beta}, a_{j\beta}] \otimes a_{i\beta} - [a_{j\beta}, a_{i\beta}, a_{m\beta}, a_{m\beta}, a_{j\beta}] \otimes a_{s\beta} \\
& + ([a_{j\beta}, a_{i\beta}, a_{m\beta}, [a_{j\beta}, a_{s\beta}]] + [a_{j\beta}, a_{s\beta}, a_{m\beta}, [a_{j\beta}, a_{i\beta}]]) \otimes a_{m\beta} \\
& + ([a_{j\beta}, a_{i\beta}, a_{m\beta}, a_{m\beta}, a_{s\beta}] + [a_{j\beta}, a_{s\beta}, a_{m\beta}, a_{m\beta}, a_{i\beta}]) \otimes a_{j\beta}, \\
& - [a_{j\beta}, a_{s\beta}, a_{m\beta}, a_{j\beta}, a_{m\beta}] \otimes a_{i\beta} - [a_{m\beta}, a_{i\beta}, a_{j\beta}, a_{m\beta}, a_{j\beta}] \otimes a_{s\beta} \\
& + ([a_{m\beta}, a_{i\beta}, a_{j\beta}, [a_{j\beta}, a_{s\beta}]] + [a_{j\beta}, a_{s\beta}, a_{m\beta}, a_{j\beta}, a_{i\beta}]) \otimes a_{m\beta} \\
& + ([a_{m\beta}, a_{i\beta}, a_{j\beta}, a_{m\beta}, a_{s\beta}] + [a_{j\beta}, a_{s\beta}, a_{m\beta}, [a_{m\beta}, a_{i\beta}]]) \otimes a_{j\beta}, \\
& - [a_{j\beta}, a_{s\beta}, a_{j\beta}, a_{m\beta}, a_{m\beta}] \otimes a_{i\beta} - [a_{m\beta}, a_{i\beta}, a_{m\beta}, a_{j\beta}, a_{j\beta}] \otimes a_{s\beta} \\
& + ([a_{j\beta}, a_{s\beta}, a_{j\beta}, [a_{m\beta}, a_{i\beta}]] + [a_{j\beta}, a_{s\beta}, a_{j\beta}, a_{m\beta}, a_{i\beta}]) \otimes a_{m\beta} \\
& + ([a_{m\beta}, a_{i\beta}, a_{m\beta}, a_{j\beta}, a_{s\beta}] + [a_{m\beta}, a_{i\beta}, a_{m\beta}, [a_{j\beta}, a_{s\beta}]]) \otimes a_{j\beta},
\end{aligned}$$

$$\begin{aligned}
 & - [a_{m\beta}, a_{s\beta}, a_{m\beta}, a_{j\beta}, a_{j\beta}] \otimes a_{i\beta} - [a_{j\beta}, a_{i\beta}, a_{j\beta}, a_{m\beta}, a_{m\beta}] \otimes a_{s\beta} \\
 & + ([a_{j\beta}, a_{i\beta}, a_{j\beta}, [a_{m\beta}, a_{s\beta}]] + [a_{j\beta}, a_{i\beta}, a_{j\beta}, a_{m\beta}, a_{s\beta}]) \otimes a_{m\beta} \\
 & + ([a_{m\beta}, a_{s\beta}, a_{m\beta}, a_{j\beta}, a_{i\beta}] + [a_{m\beta}, a_{s\beta}, a_{m\beta}, [a_{j\beta}, a_{i\beta}]]) \otimes a_{j\beta}, \\
 & - [a_{m\beta}, a_{s\beta}, a_{j\beta}, a_{m\beta}, a_{j\beta}] \otimes a_{i\beta} - [a_{j\beta}, a_{i\beta}, a_{m\beta}, a_{j\beta}, a_{m\beta}] \otimes a_{s\beta} \\
 & + ([a_{j\beta}, a_{i\beta}, a_{m\beta}, [a_{m\beta}, a_{s\beta}]] + [a_{m\beta}, a_{s\beta}, a_{j\beta}, a_{m\beta}, a_{i\beta}]) \otimes a_{j\beta} \\
 & + ([a_{j\beta}, a_{i\beta}, a_{m\beta}, a_{j\beta}, a_{s\beta}] + [a_{m\beta}, a_{s\beta}, a_{j\beta}, [a_{j\beta}, a_{i\beta}]]) \otimes a_{m\beta},
 \end{aligned}$$

and

$$\begin{aligned}
 & - [a_{m\beta}, a_{s\beta}, a_{j\beta}, a_{j\beta}, a_{m\beta}] \otimes a_{i\beta} - [a_{m\beta}, a_{i\beta}, a_{j\beta}, a_{j\beta}, a_{m\beta}] \otimes a_{s\beta} \\
 & + ([a_{m\beta}, a_{i\beta}, a_{j\beta}, a_{j\beta}, a_{s\beta}] + [a_{m\beta}, a_{s\beta}, a_{j\beta}, a_{j\beta}, a_{i\beta}]) \otimes a_{m\beta} \\
 & + ([a_{m\beta}, a_{i\beta}, a_{j\beta}, [a_{m\beta}, a_{s\beta}]] + [a_{m\beta}, a_{s\beta}, a_{j\beta}, [a_{m\beta}, a_{i\beta}]]) \otimes a_{j\beta},
 \end{aligned}$$

where $\beta \in \{(ij), (mi), (sj), (sm), (ij)(sm), 1\}$.

Let $q_i = q_m = q_s = q_t = 1$, $q_j = 2$ be fixed such that $1 \leq i < j < m < s < t \leq k$ and $q_c = 0$ for $c \notin \{i, j, m, s, t\}$. Then the elements

$$\begin{aligned}
 & [a_{t\beta}, a_{j\beta}, a_{s\beta}, a_{m\beta}, a_{j\beta}] \otimes a_{i\beta} - [a_{j\beta}, a_{i\beta}, a_{m\beta}, a_{s\beta}, a_{j\beta}] \otimes a_{t\beta} \\
 & - [a_{t\beta}, a_{j\beta}, a_{s\beta}, [a_{j\beta}, a_{i\beta}]] \otimes a_{m\beta} - [a_{i\beta}, a_{j\beta}, a_{m\beta}, [a_{j\beta}, a_{t\beta}]] \otimes a_{s\beta} \\
 & - ([a_{t\beta}, a_{j\beta}, a_{s\beta}, a_{m\beta}, a_{i\beta}] + [a_{i\beta}, a_{j\beta}, a_{m\beta}, a_{s\beta}, a_{t\beta}]) \otimes a_{j\beta}, \\
 & [a_{t\beta}, a_{j\beta}, a_{s\beta}, a_{j\beta}, a_{m\beta}] \otimes a_{i\beta} - [a_{m\beta}, a_{i\beta}, a_{j\beta}, a_{s\beta}, a_{j\beta}] \otimes a_{t\beta} \\
 & - [a_{t\beta}, a_{j\beta}, a_{s\beta}, a_{j\beta}, a_{i\beta}] \otimes a_{m\beta} - [a_{i\beta}, a_{m\beta}, a_{j\beta}, [a_{j\beta}, a_{t\beta}]] \otimes a_{s\beta} \\
 & - ([a_{t\beta}, a_{j\beta}, a_{s\beta}, [a_{m\beta}, a_{i\beta}]] + [a_{i\beta}, a_{m\beta}, a_{j\beta}, a_{s\beta}, a_{t\beta}]) \otimes a_{j\beta}, \\
 & [a_{t\beta}, a_{j\beta}, a_{j\beta}, a_{m\beta}, a_{s\beta}] \otimes a_{i\beta} - [a_{s\beta}, a_{i\beta}, a_{m\beta}, a_{j\beta}, a_{j\beta}] \otimes a_{t\beta} \\
 & - [a_{t\beta}, a_{j\beta}, a_{j\beta}, [a_{s\beta}, a_{i\beta}]] \otimes a_{m\beta} - [a_{t\beta}, a_{j\beta}, a_{j\beta}, a_{m\beta}, a_{i\beta}] \otimes a_{s\beta} \\
 & - ([a_{i\beta}, a_{s\beta}, a_{m\beta}, [a_{j\beta}, a_{t\beta}]] + [a_{i\beta}, a_{s\beta}, a_{m\beta}, a_{j\beta}, a_{t\beta}]) \otimes a_{j\beta}, \\
 & [a_{t\beta}, a_{m\beta}, a_{s\beta}, a_{j\beta}, a_{j\beta}] \otimes a_{i\beta} - [a_{j\beta}, a_{i\beta}, a_{j\beta}, a_{s\beta}, a_{m\beta}] \otimes a_{t\beta} \\
 & - [a_{i\beta}, a_{j\beta}, a_{j\beta}, a_{s\beta}, a_{t\beta}] \otimes a_{m\beta} - [a_{i\beta}, a_{j\beta}, a_{j\beta}, [a_{m\beta}, a_{t\beta}]] \otimes a_{s\beta} \\
 & - ([a_{t\beta}, a_{m\beta}, a_{s\beta}, a_{j\beta}, a_{i\beta}] + [a_{t\beta}, a_{m\beta}, a_{s\beta}, [a_{j\beta}, a_{i\beta}]]) \otimes a_{j\beta},
 \end{aligned}$$

$$\begin{aligned}
& [a_{t\beta}, a_{s\beta}, a_{j\beta}, a_{m\beta}, a_{j\beta}] \otimes a_{i\beta} - [a_{j\beta}, a_{i\beta}, a_{m\beta}, a_{j\beta}, a_{s\beta}] \otimes a_{t\beta} \\
& - [a_{t\beta}, a_{s\beta}, a_{j\beta}, [a_{j\beta}, a_{i\beta}]] \otimes a_{m\beta} - [a_{i\beta}, a_{j\beta}, a_{m\beta}, a_{j\beta}, a_{t\beta}] \otimes a_{s\beta} \\
& - ([a_{t\beta}, a_{s\beta}, a_{j\beta}, a_{m\beta}, a_{i\beta}] + [a_{i\beta}, a_{j\beta}, a_{m\beta}, [a_{s\beta}, a_{t\beta}]]) \otimes a_{j\beta},
\end{aligned}$$

$$\begin{aligned}
& [a_{t\beta}, a_{j\beta}, a_{m\beta}, a_{s\beta}, a_{j\beta}] \otimes a_{i\beta} - [a_{j\beta}, a_{i\beta}, a_{s\beta}, a_{m\beta}, a_{j\beta}] \otimes a_{t\beta} \\
& - [a_{t\beta}, a_{j\beta}, a_{m\beta}, [a_{j\beta}, a_{i\beta}]] \otimes a_{s\beta} - [a_{i\beta}, a_{j\beta}, a_{s\beta}, [a_{j\beta}, a_{t\beta}]] \otimes a_{m\beta} \\
& - ([a_{t\beta}, a_{j\beta}, a_{m\beta}, a_{s\beta}, a_{i\beta}] + [a_{i\beta}, a_{j\beta}, a_{s\beta}, a_{m\beta}, a_{t\beta}]) \otimes a_{j\beta},
\end{aligned}$$

$$\begin{aligned}
& [a_{t\beta}, a_{m\beta}, a_{j\beta}, a_{j\beta}, a_{s\beta}] \otimes a_{i\beta} - [a_{s\beta}, a_{i\beta}, a_{j\beta}, a_{j\beta}, a_{m\beta}] \otimes a_{t\beta} \\
& - [a_{t\beta}, a_{m\beta}, a_{j\beta}, a_{j\beta}, a_{i\beta}] \otimes a_{s\beta} - [a_{i\beta}, a_{s\beta}, a_{j\beta}, a_{j\beta}, a_{t\beta}] \otimes a_{m\beta} \\
& - ([a_{t\beta}, a_{m\beta}, a_{j\beta}, [a_{s\beta}, a_{i\beta}]] + [a_{i\beta}, a_{s\beta}, a_{j\beta}, [a_{m\beta}, a_{t\beta}]]) \otimes a_{j\beta},
\end{aligned}$$

$$\begin{aligned}
& [a_{t\beta}, a_{s\beta}, a_{j\beta}, a_{j\beta}, a_{m\beta}] \otimes a_{i\beta} - [a_{m\beta}, a_{i\beta}, a_{j\beta}, a_{j\beta}, a_{s\beta}] \otimes a_{t\beta} \\
& - [a_{t\beta}, a_{s\beta}, a_{j\beta}, a_{j\beta}, a_{i\beta}] \otimes a_{m\beta} - [a_{i\beta}, a_{m\beta}, a_{j\beta}, a_{j\beta}, a_{t\beta}] \otimes a_{s\beta} \\
& - ([a_{t\beta}, a_{s\beta}, a_{j\beta}, [a_{m\beta}, a_{i\beta}]] + [a_{i\beta}, a_{m\beta}, a_{j\beta}, [a_{s\beta}, a_{t\beta}]]) \otimes a_{j\beta},
\end{aligned}$$

$$\begin{aligned}
& [a_{t\beta}, a_{j\beta}, a_{m\beta}, a_{j\beta}, a_{s\beta}] \otimes a_{i\beta} - [a_{s\beta}, a_{i\beta}, a_{j\beta}, a_{m\beta}, a_{j\beta}] \otimes a_{t\beta} \\
& - [a_{t\beta}, a_{j\beta}, a_{m\beta}, a_{j\beta}, a_{i\beta}] \otimes a_{s\beta} - [a_{i\beta}, a_{s\beta}, a_{j\beta}, [a_{j\beta}, a_{t\beta}]] \otimes a_{m\beta} \\
& - ([a_{t\beta}, a_{j\beta}, a_{m\beta}, [a_{s\beta}, a_{i\beta}]] + [a_{i\beta}, a_{s\beta}, a_{j\beta}, a_{m\beta}, a_{t\beta}]) \otimes a_{j\beta},
\end{aligned}$$

$$\begin{aligned}
& [a_{t\beta}, a_{s\beta}, a_{m\beta}, a_{j\beta}, a_{j\beta}] \otimes a_{i\beta} - [a_{j\beta}, a_{i\beta}, a_{j\beta}, a_{m\beta}, a_{s\beta}] \otimes a_{t\beta} \\
& - [a_{i\beta}, a_{j\beta}, a_{j\beta}, a_{m\beta}, a_{t\beta}] \otimes a_{s\beta} - [a_{i\beta}, a_{j\beta}, a_{j\beta}, [a_{s\beta}, a_{t\beta}]] \otimes a_{m\beta} \\
& - ([a_{t\beta}, a_{s\beta}, a_{m\beta}, a_{j\beta}, a_{i\beta}] + [a_{t\beta}, a_{s\beta}, a_{m\beta}, [a_{j\beta}, a_{i\beta}]]) \otimes a_{j\beta},
\end{aligned}$$

$$\begin{aligned}
& [a_{t\beta}, a_{j\beta}, a_{j\beta}, a_{s\beta}, a_{m\beta}] \otimes a_{i\beta} - [a_{m\beta}, a_{i\beta}, a_{s\beta}, a_{j\beta}, a_{j\beta}] \otimes a_{t\beta} \\
& - [a_{t\beta}, a_{j\beta}, a_{j\beta}, [a_{m\beta}, a_{i\beta}]] \otimes a_{s\beta} - [a_{t\beta}, a_{j\beta}, a_{j\beta}, a_{s\beta}, a_{i\beta}] \otimes a_{m\beta} \\
& - ([a_{i\beta}, a_{m\beta}, a_{s\beta}, [a_{j\beta}, a_{t\beta}]] + [a_{i\beta}, a_{m\beta}, a_{s\beta}, a_{j\beta}, a_{t\beta}]) \otimes a_{j\beta},
\end{aligned}$$

and

$$\begin{aligned}
& [a_{t\beta}, a_{m\beta}, a_{j\beta}, a_{s\beta}, a_{j\beta}] \otimes a_{i\beta} - [a_{j\beta}, a_{i\beta}, a_{s\beta}, a_{j\beta}, a_{m\beta}] \otimes a_{t\beta} \\
& - [a_{t\beta}, a_{m\beta}, a_{j\beta}, [a_{j\beta}, a_{i\beta}]] \otimes a_{s\beta} - [a_{i\beta}, a_{j\beta}, a_{s\beta}, a_{j\beta}, a_{t\beta}] \otimes a_{m\beta} \\
& - ([a_{t\beta}, a_{m\beta}, a_{j\beta}, a_{s\beta}, a_{i\beta}] + [a_{i\beta}, a_{j\beta}, a_{s\beta}, [a_{m\beta}, a_{t\beta}]]) \otimes a_{j\beta},
\end{aligned}$$

form a basis of $\bigoplus_{\beta}(\ker \varphi)_{q\beta}$, where $\beta \in \{(ij), (mj), (sj), (tj), 1\}$.

For fixed i, j, m, s, t and c with $1 \leq i < j < m < s < t < c \leq k$, a basis of the multilinear fine homogenous component consists of the elements

$$\begin{aligned} & [a_c, a_{t\sigma}, a_{s\sigma}, a_{m\sigma}, a_{j\sigma}] \otimes a_i - [a_c, a_{t\sigma}, a_{s\sigma}, a_{m\sigma}, a_i] \otimes a_{j\sigma} \\ & + [a_c, a_{t\sigma}, a_{s\sigma}, [a_i, a_{j\sigma}]] \otimes a_{m\sigma} + [a_i, a_{j\sigma}, a_{m\sigma}, [a_c, a_{t\sigma}]] \otimes a_{s\sigma} \\ & - [a_i, a_{j\sigma}, a_{m\sigma}, a_{s\sigma}, a_c] \otimes a_{t\sigma} + [a_i, a_{j\sigma}, a_{m\sigma}, a_{s\sigma}, a_{t\sigma}] \otimes a_c, \end{aligned}$$

where $\sigma \in \text{Sym}\{j, m, s, t\}$.

In the special case when the equation (3.2) contains two indeterminates, as we have mentioned before, a complete description of V_n for $n \leq 6$ was given in [22]. In this case where $k = 2$, bases of $\ker \varphi_8 \cong V_7$ and $\ker \varphi_9 \cong V_8$ are given as follows:

The case $n = 7$

In this case, we have $\dim(\ker \varphi)_{(7,1)} = 0$. The elements

$$\begin{aligned} & (-[a_2, a_1, a_1, a_1, a_1, a_1, a_2] - [a_2, a_1, a_1, a_1, [a_2, a_1, a_1]] \\ & + [a_2, a_1, a_1, a_1, a_1, [a_2, a_1]]) \otimes a_1 \\ & + [a_2, a_1, a_1, a_1, a_1, a_1, a_1] \otimes a_2, \end{aligned}$$

and

$$\begin{aligned} & (-[a_1, a_2, a_2, a_2, a_2, a_2, a_1] - [a_1, a_2, a_2, a_2, [a_1, a_2, a_2]] \\ & + [a_1, a_2, a_2, a_2, a_2, [a_1, a_2]]) \otimes a_2 \\ & + [a_1, a_2, a_2, a_2, a_2, a_2, a_2] \otimes a_1, \end{aligned}$$

form a basis of $(\ker \varphi)_{(6,2)}$ and $(\ker \varphi)_{(2,6)}$, respectively.

A basis for $(\ker \varphi)_{(5,3)}$ consists of the element

$$\begin{aligned} & (-[a_2, a_1, a_1, a_2, a_1, a_1, a_2] + [a_2, a_1, a_1, a_2, a_1, [a_2, a_1]] \\ & - [a_2, a_1, a_1, a_2, [a_2, a_1, a_1]] + [a_2, a_1, a_1, a_1, a_2, [a_2, a_1]] \\ & - [a_2, a_1, a_1, a_1, a_2, a_1, a_2]) \otimes a_1 \\ & + ([a_2, a_1, a_1, a_2, a_1, a_1, a_1] + [a_2, a_1, a_1, [a_2, a_1, a_1, a_1]] \\ & + [a_2, a_1, a_1, a_1, a_2, a_1, a_1]) \otimes a_2, \end{aligned}$$

and a basis of $(\ker \varphi)_{(3,5)}$ consists of the following element

$$\begin{aligned} & (-[a_1, a_2, a_2, a_1, a_2, a_2, a_1] + [a_1, a_2, a_2, a_1, a_2, [a_1, a_2]] \\ & - [a_1, a_2, a_2, a_1, [a_1, a_2, a_2]] + [a_1, a_2, a_2, a_2, a_1, [a_1, a_2]] \\ & - [a_1, a_2, a_2, a_2, a_1, a_2, a_1]) \otimes a_2 \\ & + ([a_1, a_2, a_2, a_1, a_2, a_2, a_2] + [a_1, a_2, a_2, [a_1, a_2, a_2, a_2]] \\ & + [a_1, a_2, a_2, a_2, a_1, a_2, a_2]) \otimes a_1, \end{aligned}$$

Finally, the elements

$$\begin{aligned} & (-[a_2, a_1, a_2, a_1, [a_2, a_1, a_2]] - [a_2, a_1, a_2, a_1, a_1, a_2, a_2]) \otimes a_1 \\ & + ([a_2, a_1, a_2, a_1, a_1, a_2, a_1] + [a_2, a_1, a_2, a_1, a_1, [a_2, a_1]]) \otimes a_2, \end{aligned}$$

and

$$\begin{aligned} & ([a_1, a_2, a_2, a_2, a_1, a_1, a_2] + [a_1, a_2, a_2, a_2, a_1, [a_1, a_2]] \\ & - [a_1, a_2, a_2, a_2, [a_1, a_2, a_1]] + [a_1, a_2, a_1, a_1, a_2, a_2, a_2]) \otimes a_1 \\ & + (-[a_1, a_2, a_2, a_2, a_1, a_1, a_1] + [a_1, a_2, a_2, [a_1, a_2, a_1, a_1]] \\ & - [a_1, a_2, [a_1, a_2, a_1, a_1, a_2]] - [a_1, a_2, a_1, a_1, a_2, a_2, a_1]) \otimes a_2. \end{aligned}$$

form a basis of $(\ker \varphi)_{(4,4)}$.

The case $n = 8$

Note that $\dim(\ker \varphi)_q = 0$ for $q = (8, 1)$ and $q = (7, 2)$.

If $q = (6, 3)$, then a set consisting the element

$$\begin{aligned}
& ([a_2, a_1, a_1, a_1, a_1, a_1, a_2, a_2] + [a_2, a_1, a_1, a_1, a_1, [a_2, a_1, a_2]]) \\
& - [a_2, a_1, a_1, a_1, [a_2, a_1, a_2, a_1]] - [a_2, a_1, a_2, a_1, a_1, [a_2, a_1, a_1]] \\
& + [a_2, a_1, a_2, a_1, a_1, a_1, [a_2, a_1]] - [a_2, a_1, a_2, a_1, a_1, a_1, a_1, a_2]) \otimes a_1 \\
& + ([a_2, a_1, a_2, a_1, a_1, a_1, a_1, a_1] - [a_2, a_1, a_1, a_1, a_1, a_1, a_2, a_1] \\
& - [a_2, a_1, a_1, a_1, a_1, a_1, [a_2, a_1]]) \otimes a_2,
\end{aligned}$$

forms a basis of $(\ker \varphi)_q$. A basis of $(\ker \varphi)_{(3,6)}$ is obtained by applying the permutation automorphism $\tau_{(12)}$.

When $q = (5, 4)$, $(\ker \varphi)_q$ has a basis consisting of the following element

$$\begin{aligned}
& ([a_2, a_1, a_1, a_1, a_1, a_2, a_2, a_2] - [a_2, a_1, a_1, a_1, [a_2, a_1, a_2, a_2]]) \\
& + [a_2, a_1, a_1, [a_2, a_1, a_2, a_2, a_1]] + [a_2, a_1, a_2, a_2, a_1, a_1, [a_2, a_1]] \\
& - [a_2, a_1, a_2, a_2, a_1, a_1, a_1, a_2]) \otimes a_1 \\
& + ([a_2, a_1, a_2, a_2, a_1, a_1, a_1, a_1] - [a_2, a_1, a_1, a_1, a_1, a_2, a_2, a_1] \\
& - [a_2, a_1, a_1, a_1, a_1, a_2, [a_2, a_1]] + [a_2, a_1, a_1, a_1, a_1, [a_2, a_1, a_2]]) \otimes a_2.
\end{aligned}$$

To get a basis of $(\ker \varphi)_q$ when $q = (4, 5)$, we apply τ_β where $\beta = (12) \in S_2$.

In Appendix B.2, we obtain a basis of V_7 in the case of equations in two variables. In fact, the method that is used there takes a long time. Moreover, solving the systems of linear equations that appear in Appendix B.2 is a tedious process. Thus using the spanning set in our hands (Theorem 4.3) is more efficient and makes the task easier. However, this way is still complicated when the homogenous component of the solutions space is of a large degree or $\dim V_q$ is greater than 2. Thus the task of finding a basis for the whole solution space is left as an open problem.

Chapter 5

Waldinger's results and generalisations

By looking at the title of this chapter, it seems that it is not, in any way, related to equations over free Lie algebras. However, its content is a stepping stone on the path to proving the main results in the final chapter of this thesis. Throughout the present chapter, we state, apply and generalise Waldinger's results [29] in the case of free Lie algebras rather than free groups. As mentioned, these outcomes will be used in order to prove several results in Chapter 6.

5.1 Introduction

Let $L = L(\mathcal{A})$ be a free Lie algebra of rank $r > 1$ over a field K . For any standard Lie monomials e and h in L , the Lie monomial $[e, h]$ is either standard or can be written uniquely in standard form (as a linear combination of standard Lie monomials in $\mathcal{H}$). As a Hall basis $\mathcal{H}$ is totally ordered, whenever $[e, h]$ is written explicitly in standard form, the largest standard Lie monomial $M[e, h]$ occurring with non-zero coefficient is uniquely determined. But writing Lie products (monomials) in standard form is not always an easy task. A method to determine $M[e, h]$ was found by Waldinger where e and h are basic commutators in a free group. This result was used in order to prove the main theorem in his paper (see [29]). In this chapter, we mention and use some of these results in the language of free Lie algebras. We state one of Waldinger's lemmas ([29, Lemma IV]) which determines $M[e, h]$ in L . Then we apply and generalise this

result in several cases.

From now on, we will use a particular Hall basis in which the elements of a fixed degree $n > 1$ are ordered lexicographically, that is if $h = [h_1, h_2]$ and $h' = [h'_1, h'_2]$ are in $\mathcal{H}$ such that $h_1, h_2, h'_1, h'_2 \in \mathcal{H}$ and $\deg(h) = \deg(h') = n \geq 2$, then $h > h'$ if $h_1 > h'_1$ or $h_1 = h'_1$ and $h_2 > h'_2$. In this case we write $h >_{lex} h'$. For example, in the free Lie algebra on $\mathcal{A} = \{a_1, a_2, a_3\}$, the standard Lie monomials of degree one and degree two are ordered as follows

$$a_1 < a_2 < a_3 < [a_2, a_1] < [a_3, a_1] < [a_3, a_2].$$

A Hall basis with this given ordering is called a *special Hall basis* [22]. As $\mathcal{H}$ is a basis of L , any element $u \in L$ can be written uniquely as a linear combination of elements in $\mathcal{H}$. Define the maximal component $M(u)$ to be the largest standard Lie monomial that occurs with a non-zero coefficient in such K -linear expression of u with the convention that $M(u) = 0$ if $u = 0$. Of course, $M(h) = h$ for all $h \in \mathcal{H}$.

Throughout the thesis we refer to the next two results Lemma 5.1 and Theorem 5.2 as Waldinger's Lemma and Waldinger's Theorem, respectively.

Lemma 5.1 [29, Lemma IV] *Let e and h be elements in $\mathcal{H}$ with $h = [h_1, h_2]$ and $h_1, h_2 \in \mathcal{H}$. Suppose that $h > e$. Then*

$$M[h, e] = \begin{cases} [h, e], & \text{if } h_2 \leq e < h; \\ [M[h_1, e], h_2], & \text{if } e < h_2 < h. \end{cases}$$

Theorem 5.2 [29, Theorem p.141] *Let e, h and h' be standard Lie monomials in $\mathcal{H}$ such that $h > h'$, $e \neq h$ and $e \neq h'$. Then*

$$M[e, h] > M[e, h'].$$

We point out that our convention that $M(0) = 0$ allows us to make use of Theorem 5.2 in the case where $e = h'$. Indeed, if $e \neq h$ and $e = h'$, then we get $M[e, h] > 0$, which is trivially holds.

Remark 5.3 Note that for all $e, h \in \mathcal{H}$, the anticommutativity of the Lie product implies that

$$M[e, h] = M[h, e].$$

5.2 A generalisation of Waldinger's Theorem (1)

To prove the main theorem (Theorem 5.9) in this section, we need to go through several steps. Firstly, we determine $M[u, v]$ in the general case where u and v are arbitrary non-zero elements in L satisfying the following condition.

Lemma 5.4 *Let $u, v \in L$ such that $M(u) \neq M(v)$. Then*

$$M[u, v] = M[M(u), M(v)].$$

Proof. We prove the result when $M(u) > M(v)$, and similarly it can be proved in the case where $M(u) < M(v)$. Since any element in L can be written as a linear combination of elements from $\mathcal{H}$, we can write

$$u = \lambda M(u) + \sum_{h < M(u)} \alpha_h h,$$

and

$$v = \lambda' M(v) + \sum_{h' < M(v)} \beta_{h'} h'.$$

Then

$$\begin{aligned} [u, v] &= \lambda \lambda' [M(u), M(v)] + \sum_{h' < M(v)} \lambda \beta_{h'} [M(u), h'] \\ &\quad + \sum_{h < M(u)} \alpha_h \lambda' [h, M(v)] + \sum_{h < M(u)} \sum_{h' < M(v)} \alpha_h \beta_{h'} [h, h'] \\ &= \lambda_1 M[M(u), M(v)] + \sum_{h_1 < M_1} \gamma_{1, h_1} h_1 \end{aligned}$$

$$\begin{aligned}
 & + \lambda_2 M[M(u), \sum_{h' < M(v)} \beta_{h'} h'] + \sum_{h_2 < M_2} \gamma_{2, h_2} h_2 \\
 & + \lambda_3 M[\sum_{h < M(u)} \alpha_h h, M(v)] + \sum_{h_3 < M_3} \gamma_{3, h_3} h_3 \\
 & + \sum_{h < M(u)} \sum_{h' < M(v)} \alpha_h \beta_{h'} [h, h'], \tag{5.1}
 \end{aligned}$$

where $h, h', h_i \in \mathcal{H}$, $M_1 = M[M(u), M(v)]$, $M_2 = M[M(u), \sum_{h' < M(v)} \beta_{h'} h']$, $M_3 = M[\sum_{h < M(u)} \alpha_h h, M(v)]$, and the non-zero scalars $\alpha_h, \beta_{h'}, \gamma_{i, h_i}, \lambda, \lambda', \lambda_i \in K$ for $1 \leq i \leq 3$.

Note that

$$M[\sum_{h < M(u)} \alpha_h h, M(v)] = \max\{M[h, M(v)] : h < M(u)\}.$$

Similarly,

$$M[M(u), \sum_{h' < M(v)} \beta_{h'} h'] = \max\{M[M(u), h'] : h' < M(v)\}.$$

Since $M(u) > h$ for each h that appears in (5.1), and $M(v) \neq M(u)$, (using $M(u) > M(v)$), by Waldinger's Theorem and Remark 5.3,

$$M[M(u), M(v)] > M[h, M(v)]$$

for all h . Also, as $M(v) > h'$ for all h' , by using Waldinger's Theorem,

$$M[M(u), M(v)] > M[M(u), h'].$$

Hence the largest standard Lie monomial occurring in the given linear combination (5.1) is $M[M(u), M(v)]$. Thus

$$M[u, v] = M[M(u), M(v)].$$

□

Any $h \in \mathcal{H}$ can be written as $h = [h_0, h_1, h_2, \dots, h_s]$, where each h_i belongs to $\mathcal{H}$ for $0 \leq i \leq s$, $h_0 \in \mathcal{A}$, $h_0 > h_1$ and $h_1 \leq h_2 \leq \dots \leq h_s$ for some integer $s \geq 0$.

We refer to $h_1, h_2, \dots, h_s$ as *factors of the interior partition* of h . Moreover, h can be written uniquely in the *left-normed decomposition form* as follows

$$h = e_0 \circ (e_1^{\alpha_1} e_2^{\alpha_2} \dots e_k^{\alpha_k}) = [e_0, e_1^{\alpha_1}, e_2^{\alpha_2}, \dots, e_k^{\alpha_k}],$$

where e_i denotes the i th element in $\mathcal{H}$, $e_0 \in \mathcal{A}$ and α_i is a non-negative integer for $1 \leq i \leq k$. Of course, by the definition of standard Lie monomials in $\mathcal{H}$, $e_0 > e_j$ where j is the smallest positive integer such that $\alpha_j \neq 0$. Note that any element in L that is written in the left-normed decomposition form does not necessary belong to $\mathcal{H}$. For example, if $e_0 = a_1$, then the element

$$e_0 \circ e_8 = e_0 \circ (e_1^0 e_2^0 e_3^0 e_4^0 e_5^0 e_6^0 e_7^0 e_8) = [a_1, [a_2, a_1, a_2, a_2]]$$

in the free Lie algebra L on $\mathcal{A} = \{a_1, a_2\}$ with $a_2 > a_1$, is written in the left-normed decomposition form but it does not belong to $\mathcal{H}$. Also, for d in the PBW-basis $\mathcal{P}$ of A that is associated to $\mathcal{H}$, we write $d = e_1^{\beta_1} e_2^{\beta_2} \dots e_t^{\beta_t}$, where each β_i is a non-negative integer and $t \geq 1$.

Lemma 5.5 *Let $e \in \mathcal{H}$, $d = d_1 d_2 \dots d_k$ with $d_i \in \mathcal{H}$ and $\deg(d) = \sum_{i=1}^k \deg(d_i)$. Suppose that $\deg(e) > \deg(d)$. Then*

$$M(e \circ d) = M[e, d_1, d_2, \dots, d_k] = M[M[\dots M[M[e, d_1], d_2], \dots], d_k].$$

Proof. We prove the result by induction on k . In the case when $k = 1$, the result is trivial. If $k = 2$, then $d = d_1 d_2$. By Lemma 5.4,

$$M(e \circ d) = M[e, d_1, d_2] = M[[e, d_1], d_2] = M[M[e, d_1], M(d_2)] = M[M[e, d_1], d_2].$$

Now write $d = \bar{d} d_k$, where $\bar{d} = d_1 d_2 \dots d_{k-1}$. Suppose the result is true for $\bar{d}$, that is assume that

$$M(e \circ \bar{d}) = M[M[\dots M[M[e, d_1], d_2], \dots], d_{k-1}].$$

Then by Lemma 5.4 and by using the induction hypothesis, we get

$$\begin{aligned} M(e \circ d) &= M(e \circ (\bar{d} d_k)) \\ &= M[M(e \circ \bar{d}), d_k] \\ &= M[M[M[\dots M[M[e, d_1], d_2], \dots], d_{k-1}], d_k]. \end{aligned}$$

□

Note that in the case when $\deg(e) < \deg(d)$, Lemma 5.5 is not true. Indeed, in the free Lie algebra on $\{a, b\}$ with $a < b$, if $e = [b, a, b, b] \in \mathcal{H}$ and $d = d_1 d_2 = a[b, a, a, b, b]$. Then $[e, d_1] = [b, a, a, b, b] + [b, a, b, [b, a]]$. Here the conclusion of the lemma would give $M(e \circ d) = M[M[e, d_1], d_2] = 0$, but in reality $M(e \circ d) = [b, a, a, b, b, [b, a, b, [b, a]]] \neq 0$.

As each element in a Hall basis can be written in the left-normed decomposition form, $M(e \circ d)$ can be obtained as follows.

Lemma 5.6 *Let L be a free Lie algebra on $\mathcal{A}$. Let $e \in \mathcal{H}$ and d be an element in the PBW-basis of the universal envelope A of L such that $e = e_0 \circ e_1^{\alpha_1} e_2^{\alpha_2} \dots e_n^{\alpha_n}$, $d = e_1^{\beta_1} e_2^{\beta_2} \dots e_n^{\beta_n}$, where α_i and β_i are non-negative integers for $n \geq 1$ and $e_0 \in \mathcal{A}$. Suppose that $\deg(e) > \deg(d)$. Then*

$$M(e \circ d) = e_0 \circ (\dots e_i^{\alpha_i + \beta_i} \dots), \quad 1 \leq i \leq n.$$

Proof. Let $\beta = \sum_{i=1}^n \beta_i$ and $\alpha = \sum_{i=1}^n \alpha_i$. We prove the result by induction on β . If $\beta = 1$, then $d = e_i$ for some positive integer $i \leq n$. By Waldinger's Lemma we have

$$M(e \circ e_i) = e_0 \circ (e_1^{\alpha_1} e_2^{\alpha_2} \dots e_i^{\alpha_i + 1} \dots e_n^{\alpha_n}).$$

Thus the result holds for all α at $\beta = 1$. Suppose the result is true for all $\beta' < \beta$. Now let t be the largest positive integer such that $\beta_t > 0$. Then $d = e_1^{\beta_1} e_2^{\beta_2} \dots e_t^{\beta_t}$ where $1 \leq t \leq n$. Therefore

$$\begin{aligned} M(e \circ d) &= M(e \circ (e_1^{\beta_1} e_2^{\beta_2} \dots e_t^{\beta_t})) \\ &= M(e \circ (e_1^{\beta_1} e_2^{\beta_2} \dots e_t^{\beta_t - 1} e_t)) \\ &= M[M(e \circ (e_1^{\beta_1} e_2^{\beta_2} \dots e_t^{\beta_t - 1})), e_t] \quad (\text{by Lemma 5.4}) \\ &= M[e_0, e_1^{\alpha_1 + \beta_1}, e_2^{\alpha_2 + \beta_2}, \dots, e_t^{\alpha_t + \beta_t - 1}, e_{t+1}^{\alpha_{t+1}}, \dots, e_n^{\alpha_n}, e_t], \end{aligned}$$

where the final equality follows by the induction hypothesis. Consider the following two cases:

Case 1. Suppose $\alpha_k = 0$ for all $k > t$. Then

$$\begin{aligned} M(e \circ d) &= M[e_0, e_1^{\alpha_1 + \beta_1}, e_2^{\alpha_2 + \beta_2}, \dots, e_t^{\alpha_t + \beta_t - 1}, e_t] \\ &= [e_0, e_1^{\alpha_1 + \beta_1}, e_2^{\alpha_2 + \beta_2}, \dots, e_t^{\alpha_t + \beta_t}]. \end{aligned}$$

Case 2. Suppose there exists $k > t$ such that $\alpha_k \neq 0$. Then by the induction basis, it follows that

$$\begin{aligned} M(e \circ d) &= M[e_0, e_1^{\alpha_1+\beta_1}, e_2^{\alpha_2+\beta_2}, \dots, e_t^{\alpha_t+\beta_t-1}, \dots, e_n^{\alpha_n}, e_t] \\ &= [e_0, e_1^{\alpha_1+\beta_1}, e_2^{\alpha_2+\beta_2}, \dots, e_t^{\alpha_t+\beta_t}, e_{t+1}^{\alpha_{t+1}}, \dots, e_n^{\alpha_n}]. \end{aligned}$$

Thus in the both cases we achieve our required result. □

Note that, Lemma 5.6 is true for any d that is a product of Hall basis elements in A with $\deg(e) > \deg(d)$, that is d is of the form $e_{i_1}^{\beta_{i_1}} e_{i_2}^{\beta_{i_2}} \dots e_{i_n}^{\beta_{i_n}}$, where $e_{i_j} \in \mathcal{H}$, β_{i_j} are non-negative integers, i_j are distinct positive integers and if $j < k$, then this does not necessarily imply that $i_j < i_k$. Thus $M(e \circ (e_{i_1}^{\beta_{i_1}} e_{i_2}^{\beta_{i_2}} \dots e_{i_n}^{\beta_{i_n}}))$ is unique up to rearranging the indices. Indeed, as it can be seen, the proof of this lemma does not depend on the position of e_i as a factor in d . For example, if $e = e_0 \circ (e_1^{\alpha_1} \dots e_8^{\alpha_8})$, $d = e_1^{\beta_1} e_2^{\beta_2} e_3^{\beta_3} e_4^{\beta_4}$ and $d' = e_3^{\beta_3} e_1^{\beta_1} e_4^{\beta_4} e_2^{\beta_2}$ such that $\deg(e) > \deg(d), \deg(d')$, then

$$M(e \circ d) = M(e \circ d') = e_0 \circ (e_1^{\alpha_1+\beta_1} \dots e_4^{\alpha_4+\beta_4} e_5^{\alpha_5} \dots e_8^{\alpha_8}).$$

The following example explains how this result (Lemma 5.6) works.

Example 5.7 Let L be the free Lie algebra on $\mathcal{A} = \{a_1, a_2\}$ with $a_1 < a_2$. Let $e \in \mathcal{H}$ and $d \in \mathcal{P}$ such that

$$e = [a_2, a_1, a_2, a_2, [a_2, a_1], [a_2, a_1], [a_2, a_1, a_2], [a_2, a_1, a_2], [a_2, a_1, a_1, a_2]]$$

and

$$d = a_1 a_2 [a_2, a_1] [a_2, a_1] [a_2, a_1, a_1] [a_2, a_1, a_2].$$

To find $M(e \circ d)$ traditionally, we rewrite $e \circ d$ in the standard form, that is

$$\begin{aligned} e \circ d &= [e, a_1, a_2, [a_2, a_1], [a_2, a_1], [a_2, a_1, a_1], [a_2, a_1, a_2]] \\ &= \sum_{h \in \mathcal{H}} \alpha_h h, \end{aligned}$$

where $\alpha_h \in K$. As it can be seen, this task is really tedious, and here Lemma 5.6 shows its power. We can write

$$e = e_0 \circ (e_1 e_2^2 e_3^2 e_4^0 e_5^2 e_6^0 e_7),$$

with $e_0 = a_2$ and

$$d = e_1 e_2 e_3^2 e_4 e_5 e_6^0 e_7^0.$$

By Lemma 5.6,

$$M(e \circ d) = e_0 \circ (e_1^2 e_2^3 e_3^4 e_4 e_5^3 e_7).$$

This makes the problem much easier to solve.

For $h = [h_0, h_1, \dots, h_k]$ and $h' = [h'_0, h'_1, \dots, h'_l]$ in $\mathcal{H}$ with $\deg(h) = \deg(h')$, we say that h has a *greater interior partition* than h' and write $iPart(h) > iPart(h')$ if there exists a non-negative integer i such that $\deg(h_{k-i}) < \deg(h'_{l-i})$ and if $i > 0$, $\deg(h_{k-j}) = \deg(h'_{l-j})$ for all $j < i$. In the case where $k = l$ and $\deg(h_i) = \deg(h'_i)$ for all $1 \leq i \leq k$, we say h and h' have the *same interior partition*. In this case we write $h >_{lex} h'$ if there exists $i \geq 0$ such that $h_i >_{lex} h'_i$ and if $i > 0$, then $h_j = h'_j$ for all $j < i$.

Lemma 5.8 *Let h and h' be elements in a special Hall basis $\mathcal{H}$ of a free Lie algebra L such that $\deg(h) = \deg(h')$.*

(i) *If $iPart(h) > iPart(h')$, then $h >_{lex} h'$.*

(ii) *If $iPart(h) = iPart(h')$ and $h >_{ilex} h'$, then $h >_{lex} h'$.*

Proof. (i) Let $h = [h_0, h_1, \dots, h_k] = [\bar{h}, h_k]$ and $h' = [h'_0, h'_1, \dots, h'_l] = [\tilde{h}, h'_l]$. Since $iPart(h) > iPart(h')$, there exists $i \geq 0$ such that $\deg(h_{k-i}) < \deg(h'_{l-i})$ and if $i > 0$, $\deg(h_{k-j}) = \deg(h'_{l-j})$ for all $0 \leq j < i$. We prove the result by induction on i . At $i = 0$ we have $\deg(h_k) < \deg(h'_l)$. This implies that $\deg(\bar{h}) > \deg(\tilde{h})$ and thus $h >_{lex} h'$ in $\mathcal{H}$. Suppose that the result holds for $i = s > 0$. Now let

$$h = [[h_0, \dots, h_{k-(s+1)}, \underbrace{h_{k-s}, \dots, h_{k-1}}_{s \text{ factors}}, h_k]$$

$$h' = [[h'_0, \dots, h'_{l-(s+1)}, \underbrace{h'_{l-s}, \dots, h'_{l-1}}_{s \text{ factors}}, h'_l],$$

with $\deg(h_{k-(s+1)}) < \deg(h'_{l-(s+1)})$ and $\deg(h_{k-j}) = \deg(h'_{l-j})$ for all $0 \leq j < s + 1$. Then by the induction hypothesis $\bar{h} >_{lex} \tilde{h}$ and thus $h >_{lex} h'$.

(ii) Let $h = [h_0, h_1, \dots, h_k]$ and $h' = [h'_0, h'_1, \dots, h'_k]$ such that $\deg(h_j) = \deg(h'_j)$

for $0 \leq j \leq k$. Since $h >_{ilex} h'$, there exists $i \geq 0$ such that $h_i > h'_i$ and if $i > 0$, then $h_j = h'_j$ for all $j < i$. We prove the result by induction on k . For $k = 0$, we have $h_0 >_{lex} h'_0$ and thus $h >_{lex} h'$, as required. Suppose that for $s < k$, if

$$[h_0, h_1, \dots, h_s] >_{ilex} [h'_0, h'_1, \dots, h'_s],$$

then

$$[h_0, h_1, \dots, h_s] >_{lex} [h'_0, h'_1, \dots, h'_s].$$

Now write $h = [\bar{h}, h_k]$ and $h' = [\tilde{h}, h'_k]$. If $\bar{h} >_{ilex} \tilde{h}$, then by the induction hypothesis, $\bar{h} >_{lex} \tilde{h}$ and the result follows. Otherwise, if $\bar{h} = \tilde{h}$, then as $h >_{ilex} h'$ with $h_j = h'_j$ for all $0 \leq j < k$, we must have $h_k >_{lex} h'_k$ and therefore $h >_{lex} h'$.

□

We may define an ordering on a special Hall basis as follows. For any two elements h and h' in $\mathcal{H}$, we say $h > h'$ if (i) $\deg(h) > \deg(h')$, or (ii) $\deg(h) = \deg(h')$, and $iPart(h) > iPart(h')$, or (iii) $iPart(h) = iPart(h')$, and $h >_{ilex} h'$. Note that this order is the same as our old order according to Lemma 5.8.

For any two elements $d = d_1 d_2 \dots d_k$ and $d' = d'_1 d'_2 \dots d'_l$ in the PBW-basis $\mathcal{P}$ of A with $\deg(d) = \deg(d')$, we say that the partition of d is greater than the partition of d' and write $Part(d) > Part(d')$, if there exists an integer $i \geq 0$ such that $\deg(d_{k-i}) < \deg(d'_{l-i})$ and if $i > 0$, $\deg(d_{k-j}) = \deg(d'_{l-j})$ for all $j < i$. Otherwise, d and d' have the same partition if $k = l$ and $\deg(d_i) = \deg(d'_i)$ for all $1 \leq i \leq k$. For example, let A be the universal envelope of L as in Example 5.7. Then the element $a_1 a_1 a_1 a_2 a_2 [a_2, a_1, a_1]$ of degree 8 has a partition greater than the element $a_1 a_2 a_2 [a_2, a_1] [a_2, a_1, a_2]$ in $\mathcal{P}_8$, but both $a_1 a_2 a_2 [a_2, a_1]$ and $a_1 a_1 a_2 [a_2, a_1]$ have the same partition in $\mathcal{P}_5$.

Now we are ready to define an ordering on the PBW-basis. For any two elements d and d' in the PBW-basis $\mathcal{P}$, we say $d > d'$ if (i) $\deg(d) > \deg(d')$, or (ii) $\deg(d) = \deg(d')$, and $Part(d) > Part(d')$, or (iii) $Part(d) = Part(d')$, and $d >_{lex} d'$.

The following result is a generalisation of Waldinger's Theorem in the case where elements in $\mathcal{P}$ act on an element in $\mathcal{H}$.

Theorem 5.9 *Let $e \in \mathcal{H}$ and let both d and d' be elements in the PBW-basis $\mathcal{P}$ associated to $\mathcal{H}$, with $d > d'$ and $\deg(e) > \deg(d), \deg(d')$. Then*

$$M(e \circ d) > M(e \circ d').$$

Proof. Since $\deg(e) > \deg(d), \deg(d')$, neither $M(e \circ d)$ nor $M(e \circ d')$ is zero. Now we consider the following three cases.

Case 1. $\deg(d) > \deg(d')$. In this case $\deg(M(e \circ d)) > \deg(M(e \circ d'))$. Hence $M(e \circ d) > M(e \circ d')$, and the first case is proved.

Recall that, we may write $e = e_0 \circ (e_1^{\alpha_1} e_2^{\alpha_2} \dots e_n^{\alpha_n})$, $d = e_1^{\beta_1} e_2^{\beta_2} \dots e_n^{\beta_n}$ and $d' = e_1^{\gamma_1} e_2^{\gamma_2} \dots e_n^{\gamma_n}$, where e_i is the i th standard Lie monomial in a special Hall basis $\mathcal{H}$, the scalars α_i, β_i and γ_i are non-negative integers for $1 \leq i \leq n$ with $n \geq 1$, and e_0 is a free generator in $\mathcal{A}$. Then by Lemma 5.6, we have

$$\begin{aligned} M(e \circ d) &= e_0 \circ (e_1^{\alpha_1 + \beta_1} e_2^{\alpha_2 + \beta_2} \dots e_n^{\alpha_n + \beta_n}), \\ M(e \circ d') &= e_0 \circ (e_1^{\alpha_1 + \gamma_1} e_2^{\alpha_2 + \gamma_2} \dots e_n^{\alpha_n + \gamma_n}). \end{aligned} \tag{5.2}$$

Before continuing with the following cases, it is important to note that the standard Lie monomials (factors) e_i with $i \geq 1$ that appear in the interior partition of both $M(e \circ d)$ and $M(e \circ d')$ have an increasing ordering from the left to the right. This will be the key point in the rest of the proof.

Case 2. $\text{Part}(d) > \text{Part}(d')$. We may associate a sequence of non-negative integers $m = (m_1, m_2, \dots, m_t)$ with e such that m_i is equal to the number of standard Lie monomials of degree i appearing in the interior partition of e . We refer to the sequence m as the degree partition of e . Similarly, a degree partition associated with d (d') is a sequence $k = (k_1, k_2, \dots, k_t)$ ($l = (l_1, l_2, \dots, l_t)$) such that each k_i (l_i) represents the number of the factors of degree i that appear in d (d'), where $t = \max\{\deg(e_i) : i > 1 \text{ is the greatest integer with } \alpha_i, \beta_i \text{ or } \gamma_i \text{ is non-zero}\}$. Since

$Part(d) > Part(d')$, there exists $1 < s \leq t$ such that $k_s < l_s$ and if $s < t$, then $k_i = l_i$ for all $s < i \leq t$. Note that the degree partitions of $M(e \circ d)$ and $M(e \circ d')$ are $m + k = (m_1 + k_1, m_2 + k_2, \dots, m_t + k_t)$ and $m + l = (m_1 + l_1, m_2 + l_2, \dots, m_t + l_t)$, respectively. Since $k_s < l_s$, $m_s + k_s < m_s + l_s$ and $m_i + k_i = m_i + l_i$ for all $i > s$ if $s < t$. Therefore the interior partition of $M(e \circ d)$ is greater than the interior partition of $M(e \circ d')$ and thus the result follows by Lemma 5.8 (i).

Case 3. $d >_{lex} d'$. In this case we associate a lexicographic sequence with e , that is a sequence of non-negative integers $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_t)$. Similarly, we define $\beta = (\beta_1, \beta_2, \dots, \beta_t)$ and $\gamma = (\gamma_1, \gamma_2, \dots, \gamma_t)$ to be the lexicographic sequences associated with d and d' , respectively, where $t = \max\{i > 1 : \alpha_i, \beta_i \text{ or } \gamma_i \text{ is non-zero}\}$. Since $d >_{lex} d'$, there exists $s \geq 1$ such that $\beta_s < \gamma_s$ and if $s > 1$, we have $\beta_i = \gamma_i$ for all $1 \leq i < s$. Of course $\alpha + \beta$ is the lexicographic sequence associated with $M(e \circ d)$ and $\alpha + \gamma$ is the one that is associated with $M(e \circ d')$. As $\beta_s < \gamma_s$, we have $\alpha_s + \beta_s < \alpha_s + \gamma_s$ and $\alpha_i + \beta_i = \alpha_i + \gamma_i$ for all $i < s$ if $s > 1$. Thus $M(e \circ d) >_{ilex} M(e \circ d')$ and therefore $M(e \circ d) > M(e \circ d')$ by Lemma 5.8 (ii), as required.

□

Example 5.10 Let L be the free Lie algebra on $\{a, b\}$ over a field K such that $a < b$. Let $e = b \circ (a^3[b, a, a][b, a, a, a])$ be an element in a Hall basis of L .

Case 2: If $d = aa[b, a, a, a]$ and $d' = [b, a][b, a, a, b]$, then by (5.2) we have

$$M(e \circ d) = b \circ (a^5[b, a, a][b, a, a, a][b, a, a, a]),$$

$$M(e \circ d') = b \circ (a^3[b, a][b, a, a][b, a, a, a][b, a, a, b]).$$

Note that the degree partitions of e , d and d' are $m = (4, 0, 1, 1)$, $k = (2, 0, 0, 1)$ and $l = (0, 1, 0, 1)$, respectively. Clearly, $Part(d) > Part(d')$. Indeed, $k_2 < l_2$, $k_3 = l_3$ and $k_4 = l_4$. Therefore $m_2 + k_2 < m_2 + l_2$ and $m_i + k_i = m_i + l_i$ for $i = 3, 4$. This implies that the interior partition of $M(e \circ d)$ is greater than the interior partition of $M(e \circ d')$.

Case 3: If $d = b[b, a][b, a, a]$ and $d' = a[b, a][b, a, b]$, then $d >_{lex} d'$ and we get

$$M(e \circ d) = b \circ (a^3 b[b, a][b, a, a][b, a, a][b, a, a, a]),$$

$$M(e \circ d') = b \circ (a^4 [b, a][b, a, a][b, a, b][b, a, a, a]).$$

The lexicographic sequences associated with e , d and d' are $\alpha = (3, 0, 0, 1, 0, 1)$, $\beta = (0, 1, 1, 1, 0, 0)$ and $\gamma = (1, 0, 1, 0, 1, 0)$, respectively. Since $\beta_1 < \gamma_1$, $\alpha_1 + \beta_1 < \alpha_1 + \gamma_1$ and thus $M(e \circ d) >_{ilex} M(e \circ d')$.

Let L be a free Lie algebra of rank $r > 1$, and let A be the universal envelope of L . Then the dimension of the homogenous component A_m of degree m in A is r^m . As an application of the previous results, Corollary 5.11 shows that r^m is also equal to the dimension of $e \circ A_m$ under a certain condition.

Corollary 5.11 *For any homogenous component A_m in A and $e \in \mathcal{H}$ with $\deg(e) > m$, we have*

$$\dim(e \circ A_m) = r^m.$$

Proof. Let $\mathcal{P}_m = \{d^{(1)}, d^{(2)}, \dots, d^{(s)}\}$ be the PBW-basis of A_m where $s = r^m$. As we have already defined an ordering on the PBW-basis of the universal enveloping algebra A of L , we may suppose that $d^{(1)} < d^{(2)} < \dots < d^{(s)}$. Now let $\deg(e) = n$ and $\mathcal{H}_{m+n} = \{h_1, h_2, \dots, h_N\}$, where $N = \dim L_{m+n}$ and $h_1 < h_2 < \dots < h_N$. We claim that the set $\{e \circ d^{(l)} : 1 \leq l \leq s\}$ is a set of $s = r^m$ linearly independent elements. Recall that each element in $\mathcal{H}$ can be written uniquely in the left-normed decomposition form. Thus by Lemma 5.6, for any $d^{(k)}$ and $d^{(t)}$ ($k \neq t$) in $\mathcal{P}$ with $\deg(e) > \deg(d^{(k)})$, $\deg(d^{(t)})$, we have

$$d^{(k)} \neq d^{(t)} \text{ if and only if } M(e \circ d^{(k)}) \neq M(e \circ d^{(t)}).$$

Indeed, by rewriting each element from the set $\{e \circ d^{(l)} : 1 \leq l \leq s\}$ in the standard form (as a K -linear combination of elements from $\mathcal{H}_{m+n}$), we get

$$\begin{aligned}
 e \circ d^{(1)} &= \alpha_{1,1} h_1 + \dots + \alpha_{1,N_1} h_{N_1}, \\
 e \circ d^{(2)} &= \alpha_{2,1} h_1 + \alpha_{2,2} h_2 + \dots + \alpha_{2,N_2} h_{N_2}, \\
 &\vdots \\
 e \circ d^{(s-1)} &= \alpha_{s-1,1} h_1 + \alpha_{s-1,2} h_2 + \dots + \alpha_{s-1,N_{s-1}} h_{N_{s-1}}, \\
 e \circ d^{(s)} &= \alpha_{s,1} h_1 + \alpha_{s,2} h_2 + \alpha_{s,3} h_3 + \alpha_{s,4} h_4 + \dots + \alpha_{s,N_s} h_{N_s}, \quad (5.3)
 \end{aligned}$$

where $h_{N_l} = M(e \circ d^{(l)})$ and α_{l,N_l} is a non-zero scalar in K for $1 \leq l \leq s$.

By Theorem 5.9, for any $d^{(k)}$ and $d^{(t)}$ in $\mathcal{P}_m$ with $d^{(k)} < d^{(t)}$, we have $M(e \circ d^{(k)}) < M(e \circ d^{(t)})$. Therefore the matrix $[\alpha_{i,j}]$, where $1 \leq i \leq s$ and $1 \leq j \leq N_s$ (with the convention that $\alpha_{i,j} = 0$ if $\alpha_{i,j}$ does not appear in any identity in (5.3)) is in row echelon form and the claim follows. Hence

$$\dim(e \circ A_m) = \dim A_m = r^m.$$

□

Now we are ready to give a proof of Theorem 3.8 (see Chapter 3).

Proof. Let $(s_1, s_2, \dots, s_k)$ be a solution of equation (3.2) such that $s_i = u \circ u_A^{(i)}$ for $1 \leq i \leq k$ (see Section 3.4.1), $u \in \mathcal{H}$ with $\deg(u) = n$ and each $u_A^{(i)}$ is a unique homogenous polynomial in A_{n-1} . To show that this solution is non-zero, it is sufficient to prove that at least one of the s_i is non-zero for some $1 \leq i \leq k$. Since $u \neq 0$, at least one of the polynomials $u_A^{(i)}$ is non-zero. Let i be fixed and write

$$u_A^{(i)} = \sum_c \lambda_c c,$$

where each c is a standard associative monomial in A_{n-1} and $\lambda_c \in K$. Let

$$\mathcal{S} = \{c : c \text{ is a summand in } u_A^{(i)} \text{ and } \lambda_c \neq 0\}.$$

Since $\deg(c) = n - 1 < \deg(u)$ for all $c \in \mathcal{S}$, by Corollary 5.11, the set $u \circ \mathcal{S}$ is linearly independent in $u \circ A_{n-1}$. Therefore the element $u \circ u_A^{(i)}$ is non-zero and thus the result follows. $\square$

Waldinger's Theorem and Lemma 5.5 imply the following result.

Lemma 5.12 *Let e and h be elements in a Hall basis $\mathcal{H}$ of a free Lie algebra L with $e > h$. Let d be an arbitrary element in the PBW-basis of A associated to $\mathcal{H}$ such that $\deg(e), \deg(h) > \deg(d)$. Then*

$$M(e \circ d) > M(h \circ d).$$

Proof. Since d is in the PBW-basis of the universal envelop A of L , we can write $d = d_1 d_2 \dots d_k$, where $d_i \in \mathcal{H}$, $k \geq 1$ and $d_1 \leq d_2 \leq \dots \leq d_k$. By Lemma 5.5, we have

$$M(e \circ d) = M[M[\dots M[M[e, d_1], d_2], \dots], d_k],$$

$$M(h \circ d) = M[M[\dots M[M[h, d_1], d_2], \dots], d_k].$$

We prove the result by induction on k . For $k = 1$, since $d_1 \in \mathcal{H}$ and $\deg(e), \deg(h) > \deg(d_1)$, by Waldinger's Theorem,

$$M(e \circ d_1) = M[e, d_1] > M[h, d_1] = M(h \circ d_1).$$

Suppose that the result holds for $\tilde{d} = d_1 d_2 \dots d_{k-1}$, that is

$$\begin{aligned} M(e \circ \tilde{d}) &= M[M[\dots M[M[e, d_1], d_2], \dots], d_{k-1}] \\ &> M[M[\dots M[M[h, d_1], d_2], \dots], d_{k-1}] \\ &= M(h \circ \tilde{d}). \end{aligned}$$

Now by applying the induction hypothesis and Waldinger's Theorem, we get

$$\begin{aligned} M(e \circ d) &= M[M[\dots M[M[e, d_1], d_2], \dots], d_k] \\ &> M[M[\dots M[M[h, d_1], d_2], \dots], d_k] \\ &= M(h \circ d). \end{aligned}$$

Hence the result follows. $\square$

5.3 A generalisation of Waldinger's Theorem (2)

In the last section of this chapter, we generalise Waldinger's Theorem in the case when elements from the PBW-basis act on a non-zero element in L . Before we reach our main result, we need to know how we can determine $M(u \circ d)$, where $u \in L$ and $d \in \mathcal{P}$.

Lemma 5.13 *Let u be a non-zero homogenous element in a free Lie algebra L , and let d be an element in the PBW-basis of the universal enveloping algebra A of L with $\deg(u) > \deg(d)$. Then*

$$M(u \circ d) = M(M(u) \circ d).$$

Proof. Since $\mathcal{H}$ is a basis of L , u can be written as follows

$$u = \lambda h + \alpha_1 h_1 + \dots + \alpha_k h_k,$$

for some positive integer $k \geq 1$, where $0 \neq \lambda$, $\alpha_i \in K$, $h, h_i \in \mathcal{H}$ for $1 \leq i \leq k$ and $h = M(u)$. We write

$$\begin{aligned} u \circ d &= \lambda_0 M(h \circ d) + \sum_{h'_0 < M(h \circ d)} \alpha_{0, h'_0} h'_0 \\ &\quad + \lambda_1 M(h_1 \circ d) + \sum_{h'_1 < M(h_1 \circ d)} \alpha_{1, h'_1} h'_1 \\ &\quad + \dots + \lambda_k M(h_k \circ d) + \sum_{h'_k < M(h_k \circ d)} \alpha_{k, h'_k} h'_k, \end{aligned}$$

where $h'_i \in \mathcal{H}$ and the non-zero scalars λ_i and α_{i, h'_i} are in K for $0 \leq i \leq k$.

Since $h > h_i$ for all i , by Lemma 5.12, we get

$$M(h \circ d) > M(h_i \circ d).$$

Thus

$$M(u \circ d) = M(h \circ d) = M(M(u) \circ d),$$

as required. □

Now we are ready to prove a more general version of Waldinger's Theorem.

Theorem 5.14 *Let u be a homogenous element in L . Let d and d' be elements in the PBW-basis of A such that $d > d'$ and $\deg(u) > \deg(d), \deg(d')$. Then*

$$M(u \circ d) > M(u \circ d').$$

Proof. By applying Lemma 5.13, we have

$$M(u \circ d) = M(M(u) \circ d),$$

$$M(u \circ d') = M(M(u) \circ d').$$

Since $M(u) \in \mathcal{H}$ and $d > d'$, by Theorem 5.9, we get

$$M(M(u) \circ d) > M(M(u) \circ d').$$

Thus

$$M(u \circ d) > M(u \circ d').$$

□

Chapter 6

Radicals and coordinate algebras

In this chapter, we generalise some of the earlier results from [22, Section 6]. A large part of what follows has been obtained by using techniques similar to those used in the above mentioned paper. However, Theorem 6.12, where we obtain the radical and the coordinate algebra of the solution space of (3.2), strongly depends on Lemma 6.11, Lemma 6.10 and a new solution of Eq. (3.2) which is given in Section 6.2. To prove Lemma 6.10, we provide a new proof by using our results from Chapter 5.

6.1 Basic definitions and notation

Let $L = L(\mathcal{A})$ be the free Lie algebra on $\mathcal{A} = \{a_1, a_2, \dots, a_k, \dots\}$ over a field K , where $a_1 < a_2 < \dots < a_k < \dots$. Let $\mathcal{X}$ be a set of k indeterminates, say $x_1, x_2, \dots, x_k$. By $L[\mathcal{X}]$ we denote the free Lie algebra generated by $\mathcal{A} \cup \mathcal{X}$. For each $s = (s_1, s_2, \dots, s_k) \in L^{\oplus k}$, we define the evaluation homomorphism $\pi_s : L[\mathcal{X}] \rightarrow L$ to be the Lie homomorphism that maps a_i in $\mathcal{A}$ to itself and sends x_i to s_i for all $1 \leq i \leq k$. We introduce, as needed, the following basic notions: systems of equations over L , algebraic sets, radicals and coordinate algebras (see [10] for more details).

An element f in $L[\mathcal{X}]$ is referred to as a *polynomial* in the variables $x_1, x_2, \dots, x_k$ with coefficients from L . By putting $f = 0$, we get an *equation* over L . For simplicity, we will refer to elements in $L[\mathcal{X}]$ directly as equations. A subset S of $L[\mathcal{X}]$ is called a *system* of equations in the free Lie algebra L . An element $s = (s_1, s_2, \dots, s_k)$ is said to be a *solution* of a given equation $f \in L[\mathcal{X}]$ if $f\pi_s = 0$, and it is a solution of a system

of equations S if it is a solution of every equation $f \in S$. Let $\Sigma(S)$ denote the set of all possible solutions of a system S over L , that is

$$\Sigma(S) = \{s \in L^{\oplus k} : f\pi_s = 0 \ \forall f \in S\}.$$

A subset $U \subseteq L^{\oplus k}$ is called an *algebraic set* over the free Lie algebra L if there exists a system S in $L[\mathcal{X}]$ such that $U = \Sigma(S)$. In the case where $\Sigma(S) = \emptyset$, we say S is inconsistent system over L , otherwise we say that S is consistent. Let $S \subseteq L[\mathcal{X}]$ be a system of equations over L and let $U = \Sigma(S)$. The *radical* of U is the largest system in $L[\mathcal{X}]$ with solution set U . Thus

$$\text{Rad}(U) = \{f \in L[\mathcal{X}] : f\pi_s = 0 \ \forall s \in U\},$$

with the convention that $\text{Rad}(\emptyset) = L[\mathcal{X}]$, i.e. $L[\mathcal{X}]$ coincides with the radical of an inconsistent system. Note that an algebraic set is uniquely defined by its radical. So, for any two algebraic sets U and W , we have

$$U = W \Leftrightarrow \text{Rad}(U) = \text{Rad}(W) \text{ [10].}$$

If $S \subseteq L[\mathcal{X}]$ and $U = \Sigma(S)$ is an algebraic set in $L^{\oplus k}$, then it is clear that $S \subseteq \text{Rad}(U)$. Moreover, it is not hard to see that $\text{Rad}(U)$ is an ideal in $L[\mathcal{X}]$. Let $\text{Id}(S)$ denote the ideal in $L[\mathcal{X}]$ generated by a system S , then $\text{Id}(S) \subseteq \text{Rad}(U)$. If $\text{Id}(S) = \text{Rad}(U)$, then $\text{Id}(S)$ is called a *radical ideal*. In general, the inclusion $\text{Rad}(U) \subseteq \text{Id}(S)$ is not always satisfied. This can be seen in the following example. This example is mentioned briefly in [22]. Here we give a detailed explanation and a discussion.

Example 6.1 Let $\mathcal{A} = \{a, b\}$, $\mathcal{X} = \{x, y\}$ and let $L[\mathcal{X}]$ be the free Lie algebra with free generating set $\mathcal{A} \cup \mathcal{X}$ where $x < y < a < b$. If $S = \{[y, x]\}$, then $[y, u, [x, u]] \in \text{Rad}(U)$ for all $u \in L(\mathcal{A})$, where $U = \Sigma(S)$. But if u is a non-zero element, then $[y, u, [x, u]] \notin \text{Id}(S)$. To verify this, we write $L[\mathcal{X}] = \langle x, y, a, b \rangle \oplus L'$, where L' is the derived subalgebra of $L[\mathcal{X}]$ which is freely generated by

$$\mathcal{B} = \{[y_1, y_2, \dots, y_l] : y_1 > y_2 \leq y_3 \leq \dots \leq y_l\}$$

where $y_i \in \mathcal{A} \cup \mathcal{X}$ and $l \geq 2$ (see [23]). We may write $\mathcal{B} = \mathcal{C} \cup \mathcal{D}$, where $\mathcal{C}$ consists of all left-normed Lie monomials in $\mathcal{B}$ with $y_1 = b$ or $y_1 = a$, and

$$\mathcal{D} = \{[y, x, b_1, b_2, \dots, b_l] : x \leq b_1 \leq b_2 \leq \dots \leq b_l\}$$

where $b_i \in \mathcal{A} \cup \mathcal{X}$ and $l \geq 0$. Thus $L' = L(\mathcal{C}) \oplus D$, where $D = \text{Id}(\mathcal{D})$. We claim that D as an ideal in L' coincides with $\text{Id}([y, x])$ in $L[\mathcal{X}]$. To prove our claim we show that D is an ideal in $L[\mathcal{X}]$. Note that the ideal generated by $\mathcal{D}$ in L' is spanned by Lie monomials of the form $[g_1, g_2, \dots, g_k]$ for some positive integer k , where $g_1 \in \mathcal{D}$ and $g_i \in \mathcal{B}$ for $2 \leq i \leq k$. Thus it is enough to prove that each such Lie monomial will still belong to D after it is multiplied by c , where c is an arbitrary free generator of $L[\mathcal{X}]$. Let $c \in \mathcal{A} \cup \mathcal{X}$. Then by applying (2.3), we have

$$[[g_1, g_2, \dots, g_k], c] = [[g_1, c], g_2, \dots, g_k] + [g_1, [g_2, c], \dots, g_k] + \dots + [g_1, g_2, \dots, [g_k, c]].$$

It is clear that all the summands $[g_1, g_2, \dots, [g_i, c], \dots, g_k]$ with $1 < i \leq k$ belong to D . To reach our goal, it is enough now to show that $[g_1, c] \in D$. Indeed, we prove that $[g_1, c]$ can be written as a linear combination of elements of the form $[u_1, u_2, \dots, u_t]$, where $u_1 \in \mathcal{D}$ and $u_j \in \mathcal{B}$ for $j \neq 1$ and $t \geq 1$. Pick $g_1 = [y, x, b_1, \dots, b_l] \in \mathcal{D}$, $l \geq 0$. We shall prove this part by induction on $\deg(g_1)$. For $\deg(g_1) = 2$, we have

$$[g_1, c] = [y, x, c] \in \mathcal{D} \text{ for all } c \in \mathcal{A} \cup \mathcal{X}.$$

Suppose that the result holds for all Lie monomials of degree less than $\deg(g_1)$ in $\mathcal{D}$. If $b_l \leq c$, then

$$[g_1, c] = [[y, x, b_1, \dots, b_l], c] \in \mathcal{D}.$$

Otherwise, we write

$$[g_1, c] = [\underbrace{[y, x, b_1, \dots, b_{l-1}, c]}_{< \deg(g_1)}, b_l] + [y, x, b_1, \dots, b_{l-1}, [b_l, c]].$$

Clearly, the Lie monomial $[y, x, b_1, \dots, b_{l-1}]$ belongs to $\mathcal{D}$. Since $b_l > c$, $[b_l, c] \in \mathcal{B}$ and thus the second summand is in the required form. By the induction hypothesis, the first summand can be written as a linear combination of elements of the form $[u_1, u_2, \dots, u_t, b_l]$, where $u_1 \in \mathcal{D}$, $u_j \in \mathcal{B}$ for $j \neq 1$ and $t \geq 1$. By applying (2.3) to each summand $[u_1, u_2, \dots, u_t, b_l]$, we get

$$\sum_{i=1}^t [u_1, u_2, \dots, [u_i, b_l], \dots, u_t]. \quad (6.1)$$

Since $b_l > c$ and $b_1 \leq b_2 \leq \dots \leq b_l$, $[u_1, b_l] \in \mathcal{D}$ and $[u_j, b_l] \in \mathcal{B}$ for all $j > 1$. Therefore all the summands in (6.1) are in the required form. Thus $[g_1, c]$ belongs to $D = \text{Id}(\mathcal{D})$ and, consequently, $[[g_1, c], g_2, \dots, g_k]$ is in D . Hence the claim follows.

The following lemma can be used to determine whether a given ideal in $L[\mathcal{X}]$ is a radical ideal or not. We apply this result later in this chapter to find the radical of the solution space of Eq. (3.2).

Lemma 6.2 [22] *Let U be the algebraic set of a system $S \subseteq L[\mathcal{X}]$ and let J be an ideal in the free Lie algebra $L[\mathcal{X}]$. Then J and $\text{Rad}(U)$ are equal if and only if the following conditions hold:*

- (i) *For all elements g in J and all $s \in U$, $g\pi_s = 0$.*
- (ii) *For all g in $L[\mathcal{X}] \setminus J$ there exists an element s belonging to U such that $g\pi_s \neq 0$.*

For an algebraic set U , the *coordinate algebra* $\Gamma(U)$ is the factor algebra

$$\Gamma(U) = L[\mathcal{X}]/\text{Rad}(U).$$

In the following two sections, we will focus on a particular case in which the system is $S = [x_1, a_1] + [x_2, a_2] + \dots + [x_k, a_k]$ and U is the solution space V of equation (3.2). Clearly, in this case S is a consistent system. Indeed, S is a homogenous system consisting of one equation and $U = V$ is a linear subspace of $L^{\oplus k}$.

6.2 A particular non-zero solution

The aim of this section is to find a specific non-zero solution of equation (3.2). This solution will play an important role in order to obtain the main result in this chapter.

Lemma 6.3 *Let l and k be positive integers such that $l \geq 1$, $k \geq 2$ and let $a_1, a_2, \dots, a_k \in \mathcal{A}$. Then for $k = 2$ and $k = 3$, we have*

$$[a_2, a_1^l] = \sum_{i=1}^l (-1)^{i+1} a_1^{i-1} [a_2, a_1^{l-i}] a_1 + (-1)^l a_1^l a_2$$

and

$$-[a_2, a_1^l, a_3] = \sum_{i=1}^l (-1)^{i+1} a_3 a_1^{i-1} [a_2, a_1^{l-i}] a_1 + (-1)^l a_3 a_1^l a_2 - [a_2, a_1^l] a_3,$$

respectively. Also,

$$\begin{aligned}
(-1)^k[a_2, a_1^l, a_3, \dots, a_{k-1}, a_k] &= \sum_{i=1}^l (-1)^{i+1} a_k a_{k-1} \dots a_3 a_1^{i-1} [a_2, a_1^{l-i}] a_1 \\
&\quad + (-1)^l a_k a_{k-1} \dots a_3 a_1^l a_2 \\
&\quad - a_k a_{k-1} \dots a_4 [a_2, a_1^l] a_3 \\
&\quad + \sum_{j=4}^{k-1} (-1)^j a_k a_{k-1} \dots a_{j+1} [a_2, a_1^l, a_3, a_4, \dots, a_{j-1}] a_j \\
&\quad + (-1)^k [a_2, a_1^l, a_3, \dots, a_{k-1}] a_k,
\end{aligned}$$

for $k \geq 4$.

Proof. The result can be obtained by applying Lemma 4.2 directly. We prove it in the case when $k \geq 4$ and in a similar way the identity at $k = 2$ and $k = 3$ can be verified. We need to show that

$$\begin{aligned}
(-1)^k[a_2, a_1^l, a_3, \dots, a_k] &= (-1)^2 a_k a_{k-1} \dots a_3 [a_2, a_1^{l-1}] a_1 \\
&\quad + (-1)^3 a_k a_{k-1} \dots a_3 a_1 [a_2, a_1^{l-2}] a_1 \\
&\quad + (-1)^4 a_k a_{k-1} \dots a_3 a_1^2 [a_2, a_1^{l-3}] a_1 + \dots \\
&\quad + (-1)^{l-2} a_k a_{k-1} \dots a_3 a_1^{l-4} [a_2, a_1^3] a_1 \\
&\quad + (-1)^{l-1} a_k a_{k-1} \dots a_3 a_1^{l-3} [a_2, a_1^2] a_1 \\
&\quad + (-1)^l a_k a_{k-1} \dots a_3 a_1^{l-2} [a_2, a_1] a_1 \\
&\quad + (-1)^{l+1} a_k a_{k-1} \dots a_3 a_1^{l-1} a_2 a_1 \\
&\quad + (-1)^l a_k a_{k-1} \dots a_3 a_1^l a_2 \\
&\quad + (-1)^3 a_k a_{k-1} \dots a_4 [a_2, a_1^l] a_3 \\
&\quad + (-1)^4 a_k a_{k-1} \dots a_5 [a_2, a_1^l, a_3] a_4 \\
&\quad + (-1)^5 a_k a_{k-1} \dots a_6 [a_2, a_1^l, a_3, a_4] a_5 + \dots \\
&\quad + (-1)^{k-2} a_k a_{k-1} [a_2, a_1^l, a_3, \dots, a_{k-3}] a_{k-2} \\
&\quad + (-1)^{k-1} a_k [a_2, a_1^l, a_3, \dots, a_{k-2}] a_{k-1} \\
&\quad + (-1)^k [a_2, a_1^l, a_3, \dots, a_{k-1}] a_k,
\end{aligned}$$

where $k \geq 4$. From Lemma 4.2, we have

$$\begin{aligned}
[c_1, c_2, \dots, c_m] = & (-1)^{m-1} c_m c_{m-1} \dots c_2 c_1 \\
& + (-1)^{m-2} c_m c_{m-1} \dots c_3 c_1 c_2 \\
& + (-1)^{m-3} c_m c_{m-1} \dots c_4 [c_1, c_2] c_3 \\
& + (-1)^{m-4} c_m c_{m-1} \dots c_5 [c_1, c_2, c_3] c_4 \\
& + (-1)^{m-5} c_m c_{m-1} \dots c_6 [c_1, c_2, c_3, c_4] c_5 + \dots \\
& + (-1)^{m-(l-1)} c_m c_{m-1} \dots c_l [c_1, c_2, \dots, c_{l-2}] c_{l-1} \\
& + (-1)^{m-l} c_m c_{m-1} \dots c_{l+1} [c_1, c_2, \dots, c_{l-1}] c_l \\
& + (-1)^{m-(l+1)} c_m c_{m-1} \dots c_{l+2} [c_1, c_2, \dots, c_l] c_{l+1} \\
& + (-1)^{m-(l+2)} c_m c_{m-1} \dots c_{l+3} [c_1, c_2, \dots, c_{l+1}] c_{l+2} \\
& + (-1)^{m-(l+3)} c_m c_{m-1} \dots c_{l+4} [c_1, c_2, \dots, c_{l+2}] c_{l+3} \\
& + (-1)^{m-(l+4)} c_m c_{m-1} \dots c_{l+5} [c_1, c_2, \dots, c_{l+3}] c_{l+4} + \dots \\
& + (-1)^{m-(m-2)} c_m c_{m-1} [c_1, c_2, \dots, c_{m-3}] c_{m-2} \\
& + (-1)^{m-(m-1)} c_m [c_1, c_2, \dots, c_{m-2}] c_{m-1} \\
& + [c_1, c_2, \dots, c_{m-1}] c_m.
\end{aligned} \tag{6.2}$$

Now put

$$\begin{aligned}
c_1 &= a_2, \quad c_2 = c_3 = \dots = c_{l-1} = c_l = c_{l+1} = a_1, \quad c_{l+2} = a_3, \quad c_{l+3} = a_4, \\
\dots \quad, \quad c_{l+j} &= a_{j+1}, \quad \dots, \quad c_{m-1} = c_{l+k-2} = a_{k-1}, \quad c_m = c_{l+k-1} = a_k.
\end{aligned} \tag{6.3}$$

By substituting (6.3) in (6.2) and multiplying both sides by $(-1)^k$, we get

$$\begin{aligned}
(-1)^k [c_1, c_2, \dots, c_m] &= (-1)^k [a_2, a_1^l, a_3, \dots, a_k] \\
&= (-1)^{l+2k-2} a_k a_{k-1} \dots a_3 a_1^l a_2 \\
&\quad + (-1)^{l+2k-3} a_k a_{k-1} \dots a_3 a_1^{l-1} a_2 a_1 \\
&\quad + (-1)^{l+2k-4} a_k a_{k-1} \dots a_3 a_1^{l-2} [a_2, a_1] a_1 \\
&\quad + (-1)^{l+2k-5} a_k a_{k-1} \dots a_3 a_1^{l-3} [a_2, a_1^2] a_1 \\
&\quad + (-1)^{l+2k-6} a_k a_{k-1} \dots a_3 a_1^{l-4} [a_2, a_1^3] a_1 + \dots \\
&\quad + (-1)^{2k} a_k a_{k-1} \dots a_3 a_1^2 [a_2, a_1^{l-3}] a_1
\end{aligned}$$

$$\begin{aligned}
 & +(-1)^{2k-1}a_ka_{k-1}\dots a_3a_1[a_2, a_1^{l-2}]a_1 \\
 & +(-1)^{2k-2}a_ka_{k-1}\dots a_3[a_2, a_1^{l-1}]a_1 \\
 & +(-1)^{2k-3}a_ka_{k-1}\dots a_4[a_2, a_1^l]a_3 \\
 & +(-1)^{2k-4}a_ka_{k-1}\dots a_5[a_2, a_1^l, a_3]a_4 \\
 & +(-1)^{2k-5}a_ka_{k-1}\dots a_6[a_2, a_1^l, a_3, a_4]a_5 + \dots \\
 & +(-1)^{k+2}a_ka_{k-1}[a_2, a_1^l, a_3, \dots, a_{k-3}]a_{k-2} \\
 & +(-1)^{k+1}a_k[a_2, a_1^l, a_3, \dots, a_{k-2}]a_{k-1} \\
 & +(-1)^k[a_2, a_1^l, a_3, \dots, a_{k-1}]a_k.
 \end{aligned} \tag{6.4}$$

Note that

$$\begin{aligned}
 (-1)^{l+2k-(2i'+1)} &= (-1)^{l+(2j'+1)} = (-1)^{l-(2j'+1)} = (-1)^{l\pm 1}, \\
 (-1)^{l+2k-2i'} &= (-1)^{l+2j'} = (-1)^{l-2j'} = (-1)^l, \\
 (-1)^{2k-(2i'+1)} &= (-1)^{2j'+1} = -1, \\
 (-1)^{2k-2i'} &= (-1)^{2j'} = 1,
 \end{aligned}$$

for some integers i' and j' . By comparing all the summands and the coefficients in (6.4) with the required ones, we get the desired result.

□

Theorem 6.4 *Let l and k be positive integers such that $l \geq 1$ and $k \geq 2$. Then for a fixed l , $(s_{1,l}, s_{2,l}, \dots, s_{k,l})$ is a non-zero solution of equation (3.2) of degree n in $V_{(2l, 2, 2, \dots, 2)}$, where $n = 2k + 2l - 3$ and*

- for $k = 2$,

$$s_{1,l} = \sum_{i=1}^l (-1)^{i+1} [a_2, a_1^{l+i-1}, [a_2, a_1^{l-i}]],$$

$$s_{2,l} = (-1)^l [a_2, a_1^{2l}].$$

- For $k = 3$,

$$s_{1,l} = \sum_{i=1}^l (-1)^{i+1} [a_2, a_1^l, a_3^2, a_1^{i-1}, [a_2, a_1^{l-i}]],$$

$$s_{2,l} = (-1)^l [a_2, a_1^l, a_3^2, a_1^l],$$

$$s_{3,l} = -[a_2, a_1^l, a_3, [a_2, a_1^l]].$$

- For $k \geq 4$,

$$s_{1,l} = \sum_{i=1}^l (-1)^{i+1} [a_2, a_1^l, a_3, \dots, a_{k-1}, a_k, a_k, a_{k-1}, \dots, a_3, a_1^{i-1}, [a_2, a_1^{l-i}]],$$

$$s_{2,l} = (-1)^l [a_2, a_1^l, a_3, \dots, a_{k-1}, a_k, a_k, a_{k-1}, \dots, a_3, a_1^l],$$

$$s_{3,l} = -[a_2, a_1^l, a_3, \dots, a_{k-1}, a_k, a_k, a_{k-1}, \dots, a_4, [a_2, a_1^l]],$$

$$s_{j,l} = (-1)^j [a_2, a_1^l, a_3, \dots, a_{k-1}, a_k, a_k, a_{k-1}, \dots, a_{j+1}, [a_2, a_1^l, a_3, a_4, \dots, a_{j-1}]],$$

$$4 \leq j \leq k-1$$

$$s_{k,l} = (-1)^k [a_2, a_1^l, a_3, \dots, a_{k-1}, a_k, [a_2, a_1^l, a_3, \dots, a_{k-1}]].$$

Proof. Recall that, for all u in a free Lie algebra L we have $[u, u] = -[u, u] = 0$. Now put $u = [a_2, a_1^l, a_3, \dots, a_{k-1}, a_k]$. Then by Lemma 6.3, we get

$$\begin{aligned} 0 &= [u, u] \\ &= [a_2, a_1^l, a_3, \dots, a_{k-1}, a_k, [a_2, a_1^l, a_3, \dots, a_{k-1}, a_k]] \\ &= [s_{1,l}, a_1] + [s_{2,l}, a_2] + \dots + [s_{k,l}, a_k]. \end{aligned}$$

As $u \in \mathcal{H}$, by Theorem 3.8, the solution that we have obtained is non-zero. Similarly, for $k = 2$ and $k = 3$, put $u = [a_2, a_1^l]$ and $u = [a_2, a_1^l, a_3]$, respectively. $\square$

Recall that, for an element u in a free Lie algebra L with a hall basis $\mathcal{H}$, $M(u)$ denotes the largest element in $\mathcal{H}$ that occurs with a non-zero coefficient when we write u in standard form (as a linear combination of elements from the given basis). The following lemma is an application on Waldinger's results from Chapter 5.

Lemma 6.5 *Let $s_{j,l}$ be given as in Theorem 6.4, where $1 \leq j \leq k$. Then*

- *when $k = 2$,*

$$M(s_{1,l}) = [a_2, a_1^{2^{l-1}}, a_2],$$

$$M(s_{2,l}) = [a_2, a_1^{2^l}].$$

- *In the case where $k = 3$,*

$$M(s_{1,l}) = [a_2, a_1^{2^{l-1}}, a_2, a_3^2],$$

$$M(s_{2,l}) = [a_2, a_1^{2^l}, a_3^2],$$

$$M(s_{3,l}) = [a_2, a_1^l, a_3, [a_2, a_1^l]].$$

- *For $k \geq 4$, we have*

$$M(s_{1,l}) = [a_2, a_1^{2^{l-1}}, a_2, a_3^2, \dots, a_k^2],$$

$$M(s_{2,l}) = [a_2, a_1^{2^l}, a_3^2, \dots, a_k^2],$$

$$M(s_{3,l}) = [a_2, a_1^l, a_3, a_4^2, \dots, a_k^2, [a_2, a_1^l]],$$

$$M(s_{j,l}) = [a_2, a_1^l, a_3, \dots, a_{j-1}, a_j, a_{j+1}^2, \dots, a_k^2, [a_2, a_1^l, a_3, \dots, a_{j-1}]], \quad 4 \leq j \leq k-1$$

$$M(s_{k,l}) = [a_2, a_1^l, a_3, \dots, a_{k-1}, a_k, [a_2, a_1^l, a_3, \dots, a_{k-1}]].$$

Proof. For $k \geq 4$, note that each component $s_{j,l}$ in the given solution (Theorem 6.4) is of the form

$$\pm[a_2, a_1^l, a_3, \dots, a_{k-1}, a_k, h_1, h_2, \dots, h_{t_j}],$$

where $h_1, h_2, \dots, h_{t_j} \in \mathcal{H}$, $1 < j \leq k$ and $1 \leq t_j \leq k + l - 2$. Put $e = [a_2, a_1^l, a_3, \dots, a_k]$ and $d^{(j)} = h_1 h_2 \dots h_{t_j}$. Since $\deg(e) > \deg(d^{(j)})$, $M(s_{j,l}) = M(e \circ d^{(j)})$ can be determined by applying Lemma 5.6 for all $1 < j \leq k$. In the case when $j = 1$,

$$s_{1,l} = \sum_{i=1}^l (-1)^{i+1} w_i,$$

where each $w_i = [a_2, a_1^l, a_3, \dots, a_{k-1}, a_k, h_1, h_2, \dots, h_{t_{1_i}}]$, $k - 1 \leq t_{1_i} \leq k + l - 2$. Thus

$$M(s_{1,l}) = \max\{M(w_i) = M(e \circ d^{(1_i)}) : 1 \leq i \leq l\}.$$

The same argument can be used to prove the lemma when $k = 2$ and $k = 3$.

□

6.3 The radical and the coordinate algebra of V

The aim of this part in this chapter is to prove that $\text{Rad}(V)$ is equal to the ideal generated by $[x_1, a_1] + [x_2, a_2] + \dots + [x_k, a_k]$ in $L[\mathcal{X}]$ (Theorem 6.12), where V is the solution space of equation (3.2). To be able to do this, we need to discuss some other results.

Let L be a free Lie algebra on $\mathcal{A} = \{a_1, a_2, \dots, a_k, \dots\}$ and let $L[\mathcal{X}] = L(\mathcal{A} \cup \mathcal{X})$. Then we can write

$$L[\mathcal{X}] = L \oplus I$$

where I is the ideal generated by $x_1, x_2, \dots, x_k$ in $L[\mathcal{X}]$. By Lazard Elimination Theorem, the following elements form a free generating set for the free Lie subalgebra I of $L[\mathcal{X}]$:

$$[x_1, b_1, \dots, b_m], [x_2, b_1, \dots, b_m], \dots, [x_k, b_1, \dots, b_m] \quad (b_i \in \mathcal{A}, m \geq 0). \quad (6.5)$$

Let J be the ideal in $L[\mathcal{X}]$ generated by the left-hand side of equation (3.2), that is $J = \text{Id}([x_1, a_1] + [x_2, a_2] + \dots + [x_k, a_k])$ in $L[\mathcal{X}]$. Then the next lemma holds.

Lemma 6.6 *Let $\hat{J}$ be the ideal of I generated by the elements*

$$[x_1, a_1, b_1, \dots, b_m] + [x_2, a_2, b_1, \dots, b_m] + \dots + [x_k, a_k, b_1, \dots, b_m], \quad (6.6)$$

where $b_i \in \mathcal{A}$ and $m \geq 0$. Then $\hat{J} = J$.

Proof. Note that $[x_1, a_1] + [x_2, a_2] + \dots + [x_k, a_k] \in \hat{J}$ and the elements (6.6) are in J . Since $\hat{J}$ is generated by the elements (6.6) in I , it is spanned by all Lie products of the form $[g_1, g_2, \dots, g_l]$, where $l \geq 1$, g_1 belongs to the set of elements in (6.6) and $g_2, \dots, g_l$ are elements from (6.5). We claim that $\hat{J}$ is an ideal in $L[\mathcal{X}]$. Let a be an arbitrary element in $\mathcal{A}$. Then by applying the identity (2.3), we can write

$$[g_1, g_2, \dots, g_l, a] = [[g_1, a], g_2, \dots, g_l] + [g_1, [g_2, a], \dots, g_l] + \dots + [g_1, g_2, \dots, [g_l, a]].$$

In the all summands appearing on the right-hand side, the first entry is of the form (6.6), and all the the other ones are elements from (6.5). Thus $\hat{J}$ is an ideal in $L[\mathcal{X}]$ and the claim follows. $\square$

Corollary 6.7 *Let $m \geq 1$ and let $c_i, b_i \in \mathcal{A}$ for $1 \leq i \leq m$ such that $c_1 \neq a_1$. Then the elements*

$$x_1, x_2, \dots, x_k, [x_1, c_1, \dots, c_m], [x_2, b_1, \dots, b_m], \dots, [x_k, b_1, \dots, b_m] \quad (6.7)$$

are free generators of the free Lie algebra I/J .

Proof. Note that both sets of elements in (6.7) and (6.6) form a free generating set of the ideal I . As J coincides with the ideal in I generated by the elements (6.6), the factor algebra I/J is freely generated by the elements (6.7). $\square$

Corollary 6.8 *The factor algebra $L[\mathcal{X}]/J$ is the semidirect sum*

$$L[\mathcal{X}]/J = L \oplus I/J,$$

where I and J are the ideals in $L[\mathcal{X}]$ generated by $x_1, x_2, \dots, x_k$ and $[x_1, a_1] + [x_2, a_2] + \dots + [x_k, a_k]$, respectively.

At this stage, it is convenient to introduce the notation $\langle \mathcal{B} \rangle$ for the K -span of a set $\mathcal{B}$. Recall that $L = L(\mathcal{A})$ is an A -module, where A is the free associative algebra generated by the set $\mathcal{A}$. To prove the next lemma, it is enough to show that it is true in the case when $\mathcal{A}$ is finite, so assume that $|\mathcal{A}| = k$ where $k \geq 2$ and let A_m

be the degree m homogenous component of A . Recall that the set of all associative monomials $a_{i_1}a_{i_2}\dots a_{i_m}$, say $\mathcal{D}_m$, with $a_{i_j} \in \mathcal{A}$ form the standard basis of A_m . We may write $\mathcal{D}_m = \mathcal{D}_m^{a_1} \cup \mathcal{D}_m^{\sim a_1}$ where $\mathcal{D}_m^{a_1}$ is the set of all monomials $a_{i_1}a_{i_2}\dots a_{i_m}$ with $a_{i_1} = a_1$ and $\mathcal{D}_m^{\sim a_1} = \mathcal{D}_m \setminus \mathcal{D}_m^{a_1}$.

Suppose that $\mathcal{A}$ is an ordered set such that $a_1 < a_2 < \dots < a_k$. Let $\mathcal{H}$ denote a Hall basis of L and let $\mathcal{P}_m$ be the PBW-basis of A_m derived from $\mathcal{H}$. Then $\mathcal{P}_m = \mathcal{P}_m^{a_1} \cup \mathcal{P}_m^{\sim a_1}$, where $\mathcal{P}_m^{a_1}$ is the set of all elements in the PBW-basis of A_m such that a_1 is the first factor and $\mathcal{P}_m^{\sim a_1}$ is its complement in $\mathcal{P}_m$. Thus we can write A_m as follows

$$A_m = \langle \mathcal{D}_m^{a_1} \rangle \oplus \langle \mathcal{D}_m^{\sim a_1} \rangle = \langle \mathcal{P}_m^{a_1} \rangle \oplus \langle \mathcal{P}_m^{\sim a_1} \rangle.$$

For $l \geq 1$, let $(s_{1,l}, s_{2,l}, \dots, s_{k,l})$ be the solution of Eq. (3.2) which is given in Theorem 6.4. Then we have

$$[s_{1,l}, a_1] = -[s_{2,l}, a_2] - [s_{3,l}, a_3] - \dots - [s_{k,l}, a_k].$$

Thus

$$\begin{aligned} s_{1,l} \circ \langle \mathcal{P}_m^{a_1} \rangle &= s_{1,l} \circ \langle \mathcal{D}_m^{a_1} \rangle \\ &= [s_{1,l}, a_1] \circ A_{m-1} \\ &= -[s_{2,l}, a_2] \circ A_{m-1} - [s_{3,l}, a_3] \circ A_{m-1} - \dots - [s_{k,l}, a_k] \circ A_{m-1} \\ &\subseteq s_{2,l} \circ A_m + s_{3,l} \circ A_m + \dots + s_{k,l} \circ A_m. \end{aligned} \tag{6.8}$$

Now consider the set of elements

$$[s_{1,l}, c_1, c_2, \dots, c_m], [s_{j,l}, b_1, b_2, \dots, b_m] \quad (2 \leq j \leq k, c_i, b_i \in \mathcal{A}, c_1 \neq a_1), \tag{6.9}$$

and the set of the following elements

$$[s_{1,l}, d_1, d_2, \dots, d_t], [s_{j,l}, \bar{d}_1, \bar{d}_2, \dots, \bar{d}_{t'}] \quad (2 \leq j \leq k, d_i, \bar{d}_i \in \mathcal{H}, d_1 \neq a_1), \tag{6.10}$$

where $d_1 \leq d_2 \leq \dots \leq d_t$, $\bar{d}_1 \leq \bar{d}_2 \leq \dots \leq \bar{d}_{t'}$, $\sum_{i=1}^t \deg(d_i) = \sum_{i=1}^{t'} \deg(\bar{d}_i) = m \geq 1$, and $t, t' \leq m$. Then the elements in (6.9) and (6.10) span

$$s_{1,l} \circ \langle \mathcal{D}_m^{\sim a_1} \rangle + s_{2,l} \circ A_m + \dots + s_{k,l} \circ A_m$$

and

$$s_{1,l} \circ \langle \mathcal{P}_m^{a_1} \rangle + s_{2,l} \circ A_m + \dots + s_{k,l} \circ A_m,$$

respectively. Moreover, we have

Lemma 6.9 *The two K -spaces coincide, that is*

$$s_{1,l} \circ \langle \mathcal{D}_m^{\sim a_1} \rangle + s_{2,l} \circ A_m + \dots + s_{k,l} \circ A_m = s_{1,l} \circ \langle \mathcal{P}_m^{\sim a_1} \rangle + s_{2,l} \circ A_m + \dots + s_{k,l} \circ A_m.$$

Proof. To prove this identity, we show that

$$(1) \quad s_{1,l} \circ \langle \mathcal{D}_m^{\sim a_1} \rangle \subseteq s_{1,l} \circ \langle \mathcal{P}_m^{\sim a_1} \rangle + s_{2,l} \circ A_m + \dots + s_{k,l} \circ A_m, \text{ and}$$

$$(2) \quad s_{1,l} \circ \langle \mathcal{P}_m^{\sim a_1} \rangle \subseteq s_{1,l} \circ \langle \mathcal{D}_m^{\sim a_1} \rangle + s_{2,l} \circ A_m + \dots + s_{k,l} \circ A_m.$$

For the first part, let $s_{l,1} \circ \sum_c \lambda_c c \in s_{1,l} \circ \langle \mathcal{D}_m^{\sim a_1} \rangle$, where $c = c_1 c_2 \dots c_m \in \mathcal{D}_m^{\sim a_1}$ and $\lambda_c \in K$. In fact, it is enough to work with an arbitrary summand c from the given sum. We may write $s_{1,l} \circ (c_1 c_2 \dots c_m) = s_{1,l} \circ \sum_d \lambda_d d \in s_{1,l} \circ \langle \mathcal{P}_m \rangle$, where $d = d_1 d_2 \dots d_t \in \mathcal{P}_m$ and $\lambda_d \in K$. If $d_1 = a_1$, then by (6.8) we have

$$\begin{aligned} s_{l,1} \circ (d_1 d_2 \dots d_t) &= [s_{l,1}, a_1, d_2, \dots, d_t] \\ &= -[s_{l,2}, a_2, d_2, \dots, d_t] - \dots - [s_{l,k}, a_k, d_2, \dots, d_t] \\ &\in s_{l,2} \circ A_m + \dots + s_{l,k} \circ A_m. \end{aligned}$$

Otherwise, $s_{l,1} \circ (d_1 d_2 \dots d_t) \in s_{l,1} \circ \langle \mathcal{P}_m^{\sim a_1} \rangle$. Similarly, pick $s_{l,1} \circ \sum_d \lambda_d d \in s_{1,l} \circ \langle \mathcal{P}_m^{\sim a_1} \rangle$, where $d = d_1 d_2 \dots d_t \in \mathcal{P}_m^{\sim a_1}$ and $\lambda_d \in K$. Let $s_{1,l} \circ (d_1 d_2 \dots d_t)$ be an arbitrary summand in this sum. Then we can write $s_{1,l} \circ (d_1 d_2 \dots d_t) = s_{l,1} \circ \sum_c \lambda_c c \in s_{1,l} \circ \langle \mathcal{D}_m \rangle$, where $c = c_1 c_2 \dots c_m \in \mathcal{D}_m$ and $\lambda_c \in K$. Now if $c_1 = a_1$, then again by (6.8) we have

$$\begin{aligned} s_{l,1} \circ (c_1 c_2 \dots c_m) &= [s_{l,1}, a_1, c_2, \dots, c_m] \\ &= -[s_{l,2}, a_2, c_2, \dots, c_m] - \dots - [s_{l,k}, a_k, c_2, \dots, c_m] \\ &\in s_{l,2} \circ A_m + \dots + s_{l,k} \circ A_m. \end{aligned}$$

Otherwise, $s_{l,1} \circ (c_1 c_2 \dots c_m) \in s_{l,1} \circ \langle \mathcal{D}_m^{\sim a_1} \rangle$.

□

Lemma 6.10 *Let m and l be positive integers such that $m < l+1$. Then the elements in (6.9) are linearly independent in L .*

Proof. Since both sets of elements in (6.9) and (6.10) span the same space and they have the same cardinality, to prove that the elements (6.9) are linearly independent, it is enough to claim that the elements (6.10) are linearly independent in L . Let X_j denote the set of all elements of the form $M(M(s_{j,l}) \circ d)$ which can be determined by applying Lemma 6.5 and Lemma 5.6, where $d = d_1 d_2 \dots d_t \in \mathcal{P}_m^{\sim a_1}$ when $j = 1$ and $d = \bar{d}_1 \bar{d}_2 \dots \bar{d}_{t'} \in \mathcal{P}_m$ for $2 \leq j \leq k$. To prove our claim, we show that $\bigcup_{j=1}^k X_j$ consists of distinct elements in $\mathcal{H}$. As

$$\deg(s_{j,l}) = 2k + 2l - 3 > l + 1 > m \quad (k \geq 2),$$

Theorem 5.14 implies that each X_j contains distinct standard Lie monomials. Since $m < l + 1$, neither d_i nor $\bar{d}_{i'}$ is equal to $[a_2, a_1^l]$ or $[a_2, a_1^l, a_3, \dots, a_{j-1}]$, $4 \leq j \leq k$ for all $1 \leq i \leq t$ and $1 \leq i' \leq t'$. Indeed, for each $j \geq 4$, the Lie monomial $[a_2, a_1^l, a_3, \dots, a_{j-1}]$ is a factor in the interior partition of $M(s_{j,l})$ and thus it is a factor in the interior partition of the standard Lie monomial $M(s_{j,l} \circ d)$ in X_j for all d in $\mathcal{P}_m$. But this factor does not appear in the interior partition of $M(s_{j',l} \circ d)$ in $X_{j'}$ for all $j' \neq j$. Similarly, the factor $[a_2, a_1^l]$ which is in the interior partition of $M(s_{3,l} \circ d)$ does not appear in the interior partition of $M(s_{j',l} \circ d)$ for $j' \neq 3$. Moreover, as $d_1 \neq a_1$, $M(M(s_{1,l}) \circ d)$ with $d \in \mathcal{P}_m^{\sim a_1}$ and $M(M(s_{2,l}) \circ d)$ with $d \in \mathcal{P}_m$ belong to different fine homogenous components. Therefore, $X_j \cap X_{j'} = \emptyset \quad \forall j \neq j'$. Thus all the elements in $X_1 \cup X_2 \cup \dots \cup X_k$ are distinct elements in $\mathcal{H}$, and therefore

$$\{s_{1,l} \circ d : d \in \mathcal{P}_m^{\sim a_1}\} \cup \{s_{j,l} \circ d : d \in \mathcal{P}_m, j \neq 1\}$$

is a linearly independent set. Hence the claim follows. □

Note that in the case where $m = l + 1$, Lemma 6.10 is not true. For example, let $k = 4$, $l = 1$ and $m = l + 1 = 2$. If $d = [a_2, a_1]$ and $d' = a_2 a_3$, then by Lemma 6.5 and Lemma 5.6, we get $M(s_{1,1} \circ d) = [a_2, a_1, a_2, a_3^2, a_4^2, [a_2, a_1]] = M(s_{3,1} \circ d')$.

Lemma 6.11 *Let g be an element in $L[\mathcal{X}] \setminus J$. Then*

- (i) *if $g \in I$, then there exists a positive integer l such that $g\pi_{(s_{1,l}, \dots, s_{k,l})} \neq 0$, and*
- (ii) *if g does not belong to the ideal I , then the evaluation at zero maps g to a non-zero element in L , that is $g\pi_{(\underbrace{0, 0, \dots, 0}_{k \text{ times}})} \neq 0$.*

Proof. (i) Let $g \in I \setminus J$. Then there exists an integer $l' \geq 0$ such that g belongs to the subalgebra B of the quotient I/J that is freely generated by the elements in (6.7) with $m \leq l'$. The evaluation homomorphism $\pi_{(s_{1,l}, \dots, s_{k,l})}$ maps these elements onto the following elements

$$s_{1,l}, \dots, s_{k,l}, [s_{1,l}, c, b_1, \dots, b_m], [s_{2,l}, b_0, \dots, b_m], \dots, [s_{k,l}, b_0, \dots, b_m], \quad (6.11)$$

where $b_i, c \in \{a_1, a_2, \dots, a_k\}$, $c \neq a_1$ and $0 \leq m \leq l'$. Let $\mathcal{B}_{l,l'}$ be the set of the elements in (6.11). We claim if l is sufficiently large, then $\mathcal{B}_{l,l'}$ is reduced in L and thus it is independent. Suppose this is not true. Then there exists an element $u \in \mathcal{B}_{l,l'}$ such that

$$u = \sum_j \lambda_j w_j \quad (6.12)$$

where w_j is a Lie product in $\mathcal{B}_{l,l'} \setminus \{u\}$ with $\deg(u) = \deg(w_j)$, and $\lambda_j \in K$. Since the degrees of the elements in $\mathcal{B}_{l,l'}$ are in the range $2l + 2k - 3$ ($= \deg(s_{j,l})$), and $2l + 2k - 2 + l'$ (the degree of the elements (6.11) with $m = l'$), when l is large enough, all the w_j in (6.12) are actual elements from $\mathcal{B}_{l,l'} \setminus \{u\}$. But this contradicts Lemma 6.10 which states that $\mathcal{B}_{l,l'}$ is linearly independent. Thus any u in $\mathcal{B}_{l,l'}$ can not be in the subalgebra generated by the set $\mathcal{B}_{l,l'} \setminus \{u\}$ and the claim follows. Hence the restriction of $\pi_{(s_{1,l}, \dots, s_{k,l})}$ to B maps an independent set into an independent set in the codomain. Therefore it is injective for a sufficiently large l and thus $g\pi_{(s_{1,l}, \dots, s_{k,l})} \neq 0$.

(ii) In the case where $g \notin I$, we write $g = u + v$ where $u \in I/J$ and $0 \neq v \in L$. Then

$$g\pi_{\underbrace{(0,0,\dots,0)}_{k \text{ times}}} = u\pi_{\underbrace{(0,0,\dots,0)}_{k \text{ times}}} + v\pi_{\underbrace{(0,0,\dots,0)}_{k \text{ times}}}.$$

Since $v\pi_{\underbrace{(0,0,\dots,0)}_{k \text{ times}}} = v \neq 0$, the result follows. $\square$

Theorem 6.12 *Let $L[\mathcal{X}]$ be the free Lie algebra that is freely generated by $x_1, x_2, \dots, x_k, a_1, a_2, \dots, a_k, \dots$, and let V be the solution space of equation (3.2) over the free Lie algebra $L = L(a_1, a_2, \dots, a_k, \dots)$. Then*

- (i) $\text{Rad}(V) = \text{Id}([x_1, a_1] + [x_2, a_2] + \dots + [x_k, a_k])$ in $L[\mathcal{X}]$.

(ii) *The coordinate algebra of V is isomorphic to the semidirect sum*

$$\Gamma(V) = L \oplus \text{Id}(x_1, x_2, \dots, x_k) / \text{Id}([x_1, a_1] + [x_2, a_2] + \dots + [x_k, a_k]),$$

where $\text{Id}(x_1, x_2, \dots, x_k)$ and $\text{Id}([x_1, a_1] + [x_2, a_2] + \dots + [x_k, a_k])$ are ideals in $L[\mathcal{X}]$.

Proof. (i) To prove this equality, we apply Lemma 6.2. Firstly, since V is the solution space of equation (3.2), $[x_1, a_1] + [x_2, a_2] + \dots + [x_k, a_k] \subseteq \text{Rad}(V)$ and thus $\text{Id}([x_1, a_1] + [x_2, a_2] + \dots + [x_k, a_k]) \subseteq \text{Rad}(V)$. By Lemma 6.11 we can guarantee that for all $g \in L[\mathcal{X}] \setminus J$ and a sufficiently large l , the element $s = (s_{1,l}, \dots, s_{k,l})$ in V , which is given in Theorem 6.4, satisfies $g\pi_s \neq 0$. Therefore

$$\text{Rad}(V) = \text{Id}([x_1, a_1] + [x_2, a_2] + \dots + [x_k, a_k]),$$

as required.

(ii) This can be obtained by a straight forward application of (i) and Corollary 6.8.

□

6.4 Bilinear equations over free Lie algebras

Linear equations is the main topic of this research. However, in the last section of this thesis we consider a particular bilinear equation over a free Lie algebra L and apply the results from the previous section to determine the radical and the coordinate algebra of the solution space of

$$[x_1, x_2] + [x_3, x_4] + \dots + [x_{2k-1}, x_{2k}] = 0, \quad (6.13)$$

where each x_i is an indeterminate in $\mathcal{X} = \{x_1, x_2, \dots, x_{2k}\}$ and $k \geq 2$. Let $L[\mathcal{X}]$ be the free Lie algebra on $\mathcal{X} \cup \mathcal{A}$, where $\mathcal{A} = \{a_1, a_2, \dots, a_k, \dots\}$ is a free generating set of L , and denote by $L[x_2, x_4, \dots, x_{2k}]$ the free Lie algebra on $\mathcal{A} \cup \{x_2, x_4, \dots, x_{2k}\}$. Let $\tilde{s}_{j,l}$ denote the elements which are obtained by replacing $a_1, a_2, \dots, a_k$ by $x_2, x_4, \dots, x_{2k}$ in $s_{j,l}$ (in Theorem 6.4), respectively. If $J = \text{Id}([x_1, x_2] + [x_3, x_4] + \dots + [x_{2k-1}, x_{2k}])$, then by Lemma 6.11, for any g in $L[\mathcal{X}] \setminus J$ there exists a sufficiently large positive integer l such that $g\phi_l \neq 0$, where $\phi_l : L[\mathcal{X}] \rightarrow L[x_2, x_4, \dots, x_{2k}]$ is a Lie homomorphism such

that $b\phi_l = b$ for all b in $\mathcal{A}$, $x_j\phi_l = x_j$ for even j , and $x_j\phi_l = \tilde{s}_{\frac{j+1}{2},l}$ when j is odd. Therefore there is a homomorphism $\theta : L[x_2, x_4, \dots, x_{2k}] \rightarrow L$ such that θ maps b to itself for all $b \in \mathcal{A}$ and $g\phi_l\theta \neq 0$ in L . Indeed, the following diagram commutes

$$\begin{array}{ccc} L[\mathcal{X}] & & \\ \phi_l \downarrow & \searrow \pi_s & \\ L[x_2, \dots, x_{2k}] & \xrightarrow{\theta} & L \end{array}$$

where $\pi_s = \phi_l\theta$ is the evaluation homomorphism at the point $s = (\tilde{s}_{1,l}\theta, x_2\theta, \tilde{s}_{2,l}\theta, x_4\theta, \dots, \tilde{s}_{k,l}\theta, x_{2k}\theta) \in L^{\oplus 2k}$. From the construction of s , we can see that s is a solution of the bilinear equation (6.13) and the following theorem holds.

Theorem 6.13 *Let V be the solution space of the bilinear equation (6.13) over the free Lie algebra L . Then*

- (i) *the radical of V is equal to the ideal generated by the element $[x_1, x_2] + [x_3, x_4] + \dots + [x_{2k-1}, x_{2k}]$ in $L[\mathcal{X}]$,*
- (ii) *the coordinate algebra of V is the quotient*

$$\Gamma(V) = L[\mathcal{X}] / \text{Id}([x_1, x_2] + [x_3, x_4] + \dots + [x_{2k-1}, x_{2k}]).$$

Proof. (i) It is clear that $\text{Id}([x_1, x_2] + [x_3, x_4] + \dots + [x_{2k-1}, x_{2k}]) \subseteq \text{Rad}(V)$, and thus the result follows from the above discussion and Lemma 6.2.

(ii) As $\text{Rad}(V) = \text{Id}([x_1, x_2] + [x_3, x_4] + \dots + [x_{2k-1}, x_{2k}])$, the required result holds directly from the coordinate algebra definition. $\square$

Appendix A

Tables of dimensions

The following tables give the dimension of V_n and the dimensions of some of its corresponding V_q where $q \models n+1$, $n = 1, 2, 3, 4, 5$ and $k = 3, 4, 5$.

$k = 3$

n	1	2	3	4	5
$\dim V_n$	6	1	6	6	28

$n = 1, 2$

q	(2,0,0)	(1,1,0)	(3,0,0)	(2,1,0)
$\dim V_q$	1	1	0	0

$n = 3$

q	(4,0,0)	(3,1,0)	(2,2,0)	(2,1,1)
$\dim V_q$	0	0	1	1

$n = 4$

q	(5,0,0)	(4,1,0)	(3,1,1)	(2,2,1)
$\dim V_q$	0	0	1	1

$n = 5$

q	(6,0,0)	(3,2,1)	(4,1,1)	(2,2,2)
$\dim V_q$	0	2	1	4

$k = 4$

n	1	2	3	4	5
$\dim V_n$	10	4	20	36	146

 $n = 1, 2$

q	(2,0,0,0)	(1,1,0,0)	(2,1,0,0)	(1,1,1,0)
$\dim V_q$	1	1	0	1

 $n = 3$

q	(3,1,0,0)	(2,2,0,0)	(1,1,1,1)	(2,1,1,0)
$\dim V_q$	0	1	2	1

 $n = 4$

q	(4,1,0,0)	(3,2,0,0)	(2,2,1,0)	(2,1,1,1)
$\dim V_q$	0	0	1	3

 $n = 5$

q	(2,2,1,1)	(3,3,0,0)	(3,1,1,1)	(2,2,2,0)
$\dim V_q$	6	1	4	4

 $k = 5$

n	1	2	3	4	5
$\dim V_n$	15	10	50	126	540

 $n = 1, 2$

q	(1,1,0,0,0)	(3,0,0,0,0)	(2,1,0,0,0)	(1,1,1,0,0)
$\dim V_q$	1	0	0	1

$n = 3$

q	(4,0,0,0,0)	(2,2,0,0,0)	(1,1,1,1,0)	(2,1,1,0,0)
$\dim V_q$	0	1	2	1

 $n = 4$

q	(3,2,0,0,0)	(2,1,1,1,0)	(3,1,1,0,0)	(1,1,1,1,1)
$\dim V_q$	0	3	1	6

 $n = 5$

q	(5,1,0,0,0)	(4,2,0,0,0)	(2,1,1,1,1)	(3,2,1,0,0)
$\dim V_q$	0	1	12	2

Appendix B

More concrete solutions

B.1 Series of solutions

The following is a proof of Lemma 3.9:

Proof. We use induction on m to prove this result. At $m = 1$, we have

$$[v, u] = (-1)^0 \binom{1}{0} u^0 v u - \binom{1}{1} u v u^0 = v u - u v.$$

Suppose the result holds at m . Then

$$\begin{aligned} [v, u^{m+1}] &= [v, u^m, u] \\ &= \sum_{i=0}^m (-1)^i \binom{m}{i} (u^i v u^{m+1-i} - u^{i+1} v u^{m-i}) \\ &= \binom{m}{0} v u^{m+1} - \left(\binom{m}{0} + \binom{m}{1} \right) u v u^m \\ &\quad + \left(\binom{m}{1} + \binom{m}{2} \right) u^2 v u^{m-1} + \dots \\ &\quad + (-1)^m \left(\binom{m}{m-1} + \binom{m}{m} \right) u^m v u + (-1)^{m+1} u^{m+1} v \end{aligned}$$

$$\begin{aligned}
&= \binom{m+1}{0} vu^{m+1} - \binom{m+1}{1} uvu^m + \binom{m+1}{2} u^2vu^{m-1} - \dots \\
&\quad + (-1)^m \binom{m+1}{m} u^m vu + (-1)^{m+1} \binom{m+1}{m+1} u^{m+1}v \\
&= \sum_{i=0}^{m+1} (-1)^i \binom{m+1}{i} u^i v u^{m+1-i}.
\end{aligned}$$

□

Let L be the free Lie algebra on $\mathcal{A} = \{a, b\}$ with $a < b$ over a field K . All the solutions in this part can be verified by using Lemma 3.9, Corollary 3.10 and the anticommutativity of the Lie product.

Proposition B.1 *Let $s_{1,m}$ and $s_{2,m}$ be given as follows:*

$$\begin{aligned}
s_{1,m} &= \sum_{i=0}^{m-1} (-1)^i \binom{m}{i} [b, a^m, b^4, a^i, b, a^{m-1-i}], \\
s_{2,m} &= (-1)^m [b, a^m, b^4, a^m] + \sum_{i=0}^m (-1)^i \binom{m}{i} [b, a^m, b^2, a^i, b, a^{m-i}, b] \\
&\quad + 2 \sum_{i=0}^m (-1)^{i+1} \binom{m}{i} [b, a^m, b^3, a^i, b, a^{m-i}].
\end{aligned}$$

Then $(s_{1,m}, s_{2,m})$ is a non-zero solution of Eq. (3.12) in $V_{(2m,6)}$, for $m \geq 1$.

Proposition B.2 *Let $m \geq 2$. Then $(s_{1,m}, s_{2,m})$ is a non-zero solution of Eq. (3.12), where $s_{1,m} = [b, a^m, [b, a], s'_{1,m}]$, $s_{2,m} = [b, a^m, [b, a], s'_{2,m}]$ and $s'_{1,m}$, $s'_{2,m}$ are given as follows:*

$$\begin{aligned}
s'_{1,m} &= \sum_{i=0}^m (-1)^i \binom{m}{i} a^i b a^{m-i} b + \sum_{i=0}^{m-1} (-1)^{i+1} \binom{m}{i} b a^{i+1} b a^{m-i-1} \\
&\quad + \sum_{i=0}^{m-1} (-1)^i \binom{m}{i} a b a^i b a^{m-i-1},
\end{aligned}$$

$$s'_{2,m} = (-1)^{m+1}ba^{m+1} + (-1)^m aba^m + \sum_{i=0}^m (-1)^{i+1} \binom{m}{i} a^i ba^{m-i+1}.$$

B.2 A basis of V_7 when $k = 2$

Let L be the free Lie algebra on $\{a, b\}$ (with $a < b$) over a field K of characteristic zero. Then

$$\begin{aligned} \mathcal{H}_7 = \{ & [b, a, a, a, a, a, a], [b, a, a, a, a, a, b], [b, a, a, a, a, b, b], [b, a, a, a, b, b, b], \\ & [b, a, a, b, b, b, b], [b, a, b, b, b, b, b], [b, a, a, a, a, [b, a]], [b, a, a, a, b, [b, a]], [b, a, a, b, b, [b, a]], \\ & [b, a, b, b, b, [b, a]], [b, a, a, a, [b, a, a]], [b, a, a, a, [b, a, b]], [b, a, a, b, [b, a, a]], [b, a, a, b, [b, a, b]], \\ & [b, a, b, b, [b, a, a]], [b, a, b, b, [b, a, b]], [b, a, a, [b, a], [b, a]], [b, a, b, [b, a], [b, a]] \}. \end{aligned}$$

By applying the procedure that has been discussed earlier in Section 3.4.2, we write

$$\begin{aligned} & A[b, a, a, a, a, a, a] + B[b, a, a, a, a, a, b] + C[b, a, a, a, a, b, b, a] \\ & + D[b, a, a, a, b, b, b, a] + E[b, a, a, b, b, b, b, a] + F[b, a, b, b, b, b, b, a] \\ & + G[b, a, a, a, a, [b, a], a] + H[b, a, a, a, b, [b, a], a] + I[b, a, a, b, b, [b, a], a] \\ & + J[b, a, b, b, b, [b, a], a] + \tilde{K}[b, a, a, a, [b, a, a], a] + L[b, a, a, a, [b, a, b], a] \\ & + M[b, a, a, b, [b, a, a], a] + N[b, a, a, b, [b, a, b], a] + O[b, a, b, b, [b, a, a], a] \\ & + P[b, a, b, b, [b, a, b], a] + Q[b, a, a, [b, a], [b, a], a] + R[b, a, b, [b, a], [b, a], a] \\ & + S[b, a, a, a, a, a, a, b] + T[b, a, a, a, a, a, b, b] + U[b, a, a, a, a, b, b, b] \\ & + V[b, a, a, a, b, b, b, b] + W[b, a, a, b, b, b, b, b] + X[b, a, b, b, b, b, b, b] \\ & + Y[b, a, a, a, a, [b, a], b] + Z[b, a, a, a, b, [b, a], b] + \bar{A}[b, a, a, b, b, [b, a], b] \\ & + \bar{B}[b, a, b, b, b, [b, a], b] + \bar{C}[b, a, a, a, [b, a, a], b] + \bar{D}[b, a, a, a, [b, a, b], b] \\ & + \bar{E}[b, a, a, b, [b, a, a], b] + \bar{F}[b, a, a, b, [b, a, b], b] + \bar{G}[b, a, b, b, [b, a, a], b] \\ & + \bar{H}[b, a, b, b, [b, a, b], b] + \bar{I}[b, a, a, [b, a], [b, a], b] + \bar{J}[b, a, b, [b, a], [b, a], b] = 0, \end{aligned}$$

where $A, B, \dots, J, \tilde{K}, L, \dots, Z, \bar{A}, \dots, \bar{J} \in K$. We rewrite the left-hand side of this equation in the standard form. Then by collecting the coefficients of each standard

Lie monomial in $\mathcal{H}_8$, we get systems of linear homogenous equations in K . We use a software package (Maple [19]) to solve these systems.

$$A = X = 0,$$

$$S = G = -B, \tilde{K} = B,$$

$$C = -\overline{C}, H = M = -\frac{1}{2}\overline{C}, T = \overline{C}, L = -\frac{3}{2}\overline{C}, Q = \frac{1}{2}\overline{C}, Y = \frac{5}{2}\overline{C},$$

$$D = \frac{1}{4}\overline{D} - \frac{3}{4}\overline{I}, I = \frac{1}{4}\overline{D} - \frac{3}{4}\overline{I}, N = -\frac{3}{4}\overline{D} - \frac{3}{4}\overline{I}, O = \frac{5}{4}\overline{D} - \frac{3}{4}\overline{I}, \overline{E} = -\frac{3}{2}\overline{D} + \frac{3}{2}\overline{I},$$

$$R = -\frac{1}{2}\overline{D} + \frac{1}{2}\overline{I}, U = -\frac{1}{4}\overline{D} + \frac{3}{4}\overline{I}, Z = -\overline{D} + 3\overline{I},$$

$$V = \overline{J}, E = P = -\overline{J}, \overline{A} = \frac{9}{2}\overline{J}, \overline{F} = \frac{5}{2}\overline{J}, J = \overline{G} = -\frac{1}{2}\overline{J},$$

$$W = \overline{H}, F = -\overline{H}, \overline{B} = 4\overline{H}.$$

Put $B = \alpha$, $\overline{C} = \beta$, $\overline{I} = \gamma$, $\overline{D} = \delta$, $\overline{J} = \theta$, $\overline{H} = \sigma$. Then (s_1, s_2) is a solution of equation (3.12), where

$$\begin{aligned} s_1 = & \alpha[b, a, a, a, a, a, b] - \beta[b, a, a, a, a, b, b] + (\frac{1}{4}\delta - \frac{3}{4}\gamma)[b, a, a, a, b, b, b] \\ & - \theta[b, a, a, b, b, b, b] - \sigma[b, a, b, b, b, b, b] - \alpha[b, a, a, a, a, [b, a]] \\ & - \frac{1}{2}\beta[b, a, a, a, b, [b, a]] + (\frac{1}{4}\delta - \frac{3}{4}\gamma)[b, a, a, b, b, [b, a]] - \frac{1}{2}\theta[b, a, b, b, b, [b, a]] \\ & + (-\frac{3}{4}\delta - \frac{3}{4}\gamma)[b, a, a, b, [b, a, b]] + (\frac{5}{4}\delta - \frac{3}{4}\gamma)[b, a, b, b, [b, a, a]] - \theta[b, a, b, b, [b, a, b]] \\ & + \alpha[b, a, a, a, [b, a, a]] - \frac{3}{2}\beta[b, a, a, a, [b, a, b]] - \frac{1}{2}\beta[b, a, a, b, [b, a, a]] \\ & + \frac{1}{2}\beta[b, a, a, [b, a], [b, a]] + (-\frac{1}{2}\delta + \frac{1}{2}\gamma)[b, a, b, [b, a], [b, a]], \end{aligned}$$

$$\begin{aligned}
s_2 = & -\alpha[b, a, a, a, a, a, a] + \beta[b, a, a, a, a, a, b] + (-\tfrac{1}{4}\delta + \tfrac{3}{4}\gamma)[b, a, a, a, a, b, b] \\
& + (-\delta + 3\gamma)[b, a, a, a, b, [b, a]] + \tfrac{9}{2}\theta[b, a, a, b, b, [b, a]] + 4\sigma[b, a, b, b, b, [b, a]] \\
& + \beta[b, a, a, a, [b, a, a]] + \delta[b, a, a, a, [b, a, b]] + (-\tfrac{3}{2}\delta + \tfrac{3}{2}\gamma)[b, a, a, b, [b, a, a]] \\
& + \theta[b, a, a, a, b, b, b] + \sigma[b, a, a, b, b, b, b] + \tfrac{5}{2}\beta[b, a, a, a, a, [b, a]] \\
& + \tfrac{5}{2}\theta[b, a, a, b, [b, a, b]] - \tfrac{1}{2}\theta[b, a, b, b, [b, a, a]] + \sigma[b, a, b, b, [b, a, b]] \\
& + \gamma[b, a, a, [b, a], [b, a]] + \theta[b, a, b, [b, a], [b, a]],
\end{aligned}$$

for $\alpha, \beta, \gamma, \delta, \theta, \sigma \in K$. As we can see, $\dim V_7 = 6$ as it is given in Section 3.2.1.

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